Notes: STAT 38100 (W26) Measure-Theoretic Probability I

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1 Measurable Spaces and σ-Fields

Example 1.1 (A Nonmeasurable Event). Let $\mathbb{P} = \text{Uniform}[0, 2\pi)$. We have not formally defined $\mathbb{P}$, but for now assume the usual properties of a uniform distribution hold, including countable additivity and translation invariance. Consider the function

$$T(x) := \begin{cases} x + 1, & x + 1 < 2\pi \\ x + 1 - 2\pi, & x + 1 \geq 2\pi \end{cases}.$$

This induces an equivalence relation on $[0, 2\pi)$ by $x \sim y$ if $T^k(x) = y$ for some $k \in \mathbb{Z}$, which partitions the set $[0, 2\pi)$ into equivalence classes, say $\{A_\lambda\}_{\lambda \in \Lambda}$. For each λ, select an arbitrary $x_\lambda \in A_\lambda$ and define $X := \{x_\lambda : \lambda \in \Lambda\}$. [In doing so, we are invoking the Axiom of Choice.]

Now write $X + i := \{T(x_\lambda) : \lambda \in \Lambda\}$. Note that $\{X + i : i \in \mathbb{Z}\}$ is another partition of $[0, 2\pi)$. We have

$$1 = \mathbb{P}([0, 2\pi)) = \sum_{i \in \mathbb{Z}} \mathbb{P}(X + i) = \sum_{i \in \mathbb{Z}} \mathbb{P}(X).$$

What is $\mathbb{P}(X)$?

1.1 σ-Fields

We next construct a measure space. Fix a set Ω, called the **sample space**. Next, write 2^Ω for all subsets of Ω.

Definition 1.2. Fix Ω. We say $\mathcal{F} \subset 2^\Omega$ is a **σ-field** (or σ-algebra) if

(i) $\varnothing \in \mathcal{F}$.

(ii) If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.

(iii) If $A_n \in \mathcal{F}$, then $\bigcup_n A_n \in \mathcal{F}$, where the union is at most countable.

Example 1.3. For any Ω, the largest σ-field is 2^Ω and the smallest is $\{\varnothing, \Omega\}$.

Definition 1.4. Given $\mathcal{E} \subset 2^\Omega$, we define $\sigma(\mathcal{E})$ to be the smallest σ-field containing $\mathscr{E}$, i.e., the intersection of all σ-fields containing $\mathscr{E}$. This is called the **σ-field generated by $\mathcal{E}$**.

To see that the intersection construction is valid, we need the following observation:

Lemma 1.5. *The intersection of any collection of σ-fields is a σ-field.*

1.2 Limits of Sets

Definition 1.6. Fix a sequence of sets A_n.

- If $A_n \subset A_{n+1}$ for all n, we say the sequence is **increasing** ($A_n \uparrow$) and define the limit as $\lim_{n\to\infty} A_n := \bigcup_n A_n$.
- If $A_n \supset A_{n+1}$ for all n, we say the sequence is **decreasing** ($A_n \downarrow$) and define the limit as $\lim_{n\to\infty} A_n := \bigcap_n A_n$.

In either case, we say the sequence is **monotone**.

Definition 1.7. For an arbitrary sequence of sets A_n, we define

$$\liminf_n A_n := \bigcup_k \bigcap_{n\geq k} A_n, \quad \limsup_n A_n := \bigcap_k \bigcup_{n\geq k} A_n.$$

If $\liminf_n A_n = \limsup_n A_n$, we say the limit exists and define

$$\lim_{n\to\infty} A_n := \liminf_n A_n = \limsup_n A_n.$$

Remark 1.8. This coincides with the definitions of lim inf and lim sup for functions if we identify sets with their indicator functions. ☕

Note the following:

Proposition 1.9. *$x \in \liminf A_n$ if and only if there exists N such that for all $n \geq N$, $x \in A_n$. That is, x belongs to all but finitely many of the sets A_n.*

Similarly, $x \in \limsup A_n$ if and only if for each k, $x \in A_n$ for some $n \geq k$. That is, s belongs to infinitely many A_n's.

From this we see the

Corollary 1.10. $\liminf A_n \subset \limsup A_n$.

1.3 Other Structures on Sets

Definition 1.11.

- We say $\mathcal{P}$ is a **π-system** if it is closed under finite intersections, i.e., if $A, B \in \mathcal{P}$, then $A \cap B \in \mathcal{P}$.
- We say $\mathcal{M}$ is a **monotone class** if for each monotone sequence A_n in $\mathcal{M}$, the limit $\lim A_n \in \mathscr{M}$.
- We say $\mathcal{L}$ is a **λ-system** if
  - $\Omega \in \mathcal{L}$.
  - If $A, B \in \mathcal{L}$ and $A \subset B$, then $B \setminus A \in \mathscr{L}$.
  - If $A_n \in \mathcal{M}$, $A_n \uparrow$, then $\lim A_n \in \mathcal{L}$.
- We say $\mathcal{A}$ is a **field** (or algebra) if
  - $\varnothing \in \mathcal{A}$.
  - If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$.

– If $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$.

For each structure, given a set $\mathcal{E}$, we can define the smallest such structure containing $\mathcal{E}$ by taking intersections. Write $m(\mathcal{E})$ and $\lambda(\mathcal{E})$, respectively, for the smallest monotone class and λ-system containing $\mathscr{E}$.

Two of these structures are about finite operations, the other two allow countable operations. In some cases a mix of these two structures gives us a σ-field.

Theorem 1.12 (Monotone Class Theorem)**.** *If $\mathcal{A}$ is a field, then $m(\mathcal{A}) = \sigma(\mathcal{A})$.*

Proof. We first show $m(\mathcal{A}) \subset \sigma(\mathcal{A})$. It is sufficient to show $\sigma(\mathcal{A})$ is a monotone class. Closure under increasing limits is clear; closure under decreasing limits follows from De Morgan's law: $\cap A_n = \left(\cup A_n^{\mathsf{c}}\right)^{\mathsf{c}}$.

Next, we show $\sigma(\mathcal{A}) \subset m(\mathcal{A})$ by arguing $m(\mathcal{A})$ is a σ-field. We split this into the following steps.

Step 1 ($\varnothing \in \mathcal{A}$): This is clear.

Step 2 (Closure under complements): We need only argue that[1]

$$\mathcal{M}_0 \coloneqq \{A \in m(\mathcal{A}) : A^{\mathsf{c}} \in m(\mathcal{A})\} = m(\mathcal{A}).$$

It is clear that $\mathcal{M}_0 \subset m(\mathcal{A})$. To see the reverse inclusion, we need only show $\mathcal{M}_0 \supset \mathcal{A}$ and that $\mathcal{M}_0$ is a monotone class:

- $\mathcal{M}_0 \supset \mathcal{A}$ is obvious since $\mathcal{A}$ is closed under complements.
- Fix any monotone sequence A_n in $\mathcal{M}_0$. Since each $A_n \in m(\mathcal{A})$ we have $\lim A_n \in m(\mathcal{A})$. The complement sequence A_n^{c} is also monotone, so $\lim A_n^{\mathsf{c}} \in m(\mathcal{A})$, giving $(\lim A_n)^{\mathsf{c}} = \lim A_n^{\mathsf{c}} \in m(\mathcal{A})$. Thus $\lim A_n \in \mathcal{M}_0$.

Step 3 (Closure under finite union): It suffices to show

$$\mathcal{M}_2 \coloneqq \{A \in m(\mathcal{A}) : A \cup B \in m(\mathcal{A}) \text{ for each } B \in m(\mathcal{A})\} = m(\mathcal{A}).$$

It is clear that $\mathcal{M}_2 \subset m(\mathcal{A})$. To see the reverse inclusion, we need only show $\mathcal{M}_2 \supset \mathcal{A}$ and that $\mathcal{M}_2$ is a monotone class. The second statement is straightforward (ex.). To check the first, we need to show that for each $A \in \mathcal{A}$ and $B \in m(\mathcal{A})$, $A \cup B \in m(\mathcal{A})$. This is clearly true when we restrict $B \in \mathcal{A}$.

Let's ditch $\mathcal{M}_2$ for now and define

$$\mathcal{M}_1 \coloneqq \{A \in m(\mathcal{A}) : A \cup B \in m(\mathcal{A}) \text{ for all } B \in \mathcal{A}\}.$$

Note that the requirement of $\mathcal{M}_1$ is weaker than that of $\mathcal{M}_2$, so $\mathcal{M}_2 \subset \mathcal{M}_1$. We will show $\mathcal{M}_1 = m(\mathcal{A})$. Clearly $\mathcal{M}_1 \subset m(\mathcal{A})$. To establish the reverse inclusion, we show $\mathcal{M}_1 \supset \mathcal{A}$ and $\mathcal{M}_1$ is a monotone class. We check $\mathcal{M}_1 \supset \mathcal{A}$: for each $A, B \in \mathcal{A}$, we have $A \cup B \in \mathcal{A} \subset m(\mathcal{A})$, which gives $A \in \mathcal{M}_1$. That $\mathcal{M}_1$ is a monotone class is left as an exercise (same argument as before).

[1] This is a standard argument in measure theory.

With $\mathcal{M}_1 = m(\mathcal{A})$, let's return to showing $\mathcal{M}_2 \supset \mathcal{A}$. Fix arbitrary $A \in \mathcal{A}$ and $B \in m(\mathcal{A}) = \mathcal{M}_1$. By the definition of $\mathcal{M}_1$, we have $B \cup A \in m(\mathcal{A})$, as desired.
Step 4 (Closure under countable union): Fix countable sequence $A_n \in m(\mathcal{A})$. The partial (finite) unions $\bigcup_{k=1}^{n} A_n$ are in $m(\mathcal{A})$ by Step 3. Since $m(\mathcal{A})$ is a monotone class, the limit $\bigcup_n A_n \in m(\mathcal{A})$. □

Theorem 1.13 (Dynkin's π-λ Theorem)**.** *If $\mathcal{P}$ is a π-system, then $\lambda(\mathcal{P}) = \sigma(\mathcal{P})$.*

Proof. Same strategy as that of the Monotone Class Theorem; left as an exercise. □

1.4 Measurable Spaces

Definition 1.14. A pair $(\Omega, \mathcal{F})$ is called a **measurable space** if Ω is a set and $\mathcal{F}$ is a σ-field on Ω.

A special case is when Ω is a topological space and $\mathcal{F}$ is generated by the open sets of Ω, called the **Borel σ-field** and denoted $\mathcal{B}(\Omega)$.

Example 1.15. Consider $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$.

$$\begin{aligned}
\mathcal{B}_{\mathbb{R}} &= \sigma(\text{open sets in } \mathbb{R}) \\
&= \sigma\left(\{(a,b) : a < b, a, b \in \mathbb{R}\}\right) \\
&= \sigma\left(\{[a,\infty) : a \in \mathbb{R}\}\right) \\
&= \sigma\left(\{[a,\infty) : a \in \mathbb{Q}\}\right).
\end{aligned}$$

2 Measures

Definition 2.1. Fix $\varnothing \in \mathcal{E} \subset 2^{\Omega}$. We say $\mu : \mathcal{E} \to [0, \infty]$ is a **measure** if

- $\mu(\varnothing) = 0$.
- (**Countable additivity**) If $A_n \in \mathcal{E}$ are disjoint and $\bigcup A_n \in \mathcal{E}$, then
$$\mu\left(\bigcup A_n\right) = \sum \mu(A_n).$$

In addition:

- If $\mu(\Omega) < \infty$, then μ is called a **finite measure**.
- If $\mu(\Omega) = 1$, then μ is called a **probability measure**.
- If for each $A \in \mathcal{E}$, there exists $A_n \in \mathcal{E}$, $\mu(A_n) < \infty$ such that $\bigcup A_n \supset A$, then μ is called **σ-finite**.
- Given a measure space $(\Omega, \mathcal{F})$ and a measure μ on $\mathcal{F}$, the triple $(\Omega, \mathcal{F}, \mu)$ is called a **measure space**.

Some properties:

Proposition 2.2. *Let μ be a measure on a field $\mathcal{F}$. We have the following:*

(i) (Monotonicity): If $A \subset B$, then $\mu(A) \leq \mu(B)$.

(ii) If $A_n \uparrow$, then $\mu(\lim A_n) = \lim \mu(A_n)$.

(iii) If $A_n \downarrow$ and $\mu(A_1) < \infty$, then $\mu(\lim A_n) = \lim \mu(A_n)$.

Proof.

(i) $\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$.

(ii) Note that $A_n = A_1 \cup (A_2 \setminus A_1) \cup \cdots \cup (A_n \setminus A_{n-1})$. We have, with the convention $A_0 := \varnothing$,
$$\lim A_n = \lim_n \bigcup_{m=1}^{n} (A_m \setminus A_{m-1}) = \bigcup_{m=1}^{\infty} (A_m \setminus A_{m-1}).$$
Thus,
$$\mu(\lim A_n) = \sum_{m=1}^{\infty} \mu(A_m \setminus A_{m-1}) = \lim_{n\to\infty} \sum_{m=1}^{n} \mu(A_m \setminus A_{m-1}) = \lim_{n\to\infty} \mu(A_n).$$

(iii) Applying the previous result to $(A_1 \setminus A_n) \uparrow$ yields
$$\mu\left(\lim A_1 \setminus A_n\right) = \lim_n \mu\left(A_1 \setminus A_n\right).$$
Note that for each n, $\mu(A_1 \setminus A_n) = \mu(A_1) - \mu(A_n)$ and
$$\mu\left(\lim A_1 \setminus A_n\right) = \mu\left(A_1 \setminus \lim A_n\right) = \mu(A_1) - \mu(\lim A_n).$$
Thus we have
$$\mu(A_1) - \mu(\lim A_n) = \mu(A_1) - \lim_n \mu(A_n).$$

□

2.1 Construction of Measures

Our next goal is to construct the uniform distribution $[0, 1]$. Our strategy is as follows:

(i) We construct a measure on a field $\mathcal{A}$.

(ii) We extend the measure from $\mathcal{A}$ to 2^{Ω} as an outer measure τ.

(iii) We find a restriction of τ to a σ-field $\mathcal{F}_\tau$ containing $\mathcal{A}$ such that τ is a measure on $\mathcal{F}_\tau$.

This is the general strategy of constructing a measure on a σ-field.

Example 2.3. Let $\Omega := [0, 1]$. A natural choice of field $\mathcal{A}$ is the set of all finite unions of intervals.

Q: how to extend a measure from $\mathcal{A}$ to $\sigma(\mathcal{A})$, and is the extension unique?

Theorem 2.4 (Uniqueness of Measure Extension)**.** *If two finite measures μ and ν agree on a field $\mathcal{A}$, then they agree on $\sigma(\mathcal{A})$.*

Proof. Let the sets for which μ and ν agree be

$$\mathcal{H} := \{A \in \sigma(\mathcal{A}) : \mu(A) = \nu(A)\}.$$

It is clear that $\mathcal{A} \subset \mathcal{H} \subset \sigma(\mathcal{A})$. We will argue $\mathcal{H} = \sigma(\mathcal{A})$ by showing that $\mathcal{H}$ is a monotone class: Fix a monotone sequence $A_n \in \mathcal{H}$. We have $\mu(A_n) = \nu(A_n)$ for each n. Taking limits and using the continuity properties of measures gives $\mu(\lim A_n) = \nu(\lim A_n)$. □

Definition 2.5. We say $\tau : 2^{\Omega} \to [0, \infty]$ is an **outer measure** if

(i) $\tau(\varnothing) = 0$.

(ii) (Monotonicity): If $A \subset B$, then $\tau(A) \leq \tau(B)$.

(iii) (Countable subadditivity/union bound): For each A_n, we have $\tau\left(\bigcup A_n\right) \leq \sum \tau(A_n)$.

Theorem 2.6 (Construction of Outer Measures)**.** *Fix $\varnothing \in \mathcal{E} \subset 2^{\Omega}$ and a function $\mu : \mathcal{E} \to [0, \infty]$ such that $\mu(\varnothing) = 0$. For each $A \subset \Omega$, define*

$$\tau(A) := \inf\left\{\sum \mu(A_n) : A_n \in \mathcal{E}, A \subset \bigcup_n A_n\right\},$$

with the convention that the infimum over an empty set is ∞. Then τ is an outer measure on Ω.

Proof. We check the defining properties of an outer measure.

(i) Since $\varnothing \subset \varnothing \cup \varnothing$, we have $\tau(\varnothing) \leq \sum \mu(\varnothing) = 0$. Since also τ is nonnegative, we have $\tau(\varnothing) = 0$.

(ii) Let $A \subset B$. Any cover of B must necessarily cover A. Thus the infimum over covers of B is at least that over covers of A, giving $\tau(A) \leq \tau(B)$.

(iii) Fix A_n and small ε. For each A_n, there exists a cover $\{B_{n,k}\}_k$ with $B_{n,k} \in \mathcal{E}$ such that

$$\tau(A_n) + 2^{-n}\varepsilon \geq \sum_k \mu(B_{n,k}).$$

Note that $\cup A_n$ is covered by $\{B_{n,k}\}_{n,k}$, and so

$$\tau(\cup A_n) \leq \sum_{n,k} \mu(B_{n,k}) \leq \sum_n [\tau(A_n) + 2^{-n}\varepsilon] = \sum_n \tau(A_n) + \varepsilon.$$

We send $\varepsilon \to 0$ to conclude.

□

Next, using the outer measure τ, we construct a measure by restricting τ to a suitable σ-field:

Theorem 2.7 (Carathéodory Extension)**.** *Let τ be an outer measure on Ω. Define*

$$\mathcal{F}_\tau := \left\{A \subset \Omega : \textit{for each } B \subset \Omega, \tau(B) = \tau(B \cap A) + \tau(B \cap A^{\mathsf{c}})\right\}.$$

Then $\mathcal{F}_\tau$ is a σ-field and $(\Omega, \mathcal{F}_\tau, \tau)$ is a measure space.

Proof. First, we show $\mathcal{F}_\tau$ is a σ-field.

(i) It is clear that $\varnothing \in \mathcal{F}_\tau$.

(ii) Closure of $\mathcal{F}_\tau$ under complementation is clear by symmetry.

(iii) We show closure under finite intersection as an intermediate step. Fix $A_1, A_2 \in \mathcal{F}_\tau$ and an arbitrary $B \subset \Omega$. We have[2]

$$\begin{aligned}\tau(B) &= \tau(B \cap A_1) + \tau(B \cap A_1^{\mathsf{c}}) \\ &= \tau(B \cap A_1 \cap A_2) + \tau(B \cap A_1 \cap A_2^{\mathsf{c}}) + \tau(B \cap A_1^{\mathsf{c}} \cap A_2) + \tau(B \cap A_1^{\mathsf{c}} \cap A_2^{\mathsf{c}}).\end{aligned}$$

Thus we need only show that the latter three terms sum to $\tau(B \cap (A_1 \cap A_2)^{\mathsf{c}})$:

$$\begin{aligned}\tau(B \cap (A_1 \cap A_2)^{\mathsf{c}}) &= \tau(B \cap (A_1^{\mathsf{c}} \cup A_2^{\mathsf{c}})) \\ &= \tau(B \cap A_1^{\mathsf{c}}) + \tau(B \cap A_2^{\mathsf{c}} \cap A_1) \\ &= \tau(B \cap A_1^{\mathsf{c}} \cap A_2) + \tau(B \cap A_1^{\mathsf{c}} \cap A_2^{\mathsf{c}}) + \tau(B \cap A_2^{\mathsf{c}} \cap A_1).\end{aligned}$$

Combining (ii) with (iii) gives closure under finite union.

(iv) Closure under countable union. We need only consider sequence A_n that are disjoint, since $\cup A_n = A_1 \cup (A_2 \setminus A_1) \cup \ldots$ and each $A_n \setminus \cup_{k=1}^{n-1} A_k \in \mathcal{F}_\tau$ by (ii) and (iii). For any $B \subset \Omega$ and m, we have

$$\tau(B) = \tau\left(B \cap (\cup_{n=1}^m A_n)\right) + \tau\left(B \cap (\cup_{n=1}^m A_n)^{\mathsf{c}}\right).$$

[2]Chao: We can cut repeatedly with a good knife.

The first term on the right can be written as (using the fact that A_n are disjoint)

$$\tau\left(B \cap (\cup_{n=1}^{m} A_n)\right) = \tau\left(B \cap A_1\right) + \tau\left(B \cap \left(\cup_{n=2}^{m} A_n\right)\right) = \cdots = \sum_{n=1}^{m} \tau(B \cap A_n).$$

We thus have by monotonicity that

$$\begin{aligned}\tau(B) &= \sum_{n=1}^{m} \tau(B \cap A_n) + \tau\left(B \cap (\cup_{n=1}^{m} A_n)^{\mathsf{c}}\right) \\ &\geq \sum_{n=1}^{m} \tau(B \cap A_n) + \tau\left(B \cap (\cup_{n=1}^{\infty} A_n)^{\mathsf{c}}\right).\end{aligned}$$

Sending $m \to \infty$ gives

$$\tau(B) \geq \sum_{n=1}^{\infty} \tau(B \cap A_n) + \tau\left(B \cap (\cup_{n=1}^{\infty} A_n)^{\mathsf{c}}\right).$$

By subadditivity,

$$\tau(B) \geq \tau\left(B \cap (\cup_{n=1}^{\infty} A_n)\right) + \tau\left(B \cap (\cup_{n=1}^{\infty} A_n)^{\mathsf{c}}\right).$$

The reverse inequality also holds by subadditivity.

Next, we check that the restriction of τ to $\mathcal{F}_\tau$ is a measure.

(i) $\tau(\varnothing) = 0$ by definition of outer measure.

(ii) Fix disjoint A_n. By subadditivity, we need only show $\tau(\cup A_n) \geq \sum \tau(A_n)$. Recall that we established in the previous part that

$$\tau(B) \geq \sum_{n=1}^{\infty} \tau(B \cap A_n) + \tau\left(B \cap (\cup_{n=1}^{\infty} A_n)^{\mathsf{c}}\right), \quad \forall B \subset \Omega.$$

Taking $B = \cup A_n$ gives the desired result.

□

We compete the construction of a measure space by showing uniqueness and that $\mathcal{F}_\tau$ is not too small.

Theorem 2.8. *Suppose μ is a finite measure on a field $\mathcal{A}$. Then, there exists a unique extension of μ from $\mathcal{A}$ to $\sigma(\mathcal{A})$.*

Proof. Construct outer measure τ and restrict it to $\mathcal{F}_\tau$ as before. Previous results show that $(\Omega, \mathcal{F}_\tau, \tau)$ is a measure space, and that it must be unique. We will show that $\mathcal{A} \subset \mathcal{F}_\tau$ (and thus $\sigma(\mathcal{A}) \subset \mathcal{F}_\tau$) and that τ agrees with μ on $\mathcal{A}$.

- τ agrees with μ on $\mathcal{A}$. Any set $A \subset \mathcal{A}$ is covered by itself, so $\tau(A) \leq \mu(A)$. For the reverse inequality, fix any cover $B_n \in \mathcal{A}$ and note that

  $$\mu(A) = \mu\left(\cup(A \cap B_n)\right) \leq \sum \mu(A \cap B_n) \leq \sum \mu(B_n)$$

  Take inf over all covers to conclude.

- $\mathcal{A} \subset \mathcal{F}_\tau := \{A \subset \Omega : \forall B \subset \Omega, \tau(B) = \tau(B \cap A) + \tau(B \cap A^c)\}$. Fix B. By subadditivity, we need only show $\tau(B) \geq \tau(B \cap A) + \tau(B \cap A^c)$. For each $\varepsilon > 0$, there exists a cover B_n of B such that

$$\begin{aligned}
\tau(B) + \varepsilon &\geq \sum \mu(B_n) \\
&= \sum_n \left[\mu(B_n \cap A) + \mu(B_n \cap A^c)\right] \\
&= \sum_n \left[\tau(B_n \cap A) + \tau(B_n \cap A^c)\right] \\
&\geq \tau\left((\cup B_n) \cap A\right) + \tau\left((\cup B_n) \cap A^c\right) \\
&\geq \tau(B \cap A) + \tau(B \cap A^c).
\end{aligned}$$

  Sending $\varepsilon \to 0$ concludes the proof.

□

2.2 Complete Measure Spaces

Let's now return to our original goal of constructing the uniform distribution on $[0, 1]$. Denote as $\mathcal{A}$ the field containing all finite unions of intervals in $[0, 1]$, on which μ is defined as the total length of the intervals. Using the results discussed above, we can extend μ to $\mathcal{F} \supset \sigma(\mathcal{A}) = \mathcal{B}_{[0,1]}$.

Definition 2.9. A measure space $(\Omega, \mathcal{F}, \mu)$ is **complete** if it contains all subsets of measure zero sets. That is, for each $A \subset \mathcal{F}$ such that $\mu(A) = 0$, if $B \subset A$, then $B \in \mathcal{F}$.

Theorem 2.10. *For an arbitrary outer measure τ, the measure space $(\Omega, \mathcal{F}_\tau, \tau)$ constructed from Carathéodory's extension theorem is complete.*

Proof. Let $A \subset \Omega$ be such that $\tau(A) = 0$. Fix arbitrary $B \subset \Omega$. We have $\tau(A) \leq \tau(B \cap A) + \tau(B \cap A^c)$ by subadditivity. Conversely,

$$\tau(B) = \tau(B) + \tau(A) \geq \tau(B \cap A^c) + \tau(B \cap A).$$

□

Example 2.11. $([0, 1], \mathcal{B}([0, 1]), \mu)$ is not complete.

We use the following facts: (i) $\mathcal{B}([0, 1])$ has the same cardinality as $[0, 1]$; (ii) the cantor set C has the same cardinality as $[0, 1]$.

Since C can be written as the intersection of a decreasing sequence of sets C_n, each a finite union of closed intervals, we have $C \in \mathcal{B}([0, 1])$. Note that $\mu(C) = 0$. Since

$$\left|2^C\right| = \left|2^{[0,1]}\right| > |[0, 1]| = |\mathcal{B}([0, 1])|,$$

there exists $A \subset C$ such that $A \notin \mathcal{B}([0, 1])$. However, $\mu(A) \leq \mu(C) = 0$.

Theorem 2.12 (Completion)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Define $\tilde{\mathcal{F}} := \{A \cup N : A \in \mathcal{F}, \exists B \in \mathcal{F}$ st $\mu(B) = 0, N \subset B\}$. For every $(A \cup N) \in \tilde{\mathcal{F}}$, define $\tilde{\mu}(A \cup N) := \mu(A)$. Then $(\Omega, \tilde{\mathcal{F}}, \tilde{\mu})$ is a well-defined complete measure space. Moreover, it extends $(\Omega, \mathcal{F}, \mu)$ in the sense that $\tilde{\mu}$ agrees with μ on $\mathcal{F}$.*

Proof.

(i) $\tilde{\mathcal{F}}$ is a σ-field.

(i) $\varnothing = \varnothing \cup \varnothing \in \tilde{\mathcal{F}}$.

(ii) Suppose $A \in \mathcal{F}$ and $N \subset B \in \mathcal{F}$ with $\mu(B) = 0$. Then

$$(A \cup N)^{\mathsf{c}} = \underbrace{(A \cup B)^{\mathsf{c}}}_{\in \mathcal{F}} \cup \underbrace{[B \setminus (A \cup N)]}_{\subset B} \in \tilde{\mathcal{F}}.$$

(iii) Suppose $A_n \in \mathcal{F}$ and $N_n \subset B_n \in \mathcal{F}$ with $\mu(B_n) = 0$. Then

$$\bigcup (A_n \cup N_n) = \left(\bigcup A_n\right) \cup \underbrace{\left(\bigcup N_n\right)}_{\subset \cup B_n} \in \tilde{\mathcal{F}}.$$

(ii) $\tilde{\mu}$ is well-defined. Take $A_1, A_2 \in \mathcal{F}$, $N_1 \subset B_1 \in \mathcal{F}$, $N_2 \subset B_2 \in \mathcal{F}$ with $\mu(B_1) = \mu(B_2)$ such that $A_1 \cup N_1 = A_2 \cup N_2$.

$$\tilde{\mu}(A_1 \cup N_1) = \mu(A_1 \cup B_1) \geq \mu(A_2) = \tilde{\mu}(A_2 \cup N_2).$$

By symmetry, we have equality.

(iii) $\tilde{\mu}$ is a measure on $\tilde{\mathcal{F}}$.

(i) $\tilde{\mu}(\varnothing) = \mu(\varnothing) = 0$.

(ii) Fix disjoint $A_n \cup N_n \in \tilde{\mathcal{F}}$ with $A_n \in \mathcal{F}$ and $N_n \subset B_n \in \mathcal{F}$, $\mu(B_n) = 0$. Note that

$$\begin{aligned}\tilde{\mu}\left(\bigcup (A_n \cup N_n)\right) &= \tilde{\mu}\left(\left(\bigcup A_n\right) \cup \left(\bigcup N_n\right)\right) = \mu\left(\bigcup A_n\right) \\ &= \sum \mu(A_n) = \sum \tilde{\mu}(A_n \cup N_n).\end{aligned}$$

□

Lemma 2.13. *Suppose μ is a finite measure on a field $\mathcal{A}$ and $(\Omega, \mathcal{F}_\tau, \tau)$ is its Carathéodory extension. Then, for each $A \in \mathcal{F}_\tau$, there exists $B \in \sigma(A)$ such that $A \subset B$ and $\tau(A) = \tau(B)$.*

Proof. By the definition of outer measure, for each $m \in \mathbb{N}$ we can find $B_{m,n} \in \mathcal{A}$ such that $A \subset \cup_n B_{m,n}$ and $\tau(A) + 1/m \geq \sum_n \mu(B_{m,n})$. Consider $B := \cap_m \cup_n B_{m,n}$. Note that $A \subset B \in \sigma(\mathcal{A})$ and

$$\tau(B) \leq \tau\left(\cup_n B_{m,n}\right) \leq \sum_n \mu(B_{m,n}) \leq \tau(A) + \frac{1}{m}.$$

□

Theorem 2.14. *Suppose μ is a finite measure on a field $\mathcal{A}$ and $(\Omega, \mathcal{F}_\tau, \tau)$ is its Carathéodory extension. Then, $(\Omega, \mathcal{F}_\tau, \tau)$ is the completion of $(\Omega, \sigma(A), \tau)$.*

Proof. Write $\mathcal{F} := \sigma(\mathcal{A})$. We want to show $\tilde{\mathcal{F}} = \mathcal{F}_\tau$, where $\tilde{\mathcal{F}} := \{A \cup N : A \in \mathcal{F}, \exists B \in \mathcal{F} \text{ st } \mu(B) = 0, N \subset B\}$. Since $\mathcal{F}_\tau$ is complete and $\tilde{\mathcal{F}}$ is the smallest completion, $\tilde{\mathcal{F}} \subset \mathcal{F}_\tau$. By lemma, for each $A \in \mathcal{F}_\tau$, there exists $B \in \mathcal{F}$ such that $A \subset B$ and $\tau(A) = \tau(B)$. This implies $\tau(B \setminus A) = 0$ and thus the existence of $C \in \mathcal{F}$ such that $B \setminus A \subset C$ and $\tau(C) = 0$. We can now write

$$A = (A \cap C^{\complement}) \cup (A \cap C) = \underbrace{(B \cap C^{\complement})}_{\in \mathcal{F}} \cup \underbrace{(A \cap C)}_{\subset C} \in \tilde{\mathcal{F}}.$$

□

3 Measurable Functions

Definition 3.1. Let $f : (X, \mathcal{F}) \to (Y, \mathcal{G})$. For $B \in \mathcal{G}$, define the **preimage** of B under f as

$$f^{-1}(B) := \{x \in X : f(x) \in B\}.$$

Write $f^{-1}\mathcal{G} := \{f^{-1}(B) : B \in \mathcal{G}\}$. We say f is **measurable** if $f^{-1}\mathcal{G} \subset \mathcal{F}$.

Proposition 3.2 (Rules of Preimage)**.**

(i) $f^{-1}\varnothing = \varnothing$.

(ii) $f^{-1}(B^{\mathsf{c}}) = (f^{-1}B)^{\mathsf{c}}$

(iii) $A \subset B$ *implies* $f^{-1}A \subset f^{-1}B$.

(iv) $f^{-1}(\cup_\lambda A_\lambda) = \cup_\lambda f^{-1}A_\lambda$.

(v) $f^{-1}(\cap_\lambda A_\lambda) = \cap_\lambda f^{-1}A_\lambda$.

Lemma 3.3. $f : (X, \mathcal{F}) \to (Y, \sigma(\mathcal{E}))$ *is measurable if* $f^{-1}\mathcal{E} \subset \mathcal{F}$.

Proof. We know $\sigma(f^{-1}\mathcal{E}) \subset \mathcal{F}$ but want to show $f^{-1}(\sigma(\mathcal{E})) \subset \mathcal{F}$. It suffices to show that $f^{-1}(\sigma(\mathcal{E})) = \sigma(f^{-1}\mathcal{E})$. For $\sigma(f^{-1}\mathcal{E}) \subset f^{-1}\sigma(\mathcal{E})$, note that the latter is a σ-field containing $f^{-1}\mathcal{E}$. For $f^{-1}\sigma(\mathcal{E}) \subset \sigma(f^{-1}\mathcal{E})$, write

$$\mathcal{H} := \left\{B \subset \mathcal{Y} : f^{-1}B \in \sigma(f^{-1}\mathcal{E})\right\}.$$

Since $f^{-1}\mathcal{E} \subset \sigma(f^{-1}\mathcal{E})$, we have $\mathcal{E} \subset \mathcal{H}$. This gives $\sigma(\mathcal{E}) \subset \mathcal{H}$, since $\mathcal{H}$ can be verified to be a σ-field. Thus $f^{-1}\sigma(\mathcal{E}) \subset \sigma(f^{-1}\mathcal{E})$. □

Definition 3.4. A **random variable** is a measurable function $f : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$, where

$$\mathcal{B}_{\mathbb{R}} = \sigma\left(\{(a, \infty) : a \in \mathbb{R}\}\right) = \sigma\left(\{[a, \infty) : a \in \mathbb{R}\}\right).$$

Note that f is measurable if $\{f > a\}$ is measurable for all $a \in \mathbb{R}$ (or $\{f \geq a\}$ is measurable for all $a \in \mathbb{R}$).

Proposition 3.5.

(i) If f, g *are measurable, so is* $f + g$.

(ii) If f_k *are measurable, then so is* $\sup_k f_k$.

(iii) Suppose $f : (X, \mathcal{F}) \to (Y, \mathcal{G})$ *and* $g : (Y, \mathcal{G}) \to (Z, \mathcal{H})$ *are measurable. Then,* $g \circ f : (X, \mathcal{F}) \to (Z, \mathcal{H})$ *is measurable.*

Proof.

(i)

$$\{f + g > a\} = \bigcup_{t \in \mathbb{Q}} \left[\{f > t\} \cap \{g > a - t\}\right] \in \mathcal{F}.$$

(ii)
$$\left\{\sup_k f_k \le a\right\} = \bigcap_k \{f_k \le a\} \in \mathcal{F}.$$

(iii) This is clear.

□

Remark 3.6.

(i) Indicator functions. Consider the indicator function $\mathbb{1}_A : \Omega \to \{0, 1\}$ which takes value 1 on A and 0 otherwise. We have

$$\{\mathbb{1}_A \le a\} = \begin{cases} \varnothing, & a < 0 \\ A^{\mathsf{c}}, & 0 \le a < 1 \\ \Omega, & a \ge 1. \end{cases}$$

Thus, $\mathbb{1}_A$ is measurable if and only if $A \in \mathcal{F}$.

(ii) Simple function: Let $\{A_1, \ldots, A_n\}$ be a *finite* partition of Ω. A **simple function** is a linear combination of indicator functions on the partition:

$$f = \sum_{i=1}^{n} a_i \mathbb{1}_{A_i}, \quad A_i \in \mathbb{R}.$$

(iii) Nonnegative measurable functions: $f \ge 0$.

(iv) General measurable functions. Any measurable function f can be written as $f = f^+ - f^-$, where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$.

☕

But how does one construct/approximate nonnegative measurable functions?

Lemma 3.7. *For each nonnegative measurable function $f \ge 0$, there exists nonnegative simple functions $f_n \ge 0$ such that $f_n \uparrow f$ pointwise everywhere. If additionally, f is bounded, then the convergence is uniform.*

Proof. Define

$$f_n = \sum_{k=0}^{n2^n - 1} \frac{k}{2^n} \mathbb{1}\left\{\frac{k}{2_k} \le f < \frac{k+1}{2^n}\right\} + n\mathbb{1}\left\{f \ge n\right\}.$$

The sequence f_n is monotonic because the petitions (of the range) for larger n refine earlier partitions. We have convergence since whenever $f(\omega) \le n$,

$$|f_n(\omega) - f(\omega)| \le 2^{-n}.$$

□

4 Integral

We implicitly assume the measurability of all functions below.

(i) Defining the integral of an indicator function is straightforward:

$$\int \mathbb{1}_A \, \mathrm{d}\mu \coloneqq \mu(A).$$

(ii) For a nonnegative simple function $f = \sum_i^n a_i \mathbb{1}_{A_i}$, we define

$$\int f \, \mathrm{d}\mu \coloneqq \sum_{i=1}^n a_i \mu(A_i).$$

(iii) For a nonnegative function $f \geq 0$, we define

$$\int f \, \mathrm{d}\mu \coloneqq \sup \left\{ \inf f \, \mathrm{d}\mu : g \text{ is a nonnegative simple function and } g \leq f \right\}.$$

(iv) For a general function f, we define

$$\int f \, \mathrm{d}\mu \coloneqq \int f^+ \, \mathrm{d}\mu - \int f^- \, \mathrm{d}\mu.$$

Note that at this point, we have not shown any of the definitions above are well-defined.

Let's first show the integral of nonnegative simple functions is well-defined.

Proposition 4.1. *For a nonnegative simple function f, the value of $\int f \, \mathrm{d}\mu$ does not depend on the representation of f as a linear combination of indicator functions.*

Proof. For each representation $f = \sum_i^n a_i \mathbb{1}_{A_i}$, we can always write

$$f = \sum_j^m b_j \mathbb{1}\{f = b_j\},$$

where b_j can be taken to as the distinct values in the range of f. For each i, j, note that either $A_i \subset \{f = b_j\}$ or $A_i \cap \{f = b_j\} = \varnothing$. Now,

$$\begin{aligned}
\int f \, \mathrm{d}\mu = \sum_i a_i \mu(A_i) &= \sum_i a_i \sum_j \mu(A_i \cap \{f = b_j\}) \\
&= \sum_j \sum_i a_i \mu(A_i \cap \{f = b_j\}) \\
&= \sum_j \sum_{i : A_i \subset \{f = b_j\}} a_i \mu(A_i \cap \{f = b_j\}) \\
&= \sum_j \sum_{i : A_i \subset \{f = b_j\}} b_j \mu(A_i \cap \{f = b_j\}) \\
&= \sum_j \sum_i b_j \mu(A_i \cap \{f = b_j\}) \\
&= \sum_j b_j \mu(\{f = b_j\}).
\end{aligned}$$

□

Lemma 4.2. *Let f_n, f, g be nonnegative simple functions. Then*

(i) $\int f \, \mathrm{d}\mu \geq 0$.

(ii) $\int (af) \, \mathrm{d}\mu = a \int f \, \mathrm{d}\mu$, $a \geq 0$.

(iii) $\int (f + g) \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu$.

(iv) $f \geq g$ *implies* $\int f \, \mathrm{d}\mu \geq \int g \, \mathrm{d}\mu$.

(v) If $f_n \uparrow$ *and* $\lim f_n \geq g$, *then* $\lim \int f_n \, \mathrm{d}\mu \geq \int g \, \mathrm{d}\mu$.

Proof.

(i) Obvious.

(ii) Obvious.

(iii) If $f = \sum a_i \mathbb{1}_{A_i}$ and $g = \sum b_j \mathbb{1}_{B_j}$. We have

$$f + g = \sum_i \sum_j (a_i + b_j) \mathbb{1}_{A_i \cap B_j}.$$

Thus,

$$\begin{aligned}
\int (f + g) \, \mathrm{d}\mu &= \sum_i \sum_j (a_i + b_j) \mu(A_i \cap B_j) \\
&= \sum_i a_i \sum_j \mu(A_i \cap B_j) + \sum_j b_j \sum_i \mu(A_i \cap B_j) \\
&= \sum_i a_i \mu(A_i) + \sum_j b_j \mu(B_j) = \int f \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu.
\end{aligned}$$

(iv) Note that

$$\int g \, \mathrm{d}\mu = \int (g - f) + g \, \mathrm{d}\mu,$$

where $g - f$ is also a nonnegative simple function.

(v) For each $\alpha \in (0, 1)$, define $A_n(\alpha) := \{f_n \geq \alpha g\}$. Note that $A_n(\alpha) \uparrow \Omega$ as $n \to \infty$ (when $\alpha < 1$). Write $g = \sum_j b_j \mathbb{1}_{B_j}$. We have

$$\int f_n \, \mathrm{d}\mu \geq \int f_n \mathbb{1}_{A_n(\alpha)} \, \mathrm{d}\mu \geq \int \alpha g \mathbb{1}_{A_n(\alpha)} \, \mathrm{d}\mu = \alpha \sum_j b_j \mu(B_j \cap A_n(\alpha)).$$

Sending $n \to \infty$ gives, by continuity of the measure,

$$\lim_{n \to \infty} \int f_n \, \mathrm{d}\mu \geq \alpha \sum_j b_j \mu(B_j) = \alpha \int g \, \mathrm{d}\mu.$$

Sending $\alpha \to 1$ concludes the proof.

□

Definition 4.3. For a nonnegative function $f \geq 0$, we define

$$\int f \, \mathrm{d}\mu \coloneqq \sup\left\{\int g \, \mathrm{d}\mu : g \leq f, g \text{ is nonnegative simple}\right\}.$$

Note now that we have two definitions of the integral for nonnegative simple functions. Let's show they agree. For $f = \sum_i a_i \mathbb{1}_{A_i}$, where A_i is a partition of Ω, we write the first definition (that for simple functions) of its integral as

$$T(f) \coloneqq \sum_i a_i \mu(A_i),$$

and the second definition (that for nonnegative functions) as

$$S(f) \coloneqq \sup\left\{\int g \, \mathrm{d}\mu : g \leq f, g \text{ is nonnegative simple}\right\}.$$

Note that for each nonnegative simple g such that $g \leq f$, we have

$$T(g) \leq T(f) \implies \sup_g T(g) = S(f) \leq T(f).$$

For the reverse inequality, note that f is itself a nonnegative simple function such that $f \leq f$. Thus $S(f) \geq T(f)$.

Lemma 4.4. *Assume $f, g \geq 0$ are nonnegative functions.*

(i) Suppose a sequence of nonnegative simple functions $f_n \uparrow f$. Then, $\int f \, \mathrm{d}\mu = \lim_n \int f_n \, \mathrm{d}\mu$. Note that from this property, the linearity of integral follows.

(ii)

$$\int f \, \mathrm{d}\mu = \lim_n \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \mu\left(\frac{k}{2^n} \leq f < \frac{k+1}{2^n}\right) + n\mu\left(f \geq n\right).$$

(iii) $\int (af) \, \mathrm{d}\mu = a \int f \, \mathrm{d}\mu$, $a \geq 0$.

(iv) $\int (f+g) \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu$.

(v) $f \leq g$ implies $\int f \, \mathrm{d}\mu \leq \int g \, \mathrm{d}\mu$.

Proof. (i) By the definition of the integral for nonnegative functions, we have

$$\int f \, \mathrm{d}\mu \geq \int f_n \, \mathrm{d}\mu.$$

Taking limit on both sides gives $\int f \, \mathrm{d}\mu \geq \lim_n \int f_n \, \mathrm{d}\mu$. To establish the reverse inequality, fix any nonnegative simple function $g \leq f$. Since $\lim f_n \geq g$, a previous lemma gives

$$\lim_n \int f_n \, \mathrm{d}\mu \geq \int g \, \mathrm{d}\mu.$$

Taking sup over all such g concludes the proof.

(ii) This is a straightforward corollary of the previous part and the construction of f_n in the proof of the approximation lemma.

(iii) Find $f_n \uparrow f$ by previous results and note that $af_n \uparrow af$. Thus,

$$\int af \, \mathrm{d}\mu = \lim_n \int (af_n) \, \mathrm{d}\mu = a \lim_n \int f_n \, \mathrm{d}\mu = a \int f \, \mathrm{d}\mu.$$

(iv) Let $f_n \uparrow f$ and $g_n \uparrow g$. Then $f_n + g_n \uparrow f + g$ and we can use the linearity of integral for simple functions to conclude:

$$\begin{aligned}\int (f+g) \, \mathrm{d}\mu &= \lim_n \int (f_n + g_n) \, \mathrm{d}\mu \\ &= \lim_n \left[\int f_n \, \mathrm{d}\mu + \int g_n \, \mathrm{d}\mu \right] \\ &= \int f \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu.\end{aligned}$$

(v) For the function g, the sup is taken over a larger set.

□

For a general measurable function f, we write

$$f = f^+ - f^-.$$

If $\min\{\int f^+ \, \mathrm{d}\lambda, \int f^- \, \mathrm{d}\mu\} < \infty$, then the integral of f exists, defined as

$$\int f \, \mathrm{d}\mu := \int f^+ \, \mathrm{d}\mu - \int f^- \, \mathrm{d}\mu.$$

If $\max\{\int f^+ \, \mathrm{d}\mu, \int f^- \, \mathrm{d}\mu\} < \infty$ (equivalently, if $\int |f| \, \mathrm{d}\mu < \infty$), then we say f is **integrable**.

For $A \in \mathcal{F}$, we define

$$\int_A f \, \mathrm{d}\mu := \int f \mathbb{1}_A \, \mathrm{d}\mu.$$

For a random variable $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$, we define its **expectation** as $\mathbb{E}[X] = \int X \, \mathrm{d}\mathbb{P}$. We say the **expectation exists** if $\mathbb{E}|X| < \infty$ (i.e., X is integrable).

We will focus from now on only on integrable functions.

Lemma 4.5. *Let f, g be integrable.*

(i) If $A \in \mathcal{F}$, $\mu(A) = 0$, then $\int_A f \, \mathrm{d}\mu = 0$.

(ii) If $f \le g$ a.e. ($\mu(f > g) = 0$), then $\int f \, \mathrm{d}\mu = \int g \, \mathrm{d}\mu$.

(iii) If $f = g$ a.e., then $\int f \, \mathrm{d}\mu = \int g \, \mathrm{d}\mu$.

Proof.

(i) We start by proving this result for a nonnegative simple function $f = \sum_i a_i \mathbb{1}_{A_i}$, where A_i partitions Ω.

$$\int_A f \, \mathrm{d}\mu = \int \left(\sum a_i \mathbb{1}_{A \cap A_i} \right) \mathrm{d}\mu = \sum a_i \mu(A \cap A_i) = 0.$$

The last equality holds since each $A \cap A_i$ is a subset of a measure zero set.

Now for a measurable $f \geq 0$, there exists a sequence of simple nonnegative functions $f_n \uparrow f$, so $f_n \mathbb{1}_A \uparrow f \mathbb{1}_A$. Thus,

$$\int_A f \, \mathrm{d}\mu = \lim_n \int f_n \mathbb{1}_A \, \mathrm{d}\mu = 0.$$

Finally, for a general integrable $f = f^+ - f^-$, we have $f \mathbb{1}_A = f^+ \mathbb{1}_A - f^- \mathbb{1}_A$, and thus $\int_A f \, \mathrm{d}\mu = 0$.

(ii) Let $A := \{f \leq g\}$. We have $\mu(A^\mathsf{c}) = 0$. Let's first consider the case $f, g \geq 0$. In this case, we have

$$\begin{aligned} \int f \, \mathrm{d}\mu &= \int_A f \, \mathrm{d}\mu + \int_{A^\mathsf{c}} f \, \mathrm{d}\mu = \int f \mathbb{1}_A \, \mathrm{d}\mu \\ &\leq \int g \mathbb{1}_A \, \mathrm{d}\mu = \int g \mathbb{1}_A \, \mathrm{d}\mu + \int_{A^\mathsf{c}} g \, \mathrm{d}\mu = \int g \, \mathrm{d}\mu. \end{aligned}$$

Now, for general measurable f, g, write $f = f^+ - f^-$ and similarly for g. Note that we have $f^+ \leq g^+$ and $f^- \geq g^-$ on A. By monotonicity of the integral for nonnegative functions, we can conclude by taking differences.

(iii) Immediately follows from (ii).

□

Lemma 4.6. *Let f, g be integrable.*

(i) $\int af \, \mathrm{d}\mu = a \int f \, \mathrm{d}\mu$ *for each* $a \in \mathbb{R}$.

(ii) $\int f + g \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu$.

Proof.

(i) If $a \geq 0$, we have $(af)^+ = af^+$, $(af)^- = af^-$. Using linearity of the integral for nonnegative functions, we can conclude. If $a < 0$, note that $(af)^+ = -af^-$ and $(af)^- = -af^+$. We can again conclude using linearity of the integral for nonnegative functions.

(ii) Write $f = f^+ - f^-$, $g = g^+ - g^-$, and $f + g = (f + g)^+ - (f + g)^-$. Note that we have

$$(f + g)^+ - (f - g)^- = f^+ - f^- + g^+ - g^-,$$

or

$$(f + g)^+ + f^- + g^- = (f + g)^- + f^+ + g^+.$$

Applying linearity of integral of nonnegative functions on both sides, we have

$$\int (f+g)^+ \, \mathrm{d}\mu + \int f^- \, \mathrm{d}\mu + \int g^- \, \mathrm{d}\mu = \int (f+g)^- \, \mathrm{d}\mu + \int f^+ \, \mathrm{d}\mu + \int g^+ \, \mathrm{d}\mu.$$

Rearranging gives the desired result.

$\square$

5 Convergence of Integrals

Theorem 5.1 (Levi, Monotone Convergence Theorem)**.** *Let f_n, f be nonnegative measurable functions such that $f_n \uparrow f$ a.e. Then,*

$$\lim_n \int f_n \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu.$$

Proof. For each f_n, there exists nonnegative simple functions $f_{n,k}$ such that $f_{n,k} \uparrow f_n$ as $k \to \infty$. Now, for each k, define the nonnegative simple function

$$g_k \coloneqq \max_{1 \le n \le k} f_{n,k}.$$

Note that g_k must be monotone in k since $f_{n,k}$ is monotone in k, and each $g_k \le f_k$. We have however, that $\lim g_k \ge f_n$ for each n. Thus, $g_k \uparrow f$. A previous lemma now gives

$$\int f \, \mathrm{d}\mu = \lim_k \int g_k \, \mathrm{d}\mu.$$

Now, since each $f_n \le f$ and $g_k \le f_k$, we have

$$\lim_n \int g_n \, \mathrm{d}\mu \le \lim_n \int f_n \, \mathrm{d}\mu \le \int f \, \mathrm{d}\mu.$$

Since the first and last terms are equal, we have equality throughout. □

Theorem 5.2 (Fatou's Lemma)**.** *Suppose $f_n \ge 0$ ae. Then,*

$$\int \liminf_n f_n \, \mathrm{d}\mu \le \liminf_n \int f_n \, \mathrm{d}\mu.$$

Proof. Write $g_k \coloneqq \inf_{n \ge k} f_n$. We have $g_k \uparrow \liminf_n f_n$. By the monotone convergence theorem,

$$\lim_k \int \inf_{n \ge k} f_n \, \mathrm{d}\mu = \lim_k \int g_k \, \mathrm{d}\mu = \int \lim_k g_k \, \mathrm{d}\mu = \int \liminf_n f_n \, \mathrm{d}\mu.$$

On the other hand, since $\inf_{n \ge k} f_n \le f_n$ for each $n \ge k$, we know that

$$\int \inf_{n \ge k} f_n \, \mathrm{d}\mu \le \int_{n \ge k} f_n \, \mathrm{d}\mu, \quad \forall n \ge k.$$

Taking limit gives

$$\lim_k \int \inf_{n \ge k} f_n \, \mathrm{d}\mu \le \liminf_n \int f_n \, \mathrm{d}\mu.$$

We can conclude thus that

$$\int \liminf_n f_n \, \mathrm{d}\mu = \lim_k \int \inf_{n \ge k} f_n \, \mathrm{d}\mu \le \liminf_n \int f_n \, \mathrm{d}\mu.$$

□

Theorem 5.3 (Lebesgue, Dominated Convergence Theorem)**.** *Suppose $f_n \to f$ and $|f_n| \le g$ ae for some g with $\int |g| \, \mathrm{d}\mu < \infty$. Then,*

$$\lim_n \int f_n \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu.$$

Proof. Note that $-g \le f_n \le g$ implies $f_n + g \ge 0$. Thus Fatou's lemma gives

$$\int \liminf_n (f_n + g) \, \mathrm{d}\mu \le \liminf_n \int (f_n + g) \, \mathrm{d}\mu$$
$$\int \liminf_n f_n \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu \le \liminf_n \int f_n \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu.$$

We can cancel $\int g \, \mathrm{d}\mu$ on both sides to get

$$\int \liminf_n f_n \, \mathrm{d}\mu \le \liminf_n \int f_n \, \mathrm{d}\mu.$$

For the reverse inequality, note that $g - f_n \ge 0$ ae and so Fatou's lemma gives

$$\int \liminf_n (g - f_n) \, \mathrm{d}\mu \le \liminf_n \int (g - f_n) \, \mathrm{d}\mu$$
$$\int g \, \mathrm{d}\mu - \int \limsup_n f_n \, \mathrm{d}\mu \le \int g \, \mathrm{d}\mu - \limsup_n \int f_n \, \mathrm{d}\mu.$$

Thus,

$$\limsup_n \int f_n \, \mathrm{d}\mu \le \int \limsup_n f_n \, \mathrm{d}\mu$$

and we conclude that

$$\limsup_n \int f_n \, \mathrm{d}\mu \le \int \limsup_n f_n \, \mathrm{d}\mu = \int \liminf_n f_n \, \mathrm{d}\mu \le \liminf_n \int f_n \, \mathrm{d}\mu,$$

where the equality comes from the assumption that $f_n \to f$ ae. Since in general $\limsup \ge \liminf$, we have equality throughout. □

Corollary 5.4 (Bounded Convergence)**.** *Suppose $f_n \to f$ with $|f_n| \le M < \infty$ ae for some constant M and $\mu(\Omega) < \infty$. Then,*

$$\lim_n \int f_n \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu.$$

Proof. Choose $g \equiv M$ so that $\int |g| \, \mathrm{d}\mu = M\mu(\Omega) < \infty$ in the dominated convergence theorem. □

6 Independence

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 6.1 (Independence)**.** We say a sequence of events $A_1, \ldots, A_n$ are independent if for each $\Lambda \subset \{1, \ldots, n\}$ we have

$$\mathbb{P}\left(\cap_{i\in\Lambda} A_i\right) = \prod_{i\in\Lambda} \mathbb{P}(A_i).$$

We say the collection of subsets $\mathcal{E}_1, \ldots, \mathcal{E}_n \subset \mathcal{F}$ are independent if for each $A_1 \in \mathcal{E}_1, \ldots, A_n \in \mathcal{E}_n$ we have $A_1, \ldots, A_n$ are independent events.

Definition 6.2. Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ be a random variable. We define the **σ-algebra generated by X**, $\sigma(X)$, to be the smallest σ-field on Ω such that X is measurable from $(\Omega, \sigma(X), \mathbb{P}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. Explicitly,

$$\sigma(X) := X^{-1}\mathcal{B}_{\mathbb{R}} = \sigma\Big(\big\{\{X > a\} : a \in \mathbb{Q}\big\}\Big).$$

Example 6.3. Let $A \in \mathcal{F}$ and consider $X := \mathbb{1}_A$. Then $\sigma(A) = \{\varnothing, A, A^{\mathsf{c}}, \Omega\}$. ☍

Definition 6.4 (Independence of Random Variables)**.** We say random variables $X_1, \ldots, X_n : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ are **independent** if $\sigma(X_1), \ldots, \sigma(X_n)$ are independent. We write $X \perp\!\!\!\perp Y$ for two random variables X, Y being independent.

Proposition 6.5. *Let f, g be measurable functions. If $X \perp\!\!\!\perp Y$, then $f(X) \perp\!\!\!\perp g(Y)$.*

Proof. Note that

$$\sigma(f(X)) = (f \circ X)^{-1}\mathcal{B}_{\mathbb{R}} = X^{-1}(f^{-1}\mathcal{B}_{\mathbb{R}}) \subset X^{-1}\mathcal{B}_{\mathbb{R}} = \sigma(X).$$

Similarly, $\sigma(g(Y)) \subset \sigma(Y)$. Thus, independence follows. □

Example 6.6. Let $X \perp\!\!\!\perp Y$. For $A, B \in \mathcal{F}$, we have $\mathbb{P}(X^{-1}A \cap Y^{-1}B) = \mathbb{P}(X^{-1}A)\mathbb{P}(Y^{-1}B)$, or

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B).$$

☍

Proposition 6.7. *If $X \perp\!\!\!\perp Y$ with $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$, then*

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

Proof. First consider indicator functions $X = \mathbb{1}_A$ and $Y = \mathbb{1}_B$. We have

$$\mathbb{E}[XY] = \mathbb{E}\mathbb{1}_{A\cap B} = \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) = \mathbb{E}\mathbb{1}_A\mathbb{E}\mathbb{1}_B.$$

Next, suppose $X = \sum_i a_i \mathbb{1}_{A_i}$ and $Y = \sum_j b_j \mathbb{1}_{B_j}$ so that $XY = \sum_{i,j} a_i b_j \mathbb{1}_{A_i \cap B_j}$.

$$\begin{aligned}\mathbb{E}[XY] &= \sum_{i,j} a_i b_j \mathbb{P}(A_i \cap B_j) = \sum_{i,j} a_i b_j \mathbb{P}(A_i)\mathbb{P}(B_j) \\ &= \left(\sum_i a_i \mathbb{P}(A_i)\right)\left(\sum_j b_j \mathbb{P}(B_j)\right) = \mathbb{E}[X]\mathbb{E}[Y].\end{aligned}$$

But note that this works only when the representations of X, Y are such that each $A_i \in \sigma(X)$ and $B_j \in \sigma(Y)$. One can for example set $A_i := \{X = a_i\}$ and similarly for B_j.

Now, consider $X, Y \geq 0$. Set

$$X_n := \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \mathbb{1}\left\{\frac{k}{2^n} \leq X < \frac{k+1}{2^n}\right\} + n\mathbb{1}\{X \geq n\},$$

and similarly for Y_n. Recall that $X_n \uparrow X$ and $Y_n \uparrow Y$. Observe that $\sigma(X_n) \subset \sigma(X)$ and $\sigma(Y_n) \subset \sigma(Y)$. Thus, $X_n \perp\!\!\!\perp Y_n$ for each n. In particular, we have

$$\mathbb{E}[X_n Y_n] = \mathbb{E}[X_n]\mathbb{E}[Y_n].$$

Taking limits on both sides gives the desired result.

Finally, for general integrable X, Y, write $X = X^+ - X^-$ and similarly for Y. Then,

$$XY = X^+Y^+ + X^-Y^- - X^+Y^- - X^-Y^+.$$

We conclude by linearity of expectation, noting again that independence is preserved under taking positive and negative parts, since $\sigma(X^+) \subset \sigma(X)$, and similarly for the other terms. □

7 The Law of Large Numbers

Lemma 7.1 (Borel-Cantelli)**.** *If* $\sum_n \mathbb{P}(A_n) < \infty$*, then*

$$\mathbb{P}(A_n \text{ i.o.}) = \mathbb{P}\left(\limsup_n A_n\right) = 0.$$

Here, i.o. means infinitely often. Thus, with probability 1, only finitely many of the events A_n *occur.*

Proof. For any $\varepsilon > 0$, there exists m such that $\sum_{n=m}^{\infty} \mathbb{P}(A_n) < \varepsilon$. Observe that

$$\mathbb{P}(A_n \text{ i.o.}) = \mathbb{P}(\limsup_n A_n) \leq \mathbb{P}\left(\bigcup_{n=m}^{\infty} A_n\right) \leq \sum_{n=m}^{\infty} \mathbb{P}(A_n) < \varepsilon.$$

□

Lemma 7.2. *If* $\mathbb{E}|X|^r < \infty$ *then* $\mathbb{E}|X|^s < \infty$ *for all* $0 < s < r$.

Proof. Holder's inequality gives

$$\mathbb{E}|X|^s = \mathbb{E}\left[|X|^s \cdot 1\right] \leq \left(\mathbb{E}|X|^{s\cdot\frac{r}{s}}\right)^{\frac{s}{r}} \left(\mathbb{E}1^{\frac{r}{r-s}}\right)^{\frac{r-s}{r}} = (\mathbb{E}|X|^r)^{\frac{s}{r}} < \infty.$$

Alternatively, note that for each $t \geq 0$ we have $t^s \leq t^r + 1$ ($t^s \leq t^r$ for $t \geq 1$). Thus,

$$\mathbb{E}|X|^s \leq \mathbb{E}|X|^r + 1 < \infty.$$

□

Theorem 7.3 (Strong Law of Large Numbers)**.** *Let* $X_1, X_2, \ldots$ *be iid.*[3] *If* $\mathbb{E}|X_1| < \infty$*, then*

$$\frac{X_1 + \cdots + X_n}{n} \longrightarrow \mu := \mathbb{E}[X_1] \quad a.s.$$

We will first prove the theorem assuming bounded fourth moment, then bounded second moment, and finally only assuming finite first moment.

Proof *(SLLN, Assuming bounded fourth moment)***.** WLOG suppose $\mu = 0$. Write $S_n := \sum_{i=1}^n X_i$. It suffices to show

$$\sum_n \mathbb{P}\left(\frac{|S_n|}{n} > \varepsilon\right) < \infty, \quad \forall \varepsilon > 0,$$

since if this is true, Borel-Cantelli gives

$$\mathbb{P}\left(\limsup_n \frac{|S_n|}{n} \leq \varepsilon\right) = 1 - \mathbb{P}\left(\frac{|S_n|}{n} > \varepsilon \text{ i.o.}\right) = 1.$$

and letting $\varepsilon \to 0$ gives the desired result.

[3] We say X, Y are identically distributed if $\mathbb{P}(X \in A) = \mathbb{P}(Y \in A)$ for each measurable A.

Let's prove the summability. Note that by Chebyshev's inequality,

$$\mathbb{P}\left(\frac{|S_n|}{n} > \varepsilon\right) \leq \frac{\mathbb{E}S_n^2}{n^2\varepsilon^2}.$$

Note that by independence,

$$\mathbb{E}S_n^2 = n\mathbb{E}X_1^2 + \sum_{i\neq j}\mathbb{E}[X_iX_j] = n\mathbb{E}X_1^2 + \sum_{i\neq j}\mathbb{E}X_i\mathbb{E}X_j = n\mathbb{E}X_1^2.$$

Thus,

$$\sum_n \mathbb{P}\left(\frac{|S_n|}{n} > \varepsilon\right) \leq \frac{\mathbb{E}X_1^2}{\varepsilon^2}\sum_n \frac{1}{n}.$$

This convergence is not sufficient.

Using third moment is annoying since $S_n^3 \neq |S_n|^2$.

Let's use the fourth moment instead.

$$\mathbb{P}\left(\frac{|S_n|}{n} > \varepsilon\right) \leq \frac{\mathbb{E}S_n^4}{n^4\varepsilon^4}.$$

Expanding S_n^4 and noting that independence implies that $\mathbb{E}[X_iX_jX_kX_l] = \mathbb{E}X_i\mathbb{E}X_j\mathbb{E}X_k\mathbb{E}X_l = 0$ whenever any of the indices is unique, we are left with terms of the form X_i^4 and $X_i^2X_j^2$ (there are $\binom{4}{2}\binom{n}{2}$ terms of this form). Let $M > 0$ be such that $\mathbb{E}X_1^4 < M$. Note that for $i \neq j$, we have by Cauchy-Schwarz (or lemma above),

$$\mathbb{E}[X_i^2X_j^2] = \mathbb{E}[X_i^2]\mathbb{E}[X_j^2] \leq \sqrt{\mathbb{E}[X_i^4]\mathbb{E}[X_j^4]} < M.$$

Thus,

$$\mathbb{E}S_n^4 \leq nM + \binom{4}{2}\binom{n}{2}M \leq Cn^2M$$

for some $C > 0$. This gives the bound

$$\sum_n \mathbb{P}\left(|S_n| > n\varepsilon\right) \leq \frac{CM}{\varepsilon^4}\sum_n \frac{1}{n^2} < \infty.$$

□

Next, we relax the assumption to bounded second moment.

Lemma 7.4 (Maxmimal Inequality)**.** *Consider independent random variables $Z_1, \ldots, Z_N$. Then,*

$$\mathbb{P}\left(\max_{i\leq N}|Z_1 + \cdots + Z_i| > \varepsilon_1 + \varepsilon_2\right) \leq \beta\mathbb{P}\left(|Z_1 + \cdots + Z_N| > \varepsilon_1\right),$$

where

$$\beta = \frac{1}{\min_{i\leq N}\mathbb{P}(|Z_{i+1} + \cdots + Z_N| \leq \varepsilon_2)}.$$

Proof. Denote $S_i := Z_1 + \cdots + Z_i$ and $T_i := Z_{i+1} + \cdots + Z_N = S_N - S_i$. Define

$$\tau := \begin{cases} \text{first time } |S_i| > \varepsilon_1 + \varepsilon_2, & \text{if such } i \text{ exists,} \\ N, & \max_{i \le N} |S_i| \le \varepsilon_1 + \varepsilon_2. \end{cases}$$

We have

$$\begin{aligned} \mathbb{P}\left(\max_{i \le N} |S_i| > \varepsilon_1 + \varepsilon_2\right) &= \mathbb{P}\left(|S_\tau| > \varepsilon_1 + \varepsilon_2\right) \\ &= \sum_{i=1}^{N} \mathbb{P}\left(|S_\tau| > \varepsilon_1 + \varepsilon_2, \tau = i\right) \\ &= \sum_{i=1}^{N} \mathbb{P}\left(|S_i| > \varepsilon_1 + \varepsilon_2, \tau = i\right) \\ &= \sum_{i=1}^{N} \frac{\mathbb{P}\left(|S_i| > \varepsilon_1 + \varepsilon_2, \tau = i, |T_i| \le \varepsilon_2\right)}{\mathbb{P}\left(|T_i| \le \varepsilon_2\right)}. \end{aligned}$$

In the last step, we used independence of $(S_i, \{\tau = i\})$ and T_i, noting that $\{\tau = i\}$ is determined by $Z_1, \ldots, Z_i$ only. Thus,

$$\begin{aligned} \mathbb{P}\left(\max_{i \le N} |S_i| > \varepsilon_1 + \varepsilon_2\right) &\le \frac{1}{\min_{i \le N} \mathbb{P}(|T_i| \le \varepsilon_2)} \sum_{i=1}^{N} \mathbb{P}\left(|S_i| > \varepsilon_1 + \varepsilon_2, \tau = i, |T_i| \le \varepsilon_2\right) \\ &\le \frac{1}{\min_{i \le N} \mathbb{P}(|T_i| \le \varepsilon_2)} \sum_{i=1}^{N} \mathbb{P}\left(|S_N| > \varepsilon_1, \tau = i\right) \\ &= \frac{1}{\min_{i \le N} \mathbb{P}(|T_i| \le \varepsilon_2)} \mathbb{P}\left(|S_N| > \varepsilon_1\right). \end{aligned}$$

□

Let's return to the SLLN.

Proof *(SLLN, Assuming bounded second moment)***.** We will utilize the fact that any bound on S_n implies bounds on S_{n+1}. Define blocks $B_1, B_2, \ldots$, where $B_k := \{i \in \mathbb{N} : n_{k-1} < i \le n_k\}$, where say $n_k = 2^k$. We will show that $\max_{n \in B_k} |S_n|/n \to 0$ a.s. as $k \to \infty$, which implies that $|S_n|/n \to 0$ a.s. as $n \to \infty$. In light of Borel-Cantelli, it suffices to show that for each $\varepsilon > 0$,

$$\sum_k \mathbb{P}\left(\max_{n \in B_k} \frac{|S_n|}{n} > \varepsilon\right) < \infty.$$

The advantage of this approach is that we are summing over much fewer terms.

Note that

$$\begin{aligned} \sum_k \mathbb{P}\left(\max_{n \in B_k} \frac{|S_n|}{n} > \varepsilon\right) &\le \sum_k \mathbb{P}\left(\frac{1}{n_{k-1}} \max_{n \in B_k} |S_n| > \varepsilon\right) \\ &\le \sum_k \mathbb{P}\left(\max_{n \le n_k} |S_n| > n_{k-1}\varepsilon\right) \le \beta_k \mathbb{P}\left(|S_{n_k}| > \frac{n_{k-1}\varepsilon}{2}\right). \end{aligned}$$

The last step follows from the maximal inequality, with

$$\beta_k = \frac{1}{\min_{n \le n_k} \mathbb{P}\left(|S_{n_k} - S_n| \le \frac{n_{k-1}\varepsilon}{2}\right)}.$$

We bound β_k by Chebyshev's inequality:

$$\begin{aligned}
\beta_k^{-1} &= \min_{n \le n_k} \mathbb{P}\left(|S_{n_k} - S_n| \le \frac{n_{k-1}\varepsilon}{2}\right) \\
&= 1 - \max_{n \le n_k} \mathbb{P}\left(|S_{n_k} - S_n| > \frac{n_{k-1}\varepsilon}{2}\right) \\
&\ge 1 - \max_{n \le n_k} \frac{4\mathbb{E}|S_{n_k} - S_n|^2}{n_{k-1}^2 \varepsilon^2} \\
&\ge 1 - \max_{n \le n_k} \frac{4(n_k - n)\sigma^2}{n_{k-1}^2 \varepsilon^2} = 1 - \frac{4 n_k \sigma^2}{n_{k-1}^2 \varepsilon^2}.
\end{aligned}$$

The last term is greater than $1/2$ for large enough $k \ge K$. So we have $\beta_k \le 2$ for $k \ge K$.

This gives

$$\mathbb{P}\left(\max_{n \le n_k} |S_n| > n_{k-1}\varepsilon\right) \le 2\mathbb{P}\left(|S_{n_k}| > \frac{n_{k-1}\varepsilon}{2}\right) \le 2 \cdot \frac{4\mathbb{E}S_{n_k}^2}{n_{k-1}^2 \varepsilon^2} = \frac{8 n_k \sigma^2}{n_{k-1}^2 \varepsilon^2}$$

for $k \ge K$. Thus,

$$\sum_k \mathbb{P}\left(\max_{n \le n_k} |S_n| > n_{k-1}\varepsilon\right) \le K + \underbrace{\sum_{k \ge K} \frac{8 n_k \sigma^2}{n_{k-1}^2 \varepsilon^2}}_{O\left(\sum_k \frac{1}{2^k}\right)} < \infty.$$

□

Remark 7.5. Note that for SLLN assuming bounded second or fourth moment, we only used independence bounded moments, but not identical distribution. ☕

Before proving SLLN it its full generality, we note the following:

Theorem 7.6 (Kolmogorov SLLN)**.** *Let X_i be independent with $\mathbb{E}[X_i] = 0$ and $\sum_i \mathbb{E}X_i^2/i^2 < \infty$. Then $(X_1 + \cdots + X_n)/n \to 0$ a.s.*

Proof. HW. Same strategy as SLLN with bounded second moment. □

Now we can prove the SLLN in full generality.

Theorem 7.7. *Let X_i be iid (note that we need identical distribution here) with $\mathbb{E}X_i = 0$. Then,*

$$\frac{X_1 + \cdots + X_n}{n} \longrightarrow 0 \quad a.s.$$

Proof. The idea is truncation. Define

$$Y_i := X_i \mathbb{1}\{|X_i| \le i\}.$$

It has mean

$$\mu_i = \mathbb{E}[X_i \mathbb{1}\{|X_i| \le i\}] = \mathbb{E}[X \mathbb{1}\{|X| \le i\}] = -\mathbb{E}[X \mathbb{1}\{|X| > i\}].$$

In the last equality we used $\mathbb{E}X = 0$. Now,

$$\frac{1}{n}\sum_{i=1}^{n} X_i = \frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i) + \frac{1}{n}\sum_{i=1}^{n}(Y_i - \mu_i) + \frac{1}{n}\sum_{i=1}^{n}\mu_i.$$

We will show that each of the three terms on the RHS goes to 0 a.s.

We first consider the **third term**:

$$\begin{aligned}\left|\frac{1}{n}\sum_{i=1}^{n}\mu_i\right| &= \left|\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X\mathbb{1}\{|X| > i\}]\right| \\ &\le \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[|X|\mathbb{1}\{|X| > i\}] = \mathbb{E}\left(|X|\frac{1}{n}\sum_{i=1}^{n}\mathbb{1}\{|X| > i\}\right).\end{aligned}$$

Note that $\sum_{i\ge 1}\mathbb{1}\{|X| > i\} \le |X|$.[4] Plugging this back to our estimate of the third term:

$$\left|\frac{1}{n}\sum_{i=1}^{n}\mu_i\right| \le \mathbb{E}\left(|X|\frac{1}{n}\sum_{i=1}^{n}\mathbb{1}\{|X| > i\}\right) \le \mathbb{E}\left[|X| \cdot \min\left\{\frac{1}{n}|X|, 1\right\}\right] \longrightarrow 0,$$

where the convergence follows from dominated convergence theorem, since the term inside the expectation goes to 0 and is bounded by $|X|$, which is integrable.

Next we consider the **first term**. Note that

$$\sum_{i\ge 1}\mathbb{P}(X_i \ne Y_i) \le \sum_{i\ge 1}\mathbb{P}(|X_i| > i) = \mathbb{E}\sum_{i\ge 1}\mathbb{1}\{|X| > i\} \le \mathbb{E}|X| < \infty.$$

Now, Borel-Cantelli gives that $X_i = Y_i$ for all but finitely many i a.s., and so the first term goes to 0 a.s. More explicitly, we know there exists $\Omega_0 \subset \Omega$ with $\mathbb{P}(\Omega_0)$ such that for each $\omega \in \Omega_0$ there exists $i(\omega)$ such that

$$X_i(\omega) = Y_i(\omega), \quad \forall i > i(\omega).$$

For such an ω, we have

$$\frac{1}{n}\sum_{i=1}^{n}(X_i(\omega) - Y_i(\omega)) = \frac{1}{n}\sum_{i=1}^{i(\omega)}(X_i(\omega) - Y_i(\omega)) \longrightarrow 0.$$

4

$$\begin{aligned}\sum_{i=1}^{n}\mathbb{1}\{|X| > i\} &\le \sum_{i=1}^{\infty}\mathbb{1}\{|X| > i\} \\ &= \sum_{i=1}^{\infty}\sum_{j=i}^{\infty}\mathbb{1}\{j < |X| \le j+1\} = \sum_{j=1}^{\infty}\sum_{i=1}^{j}\mathbb{1}\{j < |X| \le j+1\} \\ &= \sum_{j=1}^{\infty} j\mathbb{1}\{j < |X| \le j+1\} \le |X|\sum_{j=1}^{\infty}\mathbb{1}\{j < |X| \le j+1\} = |X|.\end{aligned}$$

Finally we consider the **second term**. By definition of μ_i we have $\mathbb{E}[Y_i - \mu_i] = 0$. Note that

$$\begin{aligned}
\sum_i \frac{\mathbb{E}[(Y_i - \mu_i)^2]}{i^2} &\leq \sum_i \frac{\mathbb{E}[Y_i^2]}{i^2} = \sum_i \frac{1}{i^2}\mathbb{E}[X^2 \mathbb{1}\{|X| \leq i\}] \\
&= \sum_{i \geq 1} \frac{1}{i^2}\mathbb{E}\left[X^2 \sum_{j=1}^{i} \mathbb{1}\{j-1 < |X| \leq j\}\right] \\
&= \sum_{j \geq 1} \sum_{i \geq j} \frac{1}{i^2}\mathbb{E}[X^2 \mathbb{1}\{j-1 < |X| \leq j\}] \\
&\leq C \sum_{j \geq 1} \frac{1}{j}\mathbb{E}[X^2 \mathbb{1}\{j-1 < |X| \leq j\}] \\
&\leq C \sum_{j \geq 1} \mathbb{E}[|X| \mathbb{1}\{j-1 < |X| \leq j\}] \leq C\mathbb{E}|X| < \infty,
\end{aligned}$$

where the first equality comes from bounding the variance using the second moment. Kolmogorov SLLN now gives that the second term goes to 0 a.s. □

8 Weak Convergence

Consider a metric space (S, d). Denote the corresponding Borel σ-field (induced by the topology induced by the metric d) as $\mathcal{B}(S)$. We can consider the random variable

$$X : (\Omega, \mathcal{F}, \mathbb{P}) \to (S, \mathcal{B}(S)).$$

Write

$$P(X) := \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}B), \quad B \in \mathcal{B}(S).$$

This is a probability measure on $(S, \mathcal{B}(S))$. For any function $f : (S, \mathcal{B}(S)) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$, we write

$$Pf := \int f \, \mathrm{d}P(X) = \int f(X) \, \mathrm{d}\mathbb{P} = \mathbb{E}f(X).$$

The middle equality is left as an exercise.

In this case, we say P is the distribution of X and write $X \sim P$. That is, we have the

Definition 8.1 (Distribution)**.** Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (S, \mathcal{B}(S))$ be a random variable. The **distribution** of X is the probability measure P on $(S, \mathcal{B}(S))$ defined by

$$P(B) := \mathbb{P}(X \in B), \quad B \in \mathcal{B}(S).$$

8.1 Bounded Lipschitz Functions and Lower Semi-Continuity

Definition 8.2 (Bounded Lipschitz Function)**.** The set of **bounded Lipschitz functions** on S is

$$\mathrm{BL}(S) := \left\{ f : S \to \mathbb{R} : \sup_x |f(x)| < \infty, \ \sup_{x \neq y} \frac{|f(x) - f(y)|}{d(x, y)} < \infty \right\}.$$

Definition 8.3 (Weak Convergence)**.** Let P_n, P be probability measures on $(S, \mathcal{B}(S))$. We say P_n converges **weakly** to P, denoted $P_n \xrightarrow{\mathrm{d}} P$, if

$$P_n \ell \longrightarrow P\ell, \quad \forall \ell \in \mathrm{BL}(S).$$

If $X_n \sim P_n$ and $X \sim P$, we also write $X_n \xrightarrow{\mathrm{d}} X$ or $X_n \xrightarrow{\mathrm{d}} X$.[5] Note that each X_n here can live on different probability spaces. We care only about their distributions.

Definition 8.4 (Lower Semi-Continuity)**.** A function $g : S \to \mathbb{R}$ is **lower semi-continuous** (LSC) if for each $t \in \mathbb{R}$, the set $\{g > t\}$ is open in the topology of S. A function $g : S \to \mathbb{R}$ is **upper semi-continuous** (USC) if for each $t \in \mathbb{R}$, the set $\{g < t\}$ is open in the topology of S (equivalently, if $-g$ is LSC).

Remark 8.5. LSC functions are to open sets what measurable functions are to measurable sets. LSC functions extend open sets; USC functions extend closed sets. ☕

Lemma 8.6 (Approximation of LSC in BL)**.** *Let (S, d) be a metric space. Suppose $g : S \to \mathbb{R}$ is LSC and $g \geq -M$. Then there exists $\ell_n \in \mathrm{BL}(S)$ such that $\ell_n \nearrow g$.*

[5] Some use L for convergence in law.

Proof. WLOG, assume $g \geq 0$. Write $h_t = t\mathbb{1}\{g > t\}$ and note that $\sup_t h_t = g$. Each h_t is bounded but not Lipschitz.

We make them Lipschitz by defining

$$\ell_{k,t}(x) := t \wedge kd(x, \{g \leq t\}) \in \mathrm{BL}(S),$$

where $d(x, A) := \inf_{y \in A} d(x, y)$. This is a Lipschitz function as the infimum of Lipschitz functions (or prove by hand). Note that the second component is 0 on $\{g \leq t\}$. On $\{g > t\}$, the distance is strictly positive and so this component can be made arbitrarily large by taking k large enough. Thus,

$$\sup_k \ell_{k,t}(x) = t\mathbb{1}\{g(x) > t\} = h_t(x), \quad \forall t \in \mathbb{R}.$$

This gives

$$\sup_{k,t} \ell_{k,t} = g.$$

Now, we extract a countable sequence $f_1, f_2, \ldots$ such that $\sup_n f_n = g$. This is possible since we can restrict the supremum to $t \in \mathbb{Q}$ and $k \in \mathbb{N}$. We make the sequence monotone by defining

$$\ell_n := \max_{1 \leq i \leq n} f_i.$$

□

8.2 Portmanteau Theorem

Lemma 8.7. *Let (S, d) be a metric space. Suppose $P_n \xrightarrow{\mathrm{d}} P$. Then we have the following:*

(i) For any LSC and bounded from below $g : S \to \mathbb{R}$, $\liminf_n P_n g \geq Pg$.

(ii) For any USC and bounded from above $g : S \to \mathbb{R}$, $\limsup_n P_n g \leq Pg$.

(iii) For each open B, $\liminf_n P_n(B) \geq P(B)$.

(iv) For each closed C, $\limsup_n P_n(C) \leq P(C)$.

(v) For each B with $P(\partial B) = 0$, $\lim_n P_n(B) = P(B)$.

Proof. We first show (i). Fix g. There exists $\ell_i \in \mathrm{BL}(S)$ such that $\ell_i \nearrow g$. We have

$$\liminf_n P_n g \geq \liminf_n P_n \ell_i = P\ell_i, \quad \forall i,$$

where the last equality follows from weak convergence. Send $i \to \infty$. By MCT we have $P\ell_i \nearrow Pg$.

(ii) follows by applying (i) to $-g$.

(iii) follows by applying (i) to $g = \mathbb{1}_B$, which is LSC. (iv) is similar. (v) is HW. □

Definition 8.8. For function f, we define

$$\mathring{f} := \sup\{g \le f : g \text{ is LSC}\}.$$

Similarly, we define

$$\overline{f} := \inf\{g \ge f : g \text{ is USC}\}.$$

Note that this function is LSC since

$$\{\mathring{f} > t\} = \bigcup_{g \le f, g \text{ LSC}} \{g > t\}$$

is open as the union of open sets for each t. Similarly, $\overline{f}$ is USC.

Proposition 8.9. *Let B be a measurable subset of the metric space S. We have $\mathring{\mathbb{1}}_B = \mathbb{1}_{\mathring{B}}$ and $\overline{\mathbb{1}}_B = \mathbb{1}_{\overline{B}}$.*

Proof. HW. □

Note that $\mathring{f} \le f \le \overline{f}$. We have equality if f is continuous:

Theorem 8.10. *Suppose $|f| \le M$. We have $\mathring{f}(x) = \overline{f}(x)$ if and only if f is continuous at x.*

Proof. $\Rightarrow$: Fix x and arbitrary $\varepsilon > 0$. Suppose $\mathring{f}(x) = f(x)$. Then

$$U_1 := \{y : \mathring{f}(y) > f(x) - \varepsilon\}$$

is open and includes x. For any $y \in U_1$, we have $f(y) \ge \mathring{f}(y) > f(x) - \varepsilon$. Similarly, if $\overline{f}(x) = f(x)$, then

$$U_2 := \{y : \overline{f}(y) < f(x) + \varepsilon\}$$

is open and includes x. For each $y \in U_2$ we have $f(y) \le \overline{f}(y) < f(x) + \varepsilon$. In particular, for each y in the open set $U_1 \cap U_2$,

$$f(x) - \varepsilon < f(y) < f(x) + \varepsilon.$$

Thus f is continuous at x.

$\Leftarrow$: Fix x at which f is continuous. Need only show that for arbitrary $\varepsilon > 0$, we can find LSC g and USC h, $g \le f \le h$ such that $|h(x) - g(x)| < \varepsilon$. Note that there exists some open neighborhood U of x such that

$$f(x) - \varepsilon < f(y) < f(x) + \varepsilon, \quad \forall y \in U.$$

Thus,

$$f(y) \le (f(x) + \varepsilon)\mathbb{1}_U(y) + (M+1)\mathbb{1}_{U^c}(y) =: h(y), \quad \forall y,$$

where we use $M + 1$ to ensure that it is strictly larger than $f(x) + \varepsilon$ so that h is USC. Similarly,

$$f(y) \ge (f(x) - \varepsilon)\mathbb{1}_U(y) - (M+1)\mathbb{1}_{U^c}(y) =: g(y), \quad \forall y.$$

□

Note that $C_f := \{\mathring{f} = \overline{f}\} \in \mathcal{B}(S)$. (It is the preimage of $\{0\}$ of a measurable function.) So, f is continuous almost surely, $P(C_f) = 1$, if and only if $P\overline{f} = P\mathring{f}$. With this, we have the following

Theorem 8.11. *Suppose $P_n \xrightarrow{\mathrm{d}} P$. Then*

(i) $P_n f \to Pf$ *for all f that is bounded and continuous a.s.*

(ii) $P_n B \to PB$ *for all $B \in \mathcal{B}(S)$ such that $P(\partial B) = 0$.*

Proof. (i): Since $\mathring{f}$ is LSC and $\overline{f}$ USC,

$$P\mathring{f} \le \liminf P_n\mathring{f} \le \liminf P_n f \le \limsup P_n f \le \limsup P_n\overline{f} \le P\overline{f}.$$

Since $P\overline{f} = P\mathring{f}$ (see paragraph before the theorem statement), we have equality throughout and so $P_n f \to Pf$.

(ii) is a special case of (i). □

Theorem 8.12 (Portmanteau Theorem)**.** *Fix a metric space (S, d). The following are all equivalent to $P_n \xrightarrow{\mathrm{d}} P$:*

(i) $P_n\ell \to P\ell$ *for all $\ell \in \mathrm{BL}(S)$.*

(ii) $\liminf P_n g \ge Pg$ *for all bounded below LSC g.*

(iii) $\limsup P_n g \le Pg$ *for all bounded above USC g.*

(iv) $\liminf P_n(B) \ge P(B)$ *for all open B.*

(v) $\limsup P_n(C) \le P(C)$ *for all closed C.*

(vi) $P_n f \to Pf$ *for all bounded f that is continuous a.s.*

(vii) $P_n B \to PB$ *for all $B \in \mathcal{B}(S)$ with $P(\partial B) = 0$.*

Proof. In light of the work we've done, it remains only to show (vii) implies (i). We will show the stronger statement that $P_n f \to Pf$ for all bounded continuous f. WLOG assume $0 \le f \le 1$. Write

$$f(x) = \int_0^1 \mathbb{1}\{f(x) > t\}\,\mathrm{d}t.$$

We then have (using Fubini to exchange the order of integration; see next theorem):

$$\begin{aligned} Pf = \int f(x)\,\mathrm{d}P(x) &= \int \int_0^1 \mathbb{1}\{f(x) > t\}\,\mathrm{d}t\,\mathrm{d}P(x) \\ &= \int_0^1 \int \mathbb{1}\{f(x) > t\}\,\mathrm{d}P(x)\,\mathrm{d}t = \int_0^1 P\{f > t\}\,\mathrm{d}t. \end{aligned}$$

Similarly, $P_n f = \int_0^1 P_n\{f > t\}\,\mathrm{d}t$.

Now,

$$\begin{aligned}\partial\{f > t\} &= \overline{\{f = t\}} \setminus \{f \overset{\circ}{=} t\} \\ &\subset \{f \geq t\} \setminus \{f > t\} = \{f = t\},\end{aligned}$$

where the second line follows from noting that $\{f \geq t\}$ is closed and $\{f > t\}$ is open. Thus we have $P_n\{f > t\} \to P\{f > t\}$ for all t such that $P(f = t) = 0$. We will show that the set of such t has full measure in $[0, 1]$ by showing that the set of t such that $P(f = t) > 0$ is countable. Write

$$T \coloneqq \left\{t : P\{f = t\} > 0\right\} = \bigcup_n \left\{t : P\{f = t\} > \frac{1}{n}\right\}.$$

It is clear that the nth set in the union has cardinality at most n (for P is a probability measure), so T is countable.

Now, by dominated convergence theorem,

$$P_n f = \int_0^1 P_n\{f > t\} \, \mathrm{d}t \longrightarrow \int_0^1 P\{f > t\} \, \mathrm{d}t = Pf.$$

□

Consider finite measure spaces $(\Omega_i, \mathcal{F}_i, \mu_i), i = 1, 2$. The product measurable space is $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$, where

- $\Omega_1 \times \Omega_2$ is the Cartesian product,
- $\mathcal{F}_1 \otimes \mathcal{F}_2 \coloneqq \sigma(\{A_1 \times A_2 : A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2\})$.

Theorem 8.13 (Fubini)**.** *Suppose μ_1, μ_2 are finite measures in the setting above. Then,*

(i) There exists a unique μ on $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$ such that

$$\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2), \quad A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2.$$

(ii) Suppose $f : (\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2) \to (\mathbb{R}, \mathcal{B}_\mathbb{R})$ is measurable and $\int |f| \, \mathrm{d}\mu < \infty$. Then,

$$\begin{aligned}\int f \, \mathrm{d}\mu &= \int \left(\int f(\omega_1, \omega_2) \, \mathrm{d}\mu_1(\omega_1) \right) \mathrm{d}\mu_2(\omega_2) \\ &= \int \left(\int f(\omega_1, \omega_2) \, \mathrm{d}\mu_2(\omega_2) \right) \mathrm{d}\mu_1(\omega_1).\end{aligned}$$

8.3 Weak Convergence in $\mathbb{R}$

Theorem 8.14. *Let P_n, P be probability measures on $(\mathbb{R}, \mathcal{B}_\mathbb{R})$. The following are equivalent:*

(i) $P_n \ell \xrightarrow{\mathrm{d}} P\ell$ for all $\ell \in \mathrm{BL}(\mathbb{R})$.

(ii) $P_n(-\infty, t] \to P(-\infty, t]$ *for all* t *such that* $P(\{t\}) = 0$.

(iii) $P_n f \to Pf$ *for all* $f \in C^\infty(\mathbb{R})$*, where*

$$C^\infty(\mathbb{R}) := \left\{ f : \mathbb{R} \to \mathbb{R}, \|f\|_\infty < \infty, \|f^{(k)}\|_\infty, \forall k \right\}.$$

Proof. (i) $\Rightarrow$ (ii): HW. (i) $\Rightarrow$ (iii): $C^\infty(\mathbb{R}) \subset \mathrm{BL}(\mathbb{R})$.

It remains to show (iii) $\Rightarrow$ (i). Fix $\ell \in \mathrm{BL}(\mathbb{R})$ and note that ℓ is continuous. We make ℓ smooth by convolving it with a Gaussian kernel:

$$\ell_\sigma(x) := \mathbb{E}[\ell(x + \sigma Z)], \quad Z \sim \mathcal{N}(0, 1).$$

We can check that $\ell_\sigma \in C^\infty(\mathbb{R})$. The approximation error can be bounded as

$$\begin{aligned} |\ell_\sigma(x) - \ell(x)| &\le \mathbb{E}|\ell(x + \sigma Z) - \ell(x)| \\ &\le \mathbb{E}[K\sigma|Z|] = K\sigma\mathbb{E}|Z| \to 0, \end{aligned}$$

as $\sigma \to 0$, where K is the Lipschitz constant of ℓ. Thus for each ε we can find small σ such that $\sup_x |\ell_\sigma(x) - \ell(x)| \le \varepsilon$. This gives

$$|P_n\ell - P\ell| \le |P_n\ell_\sigma - P\ell_\sigma| + 2\varepsilon \longrightarrow 2\varepsilon,$$

which implies

$$\limsup_n |P_n\ell - P\ell| \le 2\varepsilon.$$

We send $\varepsilon \to 0$ to conclude. □

9 Central Limit Theorem

A rough statement of the Central Limit Theorem is as follows:

Proposition 9.1. *Let X_i be iid with mean* 0 *and variance* 1*. Then,*

$$\frac{X_1 + \cdots + X_n}{\sqrt{n}} \xrightarrow{\mathrm{d}} \mathcal{N}(0,1).$$

9.1 Proof by Lindberg's Method

We will use Lindberg's method of swapping variables. The idea is to show

$$\frac{X_1 + \cdots + X_n}{\sqrt{n}} \xrightarrow{\mathrm{d}} \frac{Z_1 + \cdots + Z_n}{\sqrt{n}},$$

where Z_i are iid $\mathcal{N}(0,1)$. Then use the fact that the RHS is $\mathcal{N}(0,1)$ by properties of normal distribution. To do this, we swap X_i one at a time with Z_i and control the error at each step using Taylor expansion.

More concretely, consider $2k$ random independent variables (interpreted as $X_i/\sqrt{k}$ and $Z_i/\sqrt{k}$):

$$\xi_1, \ldots, \xi_k$$
$$\eta_1, \ldots, \eta_k,$$

with

$$\mathbb{E}\xi_i = \mathbb{E}\eta_i, \quad \mathbb{E}\xi_i^2 = \mathbb{E}\eta_i^2, \quad \mathbb{E}\xi_i^3 < \infty.$$

Using this, we bound, for $f \in C^\infty$, $|\mathbb{E}f\left(\sum \xi_i\right) - \mathbb{E}f\left(\sum \eta_i\right)|$. At the ith step, we swap $X + Y$ with $X + Z$, where X, Y, Z are defined as

$$\begin{aligned} X &\coloneqq \eta_1 + \cdots + \eta_{i-1} + \xi_{i+1} + \cdots + \xi_k, \\ Y &\coloneqq \xi_i, \\ Z &\coloneqq \eta_i. \end{aligned}$$

We have

$$f(x+y) = f(x) + f'(x)y + \frac{1}{2}f''(x)y^2 + R(x,y),$$

where $R(x,y) = \frac{1}{6}f'''(x^*)y^3$ for some x^* between x and $x+y$. Since $f \in C^\infty$, $|R(x,y)| \le M|y|^3$. Now, using independence,

$$\begin{aligned} \mathbb{E}f(X+Y) &= \mathbb{E}f(X) + \mathbb{E}f'(X)\mathbb{E}Y + \frac{1}{2}\mathbb{E}f''(X)\mathbb{E}Y^2 + \mathbb{E}R(X,Y) \\ \mathbb{E}f(X+Z) &= \mathbb{E}f(X) + \mathbb{E}f'(X)\mathbb{E}Z + \frac{1}{2}\mathbb{E}f''(X)\mathbb{E}Z^2 + \mathbb{E}R(X,Z). \end{aligned}$$

Subtracting (and recalling that we matched moments):

$$|\mathbb{E}f(X+Y) - \mathbb{E}f(X+Z)| = |\mathbb{E}R(X,Y) - \mathbb{E}R(X,Z)| \le M\left(\mathbb{E}|Y|^3 + \mathbb{E}|Z|^3\right).$$

We bound $\mathbb{E}|Z|^3$ using $\mathbb{E}|Y|^3$:

$$\begin{aligned}\mathbb{E}|Z|^3 &= \mathbb{E}\left|(\mu+\sigma W)^3\right|, \quad W \sim \mathcal{N}(0,1)\\ &\le C_0\mathbb{E}\left[|\mu|^3+\sigma^3|W|^3\right]\\ &\le C_1|\mathbb{E}Y|^3 + C_2\left(\mathbb{E}Y^2\right)^{3/2} \le C_1\mathbb{E}|Y|^3 + C_2\mathbb{E}|Y|^3,\end{aligned}$$

where the last inequality follows from Holder inequality. Thus, repeating this swapping procedure k times, we have

$$\left|\mathbb{E}f\left(\sum \xi_i\right) - \mathbb{E}f\left(\sum \eta_i\right)\right| \le C\sum_{i=1}^n \mathbb{E}|\xi_i|^3.$$

This is the starting point of the Lindberg method. We summarize this as follows:

Proposition 9.2. *Let ξ_i, η_i be independent random variables with $\mathbb{E}\xi_i = \mathbb{E}\eta_i$, $\mathbb{E}\xi_i^2 = \mathbb{E}\eta_i^2$, and $\mathbb{E}|\xi_i|^3 < \infty$, and such that η_i's are Gaussian. Then, for each $f \in C^\infty$,*

$$\left|\mathbb{E}f\left(\sum_{i=1}^n \xi_i\right) - \mathbb{E}f\left(\sum_{i=1}^n \eta_i\right)\right| \le C\sum_{i=1}^n \mathbb{E}|\xi_i|^3.$$

Remark 9.3. Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p_n)$, with e.g., $p_n \to 0$. A classical result states that

$$\begin{aligned}\frac{\sum_{i=1}^n (X_i - p_n)}{\sqrt{np_n(1-p_n)}} &\xrightarrow{\text{d}} \mathcal{N}(0,1), \quad np_n \to \infty,\\ \frac{\sum_{i=1}^n (X_i - p_n)}{\sqrt{np_n(1-p_n)}} &\xrightarrow{\text{d}} \text{Poisson}(\lambda), \quad np_n \to \lambda.\end{aligned}$$

But what does this statement even mean if p_n changes by n? What is the distribution of say X_1? Is it Bernoulli(p_1) or Bernoulli(p_2) or others?

The precise meaning is the following:

$$\begin{aligned}n = 1 &: X_{11} \sim \text{Bernoulli}(p_1)\\ n = 2 &: X_{21}, X_{22} \overset{\text{iid}}{\sim} \text{Bernoulli}(p_2)\\ &\vdots\\ n &: X_{n1}, \dots, X_{nn} \overset{\text{iid}}{\sim} \text{Bernoulli}(p_n).\end{aligned}$$

The statement regards with the limiting distribution of P_n, where

$$\frac{\sum_{i=1}^n (X_{ni} - p_n)}{\sqrt{np_n(1-p_n)}} \sim P_n.$$

Thus, the fully general statement of Central Limit Theorem statement is about a **triangular array** of random variables:

$$\begin{gathered}\xi_{11}, \ldots, \xi_{1k(1)} \\ \vdots \\ \xi_{n1}, \ldots, \xi_{nk(n)},\end{gathered}$$

where $k(n) \to \infty$, and random variables in the same row are independent. ☕

Theorem 9.4 (Triangular Array CLT)**.** *Consider a triangular array such that for each row n,*

(i) $\sum_{i=1}^{k(n)} \mathbb{E}\xi_{n,i} \to \mu$,

(ii) $\sum_{i=1}^{k(n)} \mathbb{V}(\xi_{n,i}) \to \sigma^2$,

(iii) $\sum_{i=1}^{k(n)} \mathbb{E}|\xi_{n,i}|^3 \to 0$.

Then as $n \to \infty$,

$$\sum_{i=1}^{k(n)} \xi_{n,i} \xrightarrow{\mathrm{d}} \mathcal{N}(\mu, \sigma^2).$$

Proof. Define $\mu_n \coloneqq \sum_{i=1}^{k(n)} \mathbb{E}\xi_{n,i}$ and $\sigma_n^2 = \sum_{i=1}^{k(n)} \mathbb{V}(\xi_{n,i})$. Fix any $f \in C^\infty$. Note that

$$\begin{aligned}&\left|\mathbb{E}f\left(\sum_{i=1}^{k(n)} \xi_{n,i}\right) - \mathbb{E}f\left(\mathcal{N}(\mu, \sigma^2)\right)\right| \\ &\leq \underbrace{\left|\mathbb{E}f\left(\sum_{i=1}^{k(n)} \xi_{n,i}\right) - \mathbb{E}f\left(\mathcal{N}(\mu_n, \sigma_n^2)\right)\right|}_{\mathrm{I}} + \underbrace{\left|\mathbb{E}f\left(\mathcal{N}(\mu_n, \sigma_n^2)\right) - \mathbb{E}f\left(\mathcal{N}(\mu, \sigma^2)\right)\right|}_{\mathrm{II}}.\end{aligned}$$

Let $\eta_{n_1}, \ldots, \eta_{n,k(n)}$ be independent Gaussian that matches the first two moments of $\xi_{n_1}, \ldots, \xi_{n,k(n)}$. Note that $\sum_{i=1}^{k(n)} \eta_{n,i} \sim \mathcal{N}(\mu_n, \sigma_n^2)$.

I: By Lindberg swapping, $\mathrm{I} \leq C \sum_{i=1}^{k(n)} \mathbb{E}|\xi_{n,i}|^3 \to 0$.

II:

$$\begin{aligned}\mathrm{II} &= |\mathbb{E}f(\mu_n + \sigma_n Z) - \mathbb{E}f(\mu + \sigma Z)|, \qquad Z \sim \mathcal{N}(0,1) \\ &\leq C_1 \mathbb{E}\left[|\mu_n - \mu| + |\sigma_n - \sigma||Z|\right] \\ &\leq C_1|\mu_n - \mu| + C_2|\sigma_n - \sigma|\end{aligned}$$

where the first inequality follows from f being Lipschitz, and the last line converges to 0 by the first two assumptions. □

Remark 9.5. Some intuition for why this proof works for the normal distribution: (i) linear combinations of normal variables are normal, and (ii) normal distribution is determined by its first two moments. ☕

Example 9.6. Suppose $X_{n_1}, \ldots, X_{nk(n)} \overset{\text{iid}}{\sim} \text{Bernoulli}(p_n)$. We want to show

$$\frac{\sum_{i=1}^{k(n)} (X_{ni} - p_n)}{\sqrt{k(n)p_n(1-p_n)}} \xrightarrow{\text{d}} \mathcal{N}(0,1).$$

To do this, we check the conditions of the triangular array CLT: Write $\xi_{n,i} = (X_{ni} - p_n)/\sqrt{np_n(1-p_n)}$.

(i) $\sum_{i=1}^{n} \mathbb{E}\xi_{n,i} = 0$,

(ii) $\sum_{i=1}^{n} \mathbb{V}[\xi_{n,i}] = 1$,

(iii) Dropping the subscripts on p,

$$\sum_{i=1}^{n} \mathbb{E}|\xi_{n,i}|^3 = \sum_{i=1}^{n} \mathbb{E}\left|\frac{X_{ni} - p}{\sqrt{np(1-p)}}\right|^3 = n \cdot \frac{p(1-p)^3 + (1-p)p^3}{(np(1-p))^{3/2}} = n \cdot \frac{p(1-p)\left[(1-p)^2 + p^2\right]}{n^{3/2}\,(p(1-p))^{3/2}}$$

which has the same order as $1/\sqrt{np(1-p)}$. Thus the CLT holds when $np(1-p) \to \infty$.

Corollary 9.7 (Classical CLT)**.** *Let $X_i \overset{\text{iid}}{\sim} P$ with $\mathbb{E}[X] = 0$ and $\mathbb{V}(X) = 1$. Then,*

$$\frac{X_1 + \cdots + X_n}{\sqrt{n}} \xrightarrow{\text{d}} \mathcal{N}(0,1).$$

Proof. Define

$$\xi_{n,i} := \frac{X_i}{\sqrt{n}} \mathbb{1}\{|X_i| \le \sqrt{n}\}.$$

We check the conditions of the triangular array CLT:

(i) Note that

$$\begin{aligned}\left|\sum_{i=1}^{n} \mathbb{E}\xi_{n,i}\right| &= \left|\sqrt{n}\mathbb{E}[X\mathbb{1}\{|X| \le \sqrt{n}\}]\right| \\ &= \left|\sqrt{n}\mathbb{E}[X\mathbb{1}\{|X| > \sqrt{n}\}]\right| \le \mathbb{E}[X^2\mathbb{1}\{|X| > \sqrt{n}\}].\end{aligned}$$

By the dominated convergence theorem (with x^2 as the dominating function), the last term goes to 0.

(ii) Note that

$$\begin{aligned}\sum_{i=1}^{n} \mathbb{V}[\xi_{n,i}] &= n\,\mathbb{V}\left(\frac{X}{\sqrt{n}}\mathbb{1}\{|X| \le n\}\right) \\ &= \mathbb{V}\left(X\mathbb{1}\{|X| \le \sqrt{n}\}\right) \\ &= \mathbb{E}[X^2\mathbb{1}\{|X| \le \sqrt{n}\}] - \left(\mathbb{E}X\mathbb{1}\{|X| \le \sqrt{n}\}\right)^2.\end{aligned}$$

Note that the first term can be rewritten as $\mathbb{E}X^2 - \mathbb{E}[X^2 \mathbb{1}\{|X| > \sqrt{n}\}]$, which goes to 1 (use the previous step). The second term can be bounded by

$$|\mathbb{E}X\mathbb{1}\{|X| \le \sqrt{n}\}| = \left|\mathbb{E}X\mathbb{1}\{|X| > \sqrt{n}\}\right| \le \mathbb{E}[|X|\mathbb{1}\{|X| > \sqrt{n}\}],$$

which goes to 0 by DCT.

(iii) We have

$$\begin{aligned}\sum_{i=1}^{n} \mathbb{E}|\xi_{n,i}|^3 &= n\mathbb{E}\left[\left(\frac{|X|}{\sqrt{n}}\right)^3 \mathbb{1}\{|X| \le \sqrt{n}\}\right] \\ &= \mathbb{E}\left[\frac{|X|^3}{\sqrt{n}}\mathbb{1}\left\{\,|X| \le \sqrt{n}\,\right\}\right] \le \mathbb{E}\left[X^2 \min\left\{\frac{|X|}{\sqrt{n}}, 1\right\}\right].\end{aligned}$$

By DCT, the last term goes to 0.

Thus by the triangular array CLT, we have

$$\sum_{i=1}^{n} \xi_{n,i} \xrightarrow{\mathrm{d}} \mathcal{N}(0,1).$$

It remains to argue that truncation does not change the limit. For each $f \in C^\infty(\mathbb{R})$, we bound

$$\begin{aligned}\left|\mathbb{E}f\left(\sum_{i=1}^{n}\xi_{n,i}\right) - \mathbb{E}f\left(\sum_{i=1}^{n}\frac{X_i}{\sqrt{n}}\right)\right| &\le \left|\left[\mathbb{E}f\left(\sum_{i=1}^{n}\xi_{n,i}\right) - \mathbb{E}f\left(\sum_{i=1}^{n}\frac{X_i}{\sqrt{n}}\right)\right]\mathbb{1}\left\{\sum \xi_{n,i} \ne \sum \frac{X_i}{\sqrt{n}}\right\}\right| \\ &\le 2\|f\|_\infty \mathbb{P}\left(\left\{\sum_{i=1}^{n}\xi_{n,i} \ne \sum_{i=1}^{n}\frac{X_i}{\sqrt{n}}\right\}\right) \\ &\le 2\|f\|_\infty \sum_{i=1}^{n}\mathbb{P}\left(\xi_{n,i} \ne \frac{X_i}{\sqrt{n}}\right) = 2\|f\|_\infty \sum_{i=1}^{n}\mathbb{P}\left(|X_i| > \sqrt{n}\right) \\ &= 2\|f\|_\infty \mathbb{E}\left[n\mathbb{1}\{|X| > \sqrt{n}\}\right] \le 2\|f\|_\infty \mathbb{E}\left[X^2\mathbb{1}\{|X| > \sqrt{n}\}\right],\end{aligned}$$

which goes to 0 by DCT. Since this is true for each f, we are done by triangular inequality. □

Theorem 9.8 (Lindeberg CLT)**.** *Consider a triangular array $\{X_{n,i}\}$ such that*

(i) $\mathbb{E}X_{n,i} = 0$.

(ii) $\sum \mathbb{V}(X_{n,i}) = 1$.

(iii) ***(Lindeberg condition)*** *For each $\varepsilon > 0$,*

$$L_n(\varepsilon) := \sum_i \mathbb{E}\left[X_{n,i}^2 \mathbb{1}\{|X_{n,i}| > \varepsilon\}\right] \longrightarrow 0.$$

Then,

$$\sum_i X_{n,i} \xrightarrow{\mathrm{d}} \mathcal{N}(0,1).$$

This is almost the most general CLT for triangular arrays. There is another version of the CLT called the Feller CLT, which only requires that $L_n(\varepsilon) \to 0$ for some $\varepsilon > 0$.

Proof. Define

$$H_n(\varepsilon) := \frac{L_n(\varepsilon)}{\varepsilon^2}.$$

By assumption, for each $\varepsilon > 0$, we have $H_n(\varepsilon) \to 0$. This implies the existence of a sequence $\varepsilon_n \to 0$ such that $H_n(\varepsilon_n) \to 0$ (see next lemma).

Define

$$\xi_{n,i} := X_{n,i}\mathbb{1}\{|X_{n,i}| \le \varepsilon_n\}.$$

We will proceed as in the proof of the classical CLT to show that $\sum \xi_{n,i} \xrightarrow{\mathrm{d}} \mathcal{N}(0,1)$ and that truncation does not change the limit. We first check the conditions of the triangular array CLT.

(i) First,

$$\begin{aligned}\left|\sum \mathbb{E}\xi_{n,i}\right| &= \left|\sum \mathbb{E}X_{n,i}\mathbb{1}\{|X_{n,i}| > \varepsilon_n\}\right| \\ &\le \frac{1}{\varepsilon_n}\left|\sum \mathbb{E}X_{n,i}^2\mathbb{1}\{|X_{n,i}| > \varepsilon_n\}\right| = \varepsilon_n H_n(\varepsilon_n) \longrightarrow 0.\end{aligned}$$

(ii) Next,

$$\sum \mathbb{V}(\xi_{n,i}) = \sum \left[\mathbb{E}\xi_{n,i}^2 - \left(\mathbb{E}\xi_{n,i}\right)^2\right].$$

The first term can be rewritten as

$$\begin{aligned}\sum \mathbb{E}\xi_{n,i}^2 &= \sum \mathbb{E}\left[X_{n,i}^2\mathbb{1}\{|X_{n,i}| \le \varepsilon_n\}\right] \\ &= \sum \mathbb{E}X_{n,i}^2 - \sum \mathbb{E}\left[X_{n,i}^2\mathbb{1}\{|X_{n,i}| > \varepsilon_n\}\right] \\ &= 1 - \varepsilon_n^2 H_n(\varepsilon_n) \longrightarrow 1.\end{aligned}$$

The second term can be bounded by

$$\begin{aligned}\left(\mathbb{E}\xi_{n,i}\right)^2 &= \sum \left(\mathbb{E}\left[X_{n,i}\mathbb{1}\left\{\,|X_{n,i}| \le \varepsilon_n\,\right\}\right]\right)^2 \\ &= \sum \left(\mathbb{E}\left[X_{n,i}\mathbb{1}\left\{\,|X_{n,i}| > \varepsilon_n\,\right\}\right]\right)^2 \\ &\le \sum \mathbb{E}\left[X_{n,i}^2\mathbb{1}\left\{\,|X_{n,i}| > \varepsilon_n\,\right\}\right] = L_n(\varepsilon_n) \longrightarrow 0.\end{aligned}$$

(iii) Finally,

$$\begin{aligned}\sum \mathbb{E}|\xi_{n,i}|^3 &= \sum \mathbb{E}\left[|X_{n,i}|^3\mathbb{1}\{|X_{n,i}| \le \varepsilon_n\}\right] \\ &\le \varepsilon_n \sum \mathbb{E}\left[X_{n,i}^2\mathbb{1}\{|X_{n,i}| \le \varepsilon_n\}\right] = \varepsilon_n L_n(\varepsilon_n) \longrightarrow 0.\end{aligned}$$

This gives $\sum \xi_{n,i} \xrightarrow{\mathrm{d}} \mathcal{N}(0,1)$. Now, fix any $f \in C^\infty(\mathbb{R})$. We have

$$\begin{aligned}
\left|\mathbb{E}f\left(\sum_i \xi_{n,i}\right) - \mathbb{E}f\left(\sum_i X_{n,i}\right)\right| &\le 2\|f\|_\infty \mathbb{P}\left(\left\{\sum_i \xi_{n,i} \neq \sum_i X_{n,i}\right\}\right) \\
&\le 2\|f\|_\infty \sum_i \mathbb{P}\left(\xi_{n,i} \neq X_{n,i}\right) \\
&= 2\|f\|_\infty \sum_i \mathbb{P}\left(|X_{n,i}| > \varepsilon_n\right) \\
&\le 2\|f\|_\infty \mathbb{E}\left[\sum_i \frac{X_{n,i}^2}{\epsilon_n^2} \mathbb{1}\{|X_{n,i}| > \varepsilon_n\}\right] \\
&= 2\|f\|_\infty H_n(\varepsilon_n) \longrightarrow 0.
\end{aligned}$$

Thus,

$$\begin{aligned}
&\left|\mathbb{E}f\left(\sum_i X_{n,i}\right) - \mathbb{E}f(\mathcal{N}(0,1))\right| \\
&\le \left|\mathbb{E}f\left(\sum_i X_{n,i}\right) - \mathbb{E}f\left(\sum_i \xi_{n,i}\right)\right| + \left|\mathbb{E}f\left(\sum_i \xi_{n,i}\right) - \mathbb{E}f(\mathcal{N}(0,1))\right| \longrightarrow 0.
\end{aligned}$$

□

In the proof above, we used the following lemma:

Lemma 9.9. *Suppose H_n is such that for any $\varepsilon > 0$, we have $H_n(\varepsilon) \to 0$. Then, there exists a sequence $\varepsilon_n \to 0$ such that $H_n(\varepsilon_n) \to 0$.*

Proof. For each k, we can find large n_k such that for each $n > n_k$,

$$H_n\left(\frac{1}{k}\right) < \frac{1}{k}.$$

We may further pick n_k to be increasing in k. Set $\varepsilon_n := 1/k$ for $n_k \le n < n_{k+1}$ and we are done. □

We can rewrite the Lindeberg CLT in a more familiar form as follows:

Theorem 9.10. *Consider independent $X_{ni}, \dots, X_{nn}$ with $\mathbb{E}X_{ni} = 0$. Write*

$$S_n := \sqrt{\sum_i \mathbb{E}X_{ni}^2}.$$

If the Lindeberg condition

$$\frac{1}{S_n^2} \sum_i \mathbb{E}\left[X_{n,i}^2 \mathbb{1}\{|X_{n,i}| > \varepsilon S_n\}\right] \longrightarrow 0, \quad \forall \varepsilon > 0$$

holds, then

$$\frac{X_{n1} + \cdots + X_{nn}}{S_n} \xrightarrow{\mathrm{d}} \mathcal{N}(0,1).$$

We have also the following result, with slightly stronger assumptions that are sometimes easier to check:

Theorem 9.11 (Lyapunov CLT)**.** *Consider independent $X_{ni}, \dots, X_{nn}$ with $\mathbb{E}X_{ni} = 0$. Write*

$$S_n := \sqrt{\sum_i \mathbb{E}X_{ni}^2}.$$

If for some $\delta > 0$,

$$\frac{1}{S_n^{2+\delta}} \sum_i \mathbb{E}|X_{n,i}|^{2+\delta} \longrightarrow 0,$$

then

$$\frac{X_{n1} + \cdots + X_{nn}}{S_n} \xrightarrow{\mathrm{d}} \mathcal{N}(0,1).$$

Proof. For each $\varepsilon > 0$,

$$X_{n,i}^2 \mathbb{1}\{|X_{n,i}| > \varepsilon S_n\} = X_{n,i}^2 \mathbb{1}\{|X_{n,i}|^\delta > \varepsilon^\delta S_n^\delta\} \le \frac{|X_{n,i}|^{2+\delta}}{\varepsilon^\delta S_n^\delta}.$$

Thus,

$$\frac{1}{S_n^2} \sum_i \mathbb{E}\left[X_{n,i}^2 \mathbb{1}\{|X_{n,i}| > \varepsilon S_n\}\right] \le \frac{1}{\varepsilon^\delta S_n^{2+\delta}} \sum_i \mathbb{E}|X_{n,i}|^{2+\delta} \longrightarrow 0,$$

so the Lindeberg condition holds. □

9.2 Proof by Stein's Method

Lemma 9.12 (Stein's Identity)**.** *Let $Z \sim \mathcal{N}(0,1)$. Let f be absolutely continuous and such that $\mathbb{E}|f'(Z)| < \infty$, $\mathbb{E}|Zf(Z)| < \infty$. Then, $\mathbb{E}f'(Z) = \mathbb{E}Zf(Z)$.*

The reverse turns out also to be true, which is the key to Stein's method. Instead of checking the convergence of the distribution, we can check the convergence of the expectation of $f'(X) - Xf(X)$ for each f.

Proof *(A formal calculation)***.** Note that

$$\varphi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}, \quad \varphi'(t) = -t\varphi(t).$$

We have using integration by parts

$$\mathbb{E}Zf(Z) = \int tf(t)\varphi(t)\,\mathrm{d}t = -\left[f(t)\varphi(t)\right]_{-\infty}^{\infty} + \int f'(t)\varphi(t)\,\mathrm{d}t = \mathbb{E}f'(Z).$$

Thus provided the boundary term vanishes, we have the desired result. □

Proof *(A rigorous proof)***.** Since f is absolutely continuous, it has derivative a.e., with

$$f(b) - f(a) = \int_a^b f'(t)\,\mathrm{d}t.$$

(In this we see a connection with the definition of absolute continuity of measures: the measure induced by f is absolutely continuous with respect to the Lebesgue measure, and f' is the Radon-Nikodym derivative.) We have

$$\mathbb{E}f'(Z) = \int_{-\infty}^{\infty} f'(t)\varphi(t)\,\mathrm{d}t = \int_{-\infty}^{\infty} f'(t)\int_{-\infty}^{t}\varphi'(x)\,\mathrm{d}x\,\mathrm{d}t = \int_{-\infty}^{\infty}\varphi'(x)\int_{x}^{\infty} f'(t)\,\mathrm{d}t\,\mathrm{d}x.$$

We want to write the inner integral as $f(\infty) - f(x)$, but this works only if $f(\infty) < \infty$.

We use the following trick:

$$\begin{aligned}
\mathbb{E}f'(Z) &= \int_{-\infty}^{0} f'(t)\varphi(t)\,\mathrm{d}t + \int_{0}^{\infty} f'(t)\varphi(t)\,\mathrm{d}t \\
&= \int_{-\infty}^{0} f'(t)\int_{-\infty}^{t}\varphi'(x)\,\mathrm{d}x\,\mathrm{d}t + \int_{0}^{\infty} f'(t)\int_{t}^{\infty} -\varphi'(x)\,\mathrm{d}x\,\mathrm{d}t \\
&= \int_{-\infty}^{0}\varphi'(x)\int_{x}^{0} f'(t)\,\mathrm{d}t\,\mathrm{d}x - \int_{0}^{\infty}\varphi'(x)\int_{0}^{x} f'(t)\,\mathrm{d}t\,\mathrm{d}x \\
&= \int_{-\infty}^{0}\varphi'(x)(f(0) - f(x))\,\mathrm{d}x - \int_{0}^{\infty}\varphi'(x)(f(x) - f(0))\,\mathrm{d}x \\
&= f(0)\left(\int_{-\infty}^{0}\varphi'(x)\,\mathrm{d}x + \int_{0}^{\infty}\varphi'(x)\,\mathrm{d}x\right) - \int_{-\infty}^{0}\varphi'(x)f(x)\,\mathrm{d}x - \int_{0}^{\infty}\varphi'(x)f(x)\,\mathrm{d}x \\
&= f(0)\underbrace{\int_{-\infty}^{\infty} -x\varphi(x)\,\mathrm{d}x}_{\mathbb{E}[-Z]} + \int_{-\infty}^{\infty} xf(x)\varphi(x)\,\mathrm{d}x = \mathbb{E}Zf(Z).
\end{aligned}$$

□

For each t, let $f = f_t$ denote the solution to **Stein's equation**:

$$f'(w) - wf(w) = \mathbb{1}\{w \le t\} - \Phi(t).$$

Note that if random variable W is such that $\mathbb{E}f'(W) = \mathbb{E}Wf(W)$ for all f with sufficient regularity, we have by taking expectation above that $\mathbb{P}(W \le t) = \Phi(t)$, so $W \sim \mathcal{N}(0,1)$. That is,

Proposition 9.13. *If W is such that $\mathbb{E}f'(W) = \mathbb{E}Wf(W)$ for all f such that $\mathbb{E}|f'(Z)| < \infty$ and $\mathbb{E}|Zf(Z)| < \infty$, then $W \sim \mathcal{N}(0,1)$.*

The regularity conditions for f can of course be weakened so long as they cover solutions to the Stein's equations for all t.

We solve the DE above to find the regularity conditions needed. Note that

$$\frac{\mathrm{d}}{\mathrm{d}w}\left[e^{-w^2/2}f(w)\right] = e^{-w^2/2}f'(w) - we^{-w^2/2}f(w) = e^{-w^2/2}\left(\mathbb{1}\{w \le t\} - \Phi(t)\right).$$

Thus integrating both sides,

$$f(w) = e^{w^2/2} \int_{-\infty}^{w} e^{-x^2/2} \left(\mathbb{1}\{x \le t\} - \Phi(t)\right) \, \mathrm{d}x.$$

Now recall that we have the regularity restrictions $\mathbb{E}|f'(Z)| < \infty$ and $\mathbb{E}|Zf(Z)| < \infty$. The constant term $Ce^{w^2/2}$ must be 0 to ensure these conditions hold. It turns out also that Φ is the only cdf that satisfies the restrictions above (so we can't use the same argument to argue that W has any distribution G). To see this, swap Φ for G and note that

$$\begin{aligned}
\mathbb{E}Zf(Z) &= \int w \frac{1}{\sqrt{2\pi}} e^{-w^2/2} f(w) \, \mathrm{d}w \\
&= \int w \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{w} e^{-x^2/2} \left(\mathbb{1}\{x \le t\} - G(t)\right) \, \mathrm{d}x \, \mathrm{d}w.
\end{aligned}$$

Thus, in order to have $\mathbb{E}Zf(Z) < \infty$, we must have

$$\lim_{w \to \infty} \int_{-\infty}^{w} e^{-x^2/2} \left(\mathbb{1}\{x \le t\} - G(t)\right) \, \mathrm{d}x = \int_{-\infty}^{\infty} e^{-x^2/2} \left(\mathbb{1}\{x \le t\} - G(t)\right) \, \mathrm{d}x = 0,$$

which gives

$$G(t) = \frac{\int_{-\infty}^{\infty} \mathbb{1}\{x \le t\} e^{-x^2/2} \, \mathrm{d}x}{\int_{-\infty}^{\infty} e^{-x^2/2} \, \mathrm{d}x} = \Phi(t).$$

We now consider now a more general version of the Stein's equation:

$$f'(w) - wf(w) = h(w) - Nh,$$

where $Nh \coloneqq \mathbb{E}h(\mathcal{N}(0,1))$. Any random variable W such that $\mathbb{E}f'(W) = \mathbb{E}Wf(W)$ for all f with sufficient regularity must satisfy

$$\mathbb{E}h(W) = \mathbb{E}h(\mathcal{N}(0,1)),$$

and so $W \sim \mathcal{N}(0,1)$. If W is not normal, then we can use

$$\mathbb{E}\left[f'(W) - Wf(W)\right] = \mathbb{E}h(W) - \mathbb{E}h(\mathcal{N}(0,1))$$

to gauge how far the distribution of W is from $\mathcal{N}(0,1)$.

With the same integrating factor, the solution to the DE above is given by

$$f_h(w) = e^{w^2/2} \int_{-\infty}^{w} e^{-x^2/2} \left(h(x) - Nh\right) \, \mathrm{d}x.$$

Thus we have the following

Lemma 9.14. *Stein's equation $f_h'(w) - wf_h(w) = h(w) - Nh$, where $Nh \coloneqq \mathbb{E}h(\mathcal{N}(0,1))$ has solution*

$$f_h(w) = e^{w^2/2} \int_{-\infty}^{w} e^{-x^2/2} \left(h(x) - Nh\right) \, \mathrm{d}x.$$

Lemma 9.15. $\|f_h\|_\infty, \|f_h'\|_\infty, \|f_h''\|_\infty \le 2\|h'\|_\infty$.

Proof. Tedious. We show only that $\|f_h\|_\infty \le C\,\mathrm{Lip}(h)$. Since $e^{w^2/2}\mathbb{E}[h(Z) - Nh] = 0$, we have

$$f_h(w) = e^{w^2/2}\int_{-\infty}^{w} e^{-x^2/2}\,(h(x) - Nh)\ \mathrm{d}x = -e^{w^2/2}\int_{w}^{\infty} e^{-x^2/2}\,(h(x) - Nh)\ \mathrm{d}x.$$

Suppose $w > 0$; the same analysis applies to $w < 0$, with the other expression for $f_h(w)$. We have

$$\begin{aligned}
|f_h(w)| &\le e^{w^2/2}\int_{w}^{\infty} e^{-x^2/2}|h(x) - Nh|\ \mathrm{d}x \\
&\le e^{w^2/2}\int_{w}^{\infty} e^{-x^2/2}|h(x) - h(0)|\ \mathrm{d}x + e^{w^2/2}\int_{w}^{\infty} e^{-x^2/2}|h(0) - \underbrace{Nh}_{\mathbb{E}h(Z)}|\ \mathrm{d}x \\
&\le \mathrm{Lip}(h)e^{w^2/2}\int_{w}^{\infty} xe^{-x^2/2}\ \mathrm{d}x + \mathrm{Lip}(h)\mathbb{E}|Z|e^{w^2/2}\int_{w}^{\infty} e^{-x^2/2}\ \mathrm{d}x.
\end{aligned}$$

Now,

$$\int_{w}^{\infty} xe^{-x^2/2}\ \mathrm{d}x = \int_{w}^{\infty} -(e^{-x^2/2})'\ \mathrm{d}x = e^{-w^2/2},$$

and

$$\int_{w}^{\infty} \frac{1}{\sqrt{2\pi}}e^{-x^2/2}\ \mathrm{d}x = \mathbb{P}(Z > w) = \mathbb{P}\left(e^{tZ} > e^{tw}\right) \le \frac{\mathbb{E}e^{tZ}}{e^{tw}} = e^{-tw+t^2/2}.$$

We choose t to optimize this bound: taking $t = w$,

$$\int_{w}^{\infty} \frac{1}{\sqrt{2\pi}}e^{-x^2/2}\ \mathrm{d}x \le e^{-w^2/2}.$$

Combining these bounds,

$$|f_h(w)| \le \mathrm{Lip}(h)e^{w^2/2}\left(e^{-w^2/2} + \mathbb{E}|Z|\sqrt{2\pi}e^{-w^2/2}\right) \quad \le \mathrm{Lip}(h)\left(1 + \sqrt{2\pi}\mathbb{E}|Z|\right).$$

□

With this, we have that $\mathbb{E}f_h'(W)$ and $\mathbb{E}Wf_h(W)$ exist provided $\mathbb{E}|W| < \infty$. Thus, we may conclude the following:

Proposition 9.16. *Let $Z \sim \mathcal{N}(0,1)$. Then, for each absolutely continuous f, $\mathbb{E}f'(Z) = \mathbb{E}Zf(Z)$ provided that both sides exist. Conversely, if W is such that $\mathbb{E}|W| < \infty$ and $\mathbb{E}f'(W) = \mathbb{E}Wf(W)$ for all bounded absolutely continuous f with bounded derivative, then $W \sim \mathcal{N}(0,1)$.*

9.2.1 Wasserstein Distance

Definition 9.17. Consider two random variables $W \sim P$ and $Z \sim Q$. The **Wasserstein distance** is given by

$$\mathcal{W}(P,Q) = \mathcal{W}(W,Z) := \sup_{\mathrm{Lip}(h)\le 1} |\mathbb{E}h(W) - \mathbb{E}h(Z)|.$$

From this definition we see that convergence in Wasserstein distance implies convergence in distribution. But note from the discussion in the preceding section that

$$\begin{aligned}\mathcal{W}(P, \mathcal{N}(0,1)) &= \sup_{\mathrm{Lip}(h)\le 1} \left|\mathbb{E} f_h'(W) - \mathbb{E} W f_h(W)\right| \\ &\le \sup_{\|f\|_\infty \le 2, \|f'\|_\infty \le 2, \|f''\|_\infty \le 2} |\mathbb{E} f'(W) - \mathbb{E} W f(W)|.\end{aligned}$$

where f_h is the solution to the Stein's equation with h as the right-hand side.

9.2.2 Proof of CLT using Stein's Method

Consider independent $X_1, \dots, X_n$, each with mean 0, variance 1, and bounded third moments. Write

$$W := \frac{X_1 + \cdots + X_n}{\sqrt{n}}, \quad W_i := W - \frac{X_i}{\sqrt{n}}.$$

Note that $X_i \perp\!\!\!\perp W_i$. We have

$$\mathbb{E} W f(W) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbb{E} X_i f(W).$$

We rewrite each of the terms on the right as

$$\begin{aligned}\mathbb{E} X_i f(W) &= \mathbb{E} X_i \left(f(W) - f(W_i)\right) + \underbrace{\mathbb{E} X_i f(W_i)}_{=0} \\ &= \mathbb{E}\left[X_i\big(f(W) - f(W_i) - f'(W_i)(W - W_i)\big)\right] + \mathbb{E}\left[X_i f'(W_i)(W - W_i)\right] \\ &= \frac{1}{2}\mathbb{E} X_i f''(\xi_i)(W - W_i)^2 + \frac{1}{\sqrt{n}} \mathbb{E} X_i^2 f'(W_i) \\ &= \frac{1}{2n}\mathbb{E} X_i^3 f''(\xi_i) + \frac{1}{\sqrt{n}} \mathbb{E} f'(W_i).\end{aligned}$$

Summing over,

$$\mathbb{E} W f(W) = \frac{1}{2\sqrt{n}} \frac{1}{n} \sum_{i=1}^n \mathbb{E} X_i^3 f''(\xi_i) + \frac{1}{n} \sum_{i=1}^n \mathbb{E} f'(W_i).$$

We can now bound:

$$\begin{aligned}|\mathbb{E} f'(W) - \mathbb{E} W f(W)| &\le \frac{1}{n} \sum_{i=1}^n \mathbb{E}|f'(W) - f'(W_i)| + \frac{1}{2\sqrt{n}} \frac{1}{n} \sum_{i=1}^n \mathbb{E}\left(|X_i|^3 |f''(\xi_i)|\right) \\ &\le \frac{1}{n} \sum_{i=1}^n 2\mathbb{E}\frac{|X_i|}{\sqrt{n}} + \frac{1}{\sqrt{n}} \frac{1}{n} \sum_{i=1}^n \mathbb{E}|X_i|^3 \\ &= \frac{1}{\sqrt{n}} \frac{1}{n} \sum_{i=1}^n \left(2\mathbb{E}|X_i| + \mathbb{E}|X_i|^3\right) \le \frac{3}{\sqrt{n}} \frac{1}{n} \sum_{i=1}^n \mathbb{E}|X_i|^3.\end{aligned}$$

In the last line we use the fact that

$$\mathbb{E}|X_i| \le \left(\mathbb{E}|X_i|^3\right)^{1/3} \le \mathbb{E}|X_i|^3,$$

the second equality here following from the fact that

$$1 \le \mathbb{E}X_i^2 \le \left(\mathbb{E}|X_i|^3\right)^{2/3}.$$

Thus we have the following

Theorem 9.18. *Let $X_1, \dots, X_n$ be independent, each with mean 0 and variance 1. Then,*

$$\mathcal{W}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, \mathcal{N}(0,1)\right) \le \frac{3}{\sqrt{n}} \frac{1}{n} \mathbb{E}|X_1|^3.$$

Thus if X_i has uniformly bounded third moment, then we get CLT.

The big advantage of Stein's method is that it gives a rate of convergence, and that it can be applied to prove other limit theorems, such as Poisson approximation and large deviations.

Definition 9.19. The **Kolmogorov-Smirnov distance** is given by

$$\mathrm{KS}(P, Q) = \mathrm{KS}(W, Z) := \sup_{t \in \mathbb{R}} |P(-\infty, t] - Q(-\infty, t]|.$$

We can't directly use the same argument as above to bound the KS distance, since the indicator function is not Lipschitz. But it is easy enough to approximate the indicator with Lipschitz functions, say using a function affine between t and $t + \alpha$ and constant elsewhere to approximate $\mathbb{1}\{w \le t\}$. This function h has Lipschitz constant $1/\alpha$ and

$$h(w + \alpha) \le \mathbb{1}\{w \le t\} \le h(w) \le \mathbb{1}\{w \le t + \alpha\}.$$

Thus,

$$\begin{aligned}
\mathbb{P}(W \le t) - \mathbb{P}(Z \le t) &\le \mathbb{E}h(W) - \mathbb{P}(Z \le t) \\
&= \mathbb{E}h(W) - \mathbb{E}h(Z) + \mathbb{E}h(Z) - \mathbb{P}(Z \le t) \\
&= \mathbb{E}h(W) - \mathbb{E}h(Z) + \mathbb{P}(Z \le t + \alpha) - \mathbb{P}(Z \le t) \\
&\le \mathbb{E}h(W) - \mathbb{E}h(Z) + \alpha,
\end{aligned}$$

where in the final line we use

$$\int_t^{t+\alpha} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \, \mathrm{d}x \le \frac{\alpha}{\sqrt{2\pi}} \le \alpha.$$

Thus,

$$\mathrm{KS}(W, Z) = \sup_{t \in \mathbb{R}} |\mathbb{P}(W \le t) - \mathbb{P}(Z \le t)| \le \frac{1}{\alpha} \mathcal{W}(W, Z) + \alpha, \quad \forall \alpha > 0,$$

which gives

Proposition 9.20.

$$\mathrm{KS}(W, Z) \leq 2\sqrt{\mathcal{W}(W, Z)}.$$

Corollary 9.21. *Let $X_1, \ldots, X_n$ be independent, each with mean* 0 *and variance* 1. *Then,*

$$\mathrm{KS}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, \mathcal{N}(0, 1)\right) \leq 2\sqrt{\frac{3}{\sqrt{n}} \frac{1}{n} \sum_i \mathbb{E}|X_i|^3}.$$

Berry-Esseen gives a tighter bound for the KS distance:

Proposition 9.22 (Berry-Esseen)**.** *Let $X_1, \ldots, X_n$ be independent, each with mean* 0 *and variance* 1. *Then,*

$$\mathrm{KS}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, \mathcal{N}(0, 1)\right) \leq \frac{C}{\sqrt{n}} \frac{1}{n} \sum_i \mathbb{E}|X_i|^3,$$

for some universal constant C.

10 Poisson Limit Theorems

Also called the **law of small numbers**, since Poisson distribution describes the number of occurrences of rare events.

The central result is

$$\text{Binomial}(n, p) \xrightarrow{\text{d}} \text{Poisson}(\lambda), \quad \text{when } np \to \lambda.$$

10.1 Coupling

Consider random variables

$$\begin{aligned} X &: (\Omega_1, \mathcal{F}, \mathbb{P}_1) \longrightarrow (\mathbb{R}, \mathcal{B}), \\ Y &: (\Omega_2, \mathcal{F}, \mathbb{P}_2) \longrightarrow (\mathbb{R}, \mathcal{B}). \end{aligned}$$

Definition 10.1. We say $(\hat{X}, \hat{Y})$ is a **coupling** of (X, Y) if $\hat{X}$ and $\hat{Y}$ are on the same space, $\hat{X}$ has the same distribution as X, and $\hat{Y}$ has the same distribution as Y.

Example 10.2. Let $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(1, 1)$. A coupling of (X, Y) is given by $(X, X + 1)$.

Definition 10.3. $(\hat{X}, \hat{Y})$ is a **maximal coupling** of (X, Y) if $\mathbb{P}(\hat{X} = \hat{Y})$ is maximized.

Theorem 10.4 (Maximal coupling, discrete case)**.** *Let X, Y be discrete, and wlog, take values in $\mathbb{N}$. Write $p_i := \mathbb{P}(X = i)$ and $q_i := \mathbb{P}(Y = i)$. The maximal coupling of (X, Y) satisfies*

$$\mathbb{P}(\hat{X} \neq \hat{Y}) = \sum_i \min\{p_i, q_i\} = \sum_i p_i \wedge q_i.$$

Proof. Step 1: The upper bound.

Write $\alpha := \sum_i p_i \wedge q_i$ and let $(\hat{X}, \hat{Y})$ be a coupling. Define $A := \{i : p_i < q_i\}$. We have

$$\begin{aligned} \mathbb{P}(\hat{X} = \hat{Y}) &= \mathbb{P}(\hat{X} = \hat{Y} \in A) + \mathbb{P}(\hat{X} = \hat{Y} \in A^c) \\ &\leq \mathbb{P}(\hat{X} \in A) + \mathbb{P}(\hat{Y} \in A^c) \\ &= \mathbb{P}(X \in A) + \mathbb{P}(Y \in A^c) = \sum_{i \in A} p_i + \sum_{i \in A^c} q_i = \alpha. \end{aligned}$$

Step 2: We show that equality can be achieved. Write

$$b_i := \frac{p_i \wedge q_i}{\alpha}, \quad c_i := \frac{p_i - p_i \wedge q_i}{1 - \alpha}, \quad d_i = \frac{q_i - p_i \wedge q_i}{1 - \alpha}.$$

One can verify by direct computation that b, c, d are probability distributions.

We construct the coupling as follows: Let B, C, D, I be independent random variables such that B, C, D has distributions b, c, d respectively and $I \sim \text{Bernoulli}(\alpha)$. Then define

$$(\hat{X}, \hat{Y}) = \begin{cases} (B, B), & I = 0 \\ (C, D), & I = 1 \end{cases}.$$

We verify that $(\hat{X}, \hat{Y})$ is indeed a coupling of (X, Y): for each i,

$$\begin{aligned}\mathbb{P}(\hat{X} = i) &= \mathbb{P}(I = 0)\mathbb{P}(\hat{X} = i | I = 0) + \mathbb{P}(I = 1)\mathbb{P}(\hat{X} = i | I = 1) \\ &= \mathbb{P}(I = 0)\mathbb{P}(B = i) + \mathbb{P}(I = 1)\mathbb{P}(C = i) \\ &= \alpha b_i + (1 - \alpha)c_i = p_i,\end{aligned}$$

and similarly for Y. Moreover, since $\hat{X} = \hat{Y}$ whenever $I = 0$, we have

$$\mathbb{P}(\hat{X} = \hat{Y}) \geq \mathbb{P}(I = 1) = \alpha.$$

□

Remark 10.5. The intuition here is that we want to maximize overlap at each x, and the best we can do is to overlap a mass of $p_i \wedge q_i$. The assumption of discreteness is not essential; the same proof applies in the continuous case: ☕

Theorem 10.6 (Maximal coupling, continuous case)**.** *Let P and Q be probability measures on $\mathbb{R}$ with densities p and q with respect to Lebesgue measure. Then the maximal coupling $(\hat{X}, \hat{Y})$ satisfies $\mathbb{P}(\hat{X} = \hat{Y}) = \int p(x) \wedge q(x)\, \mathrm{d}x$.*

Proof. Define $r(x) = p(x) \wedge q(x)$ and $m = \int r(x)\, \mathrm{d}x$. We construct the coupling as follows: With probability m, draw Z with density $r(x)/m$ and set $\hat{X} = \hat{Z}$. With probability $1 - m$, draw $\hat{X}$ and $\hat{Y}$ independently with densities $(p(x) - r(x))/(1 - m)$ and $(q(x) - r(x))/(1 - m)$ respectively. Then,

$$\mathbb{P}(\hat{X} \in A) = m \int_A \frac{r(x)}{m}\, \mathrm{d}x + (1 - m) \int_A \frac{p(x) - r(x)}{1 - m}\, \mathrm{d}x = \int_A p(x)\, dx,$$

so $\hat{X} \sim P$. Similarly $\hat{Y} \sim Q$. By construction we have $\mathbb{P}(\hat{X} = \hat{Y}) = m = \int p(x) \wedge q(x)\, \mathrm{d}x$.

A similar argument as in the first part of the previous proof shows that this must be a maximal coupling. □

10.2 Total Variation Distance

Definition 10.7. Let P, Q be probability measures on $(\Omega, \mathcal{F})$. The **total variation distance** is given by

$$\mathrm{TV}(P, Q) := \sup_{B \in \mathcal{F}} |P(B) - Q(B)|.$$

Note that convergence in TV implies convergence in cdf and thus weak convergence.

Proposition 10.8. *Let P, Q be measures and let $(\hat{X}, \hat{Y})$ be a maximal coupling of (P, Q). Then, $\mathrm{TV}(P, Q) = \mathbb{P}(\hat{X} \neq \hat{Y})$.*

Proof. We show the discrete case. It suffices to show

$$\mathrm{TV}(P, Q) = 1 - \sum_i p_i \wedge q_i.$$

Note that the quantity $\sum_i p_i \wedge q_i$ is a measure of how close P and Q are, called the **total variation affinity**. Let $A := \{i : p_i < q_i\}$. We have by definition

$$\mathrm{TV}(P, Q) \geq |P(A) - Q(A)| = Q(A) - P(A).$$

We show the opposite inequality. For each B,

$$\begin{aligned}
|P(B) - Q(B)| &= \left|\sum_{i \in B} p_i - q_i\right| = \left|\sum_{i \in B \cap A} (p_i - q_i) + \sum_{i \in B \cap A^c} (p_i - q_i)\right| \\
&= \left|\sum_{i \in B \cap A^c} (p_i - q_i) - \sum_{i \in B \cap A} (q_i - p_i)\right| \\
&\leq \max\left\{\sum_{i \in B \cap A^c} (p_i - q_i), \sum_{i \in B \cap A} (q_i - p_i)\right\} \\
&\leq \max\left\{\sum_{i \in A^c} (p_i - q_i), \sum_{i \in A} (q_i - p_i)\right\},
\end{aligned}$$

where both of the terms in the max are equal to $Q(A) - P(A)$. Taking sup over B gives

$$\mathrm{TV}(P, Q) \leq Q(A) - P(A).$$

We have thus shown that $\mathrm{TV}(P, Q) = Q(A) - P(A)$. Now, since

$$\sum_i p_i \wedge q_i = \sum_{i \in A} p_i + \sum_{i \in A^c} q_i = P(A) + Q(A^c),$$

we have

$$\mathrm{TV}(P, Q) + \sum_i p_i \wedge q_i = 1,$$

from which the desired result follows. □

10.3 Poisson Limit Theorem

Theorem 10.9 (Le Cam)**.** *Let $X_i \sim \mathrm{Bernoulli}(p_i)$ be independent for $i = 1, \ldots, n$. Define $\lambda := \sum_{i=1}^n p_i$. Then,*

$$\mathrm{TV}\left(\sum_{i=1}^n X_i, \mathrm{Poisson}(\lambda)\right) \leq \sum_{i=1}^n p_i^2.$$

Example 10.10. If $p_i = \lambda/n$, we have

$$\mathrm{TV}\left(\mathrm{Binomial}\left(n, \frac{\lambda}{n}\right), \mathrm{Poisson}(\lambda)\right) \leq \frac{\lambda^2}{n} \longrightarrow 0,$$

which gives the Poisson limit theorem. Le Cam's thus not only gives an explicit rate of convergence, but also gives convergence in total variation, which is stronger than convergence in distribution. However, it is hard to relax the independence assumption with Le Cam's method. For that, we present a different proof using Stein's method.

Proof. Let $Z_i \sim \text{Poisson}(p_i)$ be independent for $i = 1, \ldots, n$ so that $\sum Z_i \sim \text{Poisson}(\lambda)$. Let $(\hat{X}_i, \hat{Z}_i)$ be a maximal coupling of (X_i, Z_i). We have

$$\mathbb{P}(\hat{X}_i = \hat{Z}_i) = \sum_k \mathbb{P}(X_i = k) \wedge \mathbb{P}(Z_i = k).$$

Recall that

$$\mathbb{P}(Z_i = k) = \frac{p_i^k}{k!} e^{-p_i}, \quad \mathbb{P}(X_i = k) = \begin{cases} p_i, & k = 1 \\ 1 - p_i, & k = 0 \end{cases}.$$

Thus,

$$\begin{aligned} \mathbb{P}(\hat{X}_i = \hat{Z}_i) &= \min(1 - p_i, e^{-p_i}) + \min(p_i, p_i e^{-p_i}) \\ &= 1 - p_i + p_i e^{-p_i} \geq 1 - p_i + p_i(1 - p_i) = 1 - p_i^2. \end{aligned}$$

Now note that $(\sum \hat{X}_i, \sum \hat{Z}_i)$ is a coupling of $(\sum X_i, \sum Z_i)$ (the independence assumption is used here), so

$$\text{TV}\left(\sum X_i, \text{Poisson}(\lambda)\right) \leq \mathbb{P}\left(\sum X_i \neq \sum \hat{Z}_i\right) \leq \sum_{i=1}^n \mathbb{P}(\hat{X}_i \neq \hat{Z}_i) \leq \sum_i p_i^2.$$

□

10.4 Stein's Method

Lemma 10.11 (Stein-Chen)**.** *$Z \sim \text{Poisson}(\lambda)$ if and only if $\mathbb{E}Zf(Z) = \lambda \mathbb{E}f(Z+1)$ for all f such that both expectations are finite.*

Proof. $\Rightarrow$: Recall

$$\mathbb{P}(Z = k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, \ldots.$$

Thus,

$$\begin{aligned} \mathbb{E}Zf(Z) &= \sum_{k \geq 0} k f(k) \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k \geq 1} k f(k) \frac{\lambda^k}{k!} e^{-\lambda} \\ &= \lambda \sum_{k \geq 1} f(k) \frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda} = \lambda \sum_{k \geq 0} f(k+1) \frac{\lambda^k}{k!} e^{-\lambda} = \lambda \mathbb{E}f(Z+1). \end{aligned}$$

$\Leftarrow$: First let $Z \sim \text{Poisson}(\lambda)$. Let A be any subset of $\mathbb{N}$. The **Stein-Chen equation** is

$$\lambda f(k+1) - k f(k) = \mathbb{1}\{k \in A\} - \mathbb{P}(Z \in A), \quad k = 0, 1, \ldots.$$

Multiplying both sides by $\lambda^k / k!$, we get

$$\frac{\lambda^{k+1}}{k!} f(k+1) - \frac{\lambda^k}{(k-1)!} f(k) = \frac{\lambda^k}{k!} \big(\mathbb{1}\{k \in A\} - \mathbb{P}(Z \in A)\big), \quad k = 1, 2, \ldots$$

Writing $g(k+1) := \lambda^{k+1} f(k+1)/k!$, we have

$$g(k+1) - g(k) = \frac{\lambda^k}{k!}\big(\mathbb{1}\{k \in A\} - \mathbb{P}(Z \in A)\big), \quad k = 0, 1, 2, \ldots.$$

For case $k = 0$, we have the initial condition

$$\lambda f(1) = g(1) = \mathbb{1}\{0 \in A\} - \mathbb{P}(Z \in A).$$

Thus,

$$g(k+1) = \sum_{j=0}^{k} \frac{\lambda^j}{j!} \left(\mathbb{1}\{\, j \in A \,\} - \mathbb{P}(Z \in A)\right)$$

and so

$$f(k+1) = f_A(k+1) = \frac{k!}{\lambda^{k+1}} \sum_{j=0}^{k-1} \frac{\lambda^j}{j!} \left(\mathbb{1}\{\, j \in A \,\} - \mathbb{P}(Z \in A)\right).$$

We have the same question as before: can we not replace Z by any other random variable W in the equation and get the same conclusion? The answer is the same as before: the Poisson density is the only one that satisfies the regularity conditions.

Now, let W denote any random variable and suppose the Stein-Chen identity $\mathbb{E}Wf(W) = \lambda \mathbb{E} f(W+1)$ holds for all f such that both expectations are finite. Then by taking expectations on both sides, we have

$$0 = \mathbb{E}\left(\lambda f_A(W+1) - W f_A(W)\right) = \mathbb{P}(W \in A) - \mathbb{P}(Z \in A).$$

Since A is arbitrary, we have $W \sim \text{Poisson}(\lambda)$. □

Lemma 10.12. *For each A, the solution $f_A(k+1) = \frac{k!}{\lambda^{k+1}} \sum_{j=0}^{k} \frac{\lambda^j}{j!} \left(\mathbb{1}\{\, j \in A \,\} - \mathbb{P}(Z \in A)\right)$ to the Stein-Chen equation is Lipschitz:*

$$|f_A(k) - f_A(\ell)| \le \min\left\{1, \frac{1}{\lambda}\right\} |k - \ell|, \quad k, \ell = 0, 1, \ldots.$$

Proof. Note that $f_A(k+1) = \sum_{i \in A} f_i(k+1)$ where

$$\begin{aligned}
f_i(k+1) &= \frac{k!}{\lambda^{k+1}} \sum_{j=0}^{k} \frac{\lambda^j}{j!} \left(\mathbb{1}\{\, j = i \,\} - \mathbb{P}(Z = i)\right) \\
&= \frac{k!}{\lambda^{k+1} e^{-\lambda}} \left(\sum_{j=0}^{k} \frac{\lambda^j}{j!} e^{-\lambda} \mathbb{1}\{\, j = i \,\} - \sum_{j=0}^{k} \frac{\lambda^j}{j!} e^{-\lambda} \mathbb{P}(Z = i) \right) \\
&= \frac{\mathbb{P}(Z = i)}{\lambda \mathbb{P}(Z = k)} \left(\sum_{j=0}^{k} \mathbb{1}\{\, j = i \,\} - \sum_{j=0}^{k} \frac{\lambda^j}{j!} e^{-\lambda} \right) \\
&= \frac{\mathbb{P}(Z = i)}{\lambda \mathbb{P}(Z = k)} \left(\mathbb{1}\{\, i \le k \,\} - \mathbb{P}(Z \le k)\right).
\end{aligned}$$

We claim that

(i) $f_k(k+1) \geq f_k(k)$

(ii) For $i \neq k$, $f_i(k+1) \leq f_i(k)$.

Using these two claims in order, we have

$$\begin{aligned} f_A(k+1) - f_A(k) &\leq f_{A\cup\{k\}}(k+1) - f_{A\cup\{k\}}(k) \\ &\leq f_k(k+1) - f_k(k) \\ &= \frac{1-\mathbb{P}(Z \leq k)}{\lambda} + \frac{\mathbb{P}(Z=k)\mathbb{P}(Z \leq k-1)}{\lambda \mathbb{P}(Z=k-1)} \\ &= \frac{1-\mathbb{P}(Z \leq k)}{\lambda} + \frac{\mathbb{P}(Z \leq k-1)}{\lambda}\frac{\lambda}{k}. \end{aligned}$$

The second term can be rewritten as

$$\begin{aligned} \frac{\mathbb{P}(Z \leq k-1)}{k} &= \frac{1}{k}\sum_{j=0}^{k-1}\frac{j+1}{\lambda}e^{-\lambda}\frac{\lambda^{j+1}}{(j+1)!} = \frac{1}{\lambda}\sum_{j=0}^{k-1}\frac{j+1}{k}e^{-\lambda}\frac{\lambda^{j+1}}{(j+1)!} \\ &\leq \frac{1}{\lambda}\sum_{j=0}^{k-1}e^{-\lambda}\frac{\lambda^{j+1}}{(j+1)!} = \frac{1}{\lambda}\sum_{j=1}^{k}\frac{\lambda^j}{j!}e^{-\lambda} = \frac{\mathbb{P}(1 \leq Z \leq k)}{\lambda}. \end{aligned}$$

Thus,

$$f_A(k+1) - f_A(k) \leq \frac{1-\mathbb{P}(Z=0)}{\lambda} = \frac{1-e^{-\lambda}}{\lambda} \leq \min\left\{1, \frac{1}{\lambda}\right\}.$$

This is one side of the Lipschitz condition. For the other side, note only that $f_A(k) + f_{A^c}(k) = 0$ implies $f_A(k) - f_A(k+1) = f_{A^c}(k+1) - f_{A^c}(k)$, so the previous result gives

$$f_A(k) - f_A(k+1) \leq \min\left\{1, \frac{1}{\lambda}\right\}.$$

Using triangle inequality, we have

$$|f_A(k) - f_A(\ell)| \leq \min\left\{1, \frac{1}{\lambda}\right\}|k-\ell|.$$

It remains to verify the two claims.

(i) $f_k(k+1) \geq f_k(k)$:

$$f_k(k+1) - f_k(k) = \frac{1-\mathbb{P}(Z \leq k)}{\lambda} + \frac{\mathbb{P}(Z=k)\mathbb{P}(Z \leq k-1)}{\lambda\mathbb{P}(Z=k-1)} > 0.$$

(ii) $f_i(k+1) \leq f_i(k)$ for $i \neq k$: When $i < k$,

$$f_i(k+1) = \frac{\mathbb{P}(Z=i)\mathbb{P}(Z>k)}{\lambda\mathbb{P}(Z=k)}, \quad f_i(k) = \frac{\mathbb{P}(Z=i)\mathbb{P}(Z>k-1)}{\lambda\mathbb{P}(Z=k-1)}.$$

To check their relative size, consider the function

$$k \mapsto \frac{\mathbb{P}(Z > k)}{\mathbb{P}(Z = k)} = \sum_{j>k} \frac{\lambda^j}{j!} \frac{k!}{\lambda^k} = \sum_{j>k} \lambda^{j-k} \frac{k!}{j!} = \sum_{j>0} \lambda^j \frac{k!}{(k+j)!}$$

is decreasing in k.

Now consider case $i > k$. We have

$$f_i(k+1) = -\frac{\mathbb{P}(Z = i)\mathbb{P}(Z \le k)}{\lambda \mathbb{P}(Z = k)}, \quad f_i(k) = -\frac{\mathbb{P}(Z = i)\mathbb{P}(Z \le k-1)}{\lambda \mathbb{P}(Z = k-1)}.$$

Similarly, note that the function

$$k \mapsto \frac{\mathbb{P}(Z \le k)}{\mathbb{P}(Z = k)} = \sum_{j \le k} \frac{\lambda^j}{j!} \frac{k!}{\lambda^k} = \sum_{j \le k} \lambda^{j-k} \frac{k!}{j!} = \sum_{\ell \le k} \lambda^{-\ell} \frac{k!}{(k-\ell)!}$$

is increasing in k (both the range and the summand increase).

Thus in either case we've shown $f_i(k+1) \le f_i(k)$ for $i \ne k$.

□

With the lemmas above, we conclude the following:

Proposition 10.13. *If $Z \sim \text{Poisson}(\lambda)$, then*

$$\begin{aligned}
\text{TV}(W, Z) &= \sup_A |\mathbb{P}(W \in A) - \mathbb{P}(Z \in A)| \\
&= \sup_A |\mathbb{E}\left(W f_A(W) - \lambda f_A(W+1)\right)| \\
&\le \sup_{\text{Lip}(f) \le \min\{1, 1/\lambda\}} |\mathbb{E}\left(W f(W) - \lambda f(W+1)\right)|.
\end{aligned}$$

Theorem 10.14 (Stein-Chen)**.** *Let $X_i \sim \text{Bernoulli}(p_i)$ be independent for $i = 1, \dots, n$ and $\lambda := \sum_{i=1}^n p_i$. Then,*

$$\text{TV}\left(\sum_{i=1}^n X_i, \text{Poisson}(\lambda)\right) \le \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^n p_i^2,$$

Proof. Write $W := X_1 + \cdots + X_n$ and $W = X_i + W_i$. Let $\lambda := \sum_{i=1}^n p_i$. We have

$$\begin{aligned}
\lambda \mathbb{E} f(W+1) &= \sum_{i=1}^n p_i \mathbb{E} f(W+1), \\
\mathbb{E} W f(W) &= \sum_{i=1}^n \mathbb{E} X_i f(W) = \sum_{i=1}^n \mathbb{E} X_i f(W_i + X_i).
\end{aligned}$$

Each term $\mathbb{E} X_i f(W_i + X_i)$ can be rewritten as

$$\mathbb{E}\big(X_i f(W_i + X_i)\big) = \mathbb{E}\mathbb{E}\big(X_i f(W_i + X_i) | W_i\big) = \mathbb{E} p_i f(W_i + 1).$$

(Alternatively, since X_i is binary, we have $X_i f(W_i + X_i) \equiv X_i f(W_i + 1)$. Take expectations to get $\mathbb{E}(X_i f(W_i + X_i)) = \mathbb{E}X_i f(W_i + X_i) = p_i \mathbb{E} f(W_i + 1)$.) Using this, we have

$$\begin{aligned}|\lambda \mathbb{E} f(W+1) - \mathbb{E} W f(W)| &\leq \sum_{i=1}^n p_i \mathbb{E}|f(W+1) - f(W_i+1)| \\ &\leq \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^n p_i \mathbb{E} X_i = \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^n p_i^2.\end{aligned}$$

□

We next generalize this result to accommodate potential dependency.

Definition 10.15 (Dependency Neighborhood)**.** For each i, let $i \in N_i \subseteq \{1, \ldots, n\}$ be such that X_i is independent of $\{X_j : j \notin N_i\}$. Then, we call N_i a **dependency neighborhood** of X_i.

Example 10.16. In the independent case, we can take $N_i = \{i\}$.

We have the following generalization of the Stein-Chen theorem:

Theorem 10.17. *Let $X_i \sim \text{Bernoulli}(p_i)$ be random variables with dependency neighborhoods N_i for $i = 1, \ldots, n$. Let $\lambda := \sum_{i=1}^n p_i$. Then,*

$$\text{TV}\left(\sum_{i=1}^n X_i, \text{Poisson}(\lambda)\right) \leq \min\left\{1, \frac{1}{\lambda}\right\}\left(\sum_{i=1}^n \sum_{j \in N_i \setminus \{i\}} \mathbb{E} X_i X_j + \sum_{i=1}^n \sum_{j \in N_i} p_i p_j\right).$$

Proof. Write

$$\begin{aligned}W &:= X_i + \sum_{j \in N_i \setminus \{i\}} X_j + \sum_{j \notin N_i} X_j \\ &=: X_i + \sum_{j \in N_i \setminus \{i\}} X_j + V_i,\end{aligned}$$

so that $X_i \perp\!\!\!\perp V_i$. Now,

$$\begin{aligned}\mathbb{E} W f(W) - \lambda \mathbb{E} f(W+1) &= \sum_{i=1}^n (\mathbb{E} X_i f(W) - p_i \mathbb{E} f(W+1)) \\ &= \sum_{i=1}^n (\mathbb{E} X_i f(W) - \mathbb{E} X_i f(V_i+1)) + \sum_{i=1}^n (\mathbb{E} X_i f(V_i+1) - p_i \mathbb{E} f(W+1)) \\ &= \sum_{i=1}^n (\mathbb{E} X_i f(W) - \mathbb{E} X_i f(V_i+X_i)) + \sum_{i=1}^n (\mathbb{E} p_i f(V_i+1) - p_i \mathbb{E} f(W+1)) \\ &=: \text{I} + \text{II},\end{aligned}$$

where the second to last line follows from noting that $X_i f_i(V_i + 1) \equiv X_i f(V_i + X_i)$ and $\mathbb{E} X_i f(V_i + 1) = p_i \mathbb{E} f(V_i + 1)$. We bound the two terms separately:

$$\text{I} \leq \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^n \mathbb{E} X_i \sum_{j \in N_i \setminus \{i\}} X_j = \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^n \sum_{j \in N_i \setminus \{i\}} \mathbb{E} X_i X_j.$$

$$\mathrm{II} \le \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} p_i \mathbb{E} \sum_{j \in N_i} X_j = \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} \sum_{j \in N_i} p_i p_j.$$

□

Example 10.18. An **Erdös-Rényi graph** $G(n, p)$ is a random graph on n vertices, summarized by an adjacency matrix $A_{ij} \in \{0, 1\}$, $1 \le i < j \le n$, where $A_{ij} \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$. The graph is called **sparse** if $p \sim 1/n$ or $p \sim \log n/n$.

We are interested in the number of triangles in $G(n, p)$:

$$T := \sum_{1 \le i < j < k \le n} A_{ij} A_{jk} A_{ki}.$$

Write $X_\alpha := A_{ij} A_{jk} A_{ki} \sim \text{Bernoulli}(p_\alpha)$, where $\alpha = (i, j, k)$, and write $\lambda := \sum_\alpha p_\alpha$. We have $|N_\alpha| \le 3(n-3) + 1$, where $3(n-3)$ counts the number of triangles that share an edge with α and 1 counts α itself (note that sharing two edges implies sharing all three). Then,

$$\underbrace{\sum_{\alpha}}_{O(n^3)} \underbrace{\sum_{\beta \in N_\alpha}}_{O(n)} \underbrace{p_\alpha p_\beta}_{O(p^6)} = O(n^4 p^6)$$

$$\underbrace{\sum_{\alpha}}_{O(n^3)} \underbrace{\sum_{\beta \in N_\alpha \setminus \{\alpha\}}}_{O(n)} \underbrace{\mathbb{E} X_\alpha X_\beta}_{O(p^5)} = O(n^4 p^5)$$

$$\lambda = \binom{n}{3} p^3 = O(n^3 p^3),$$

for the second line, p^5 comes from noting that α and β shares an edge, so there are five edges in total. Thus the previous theorem gives

$$\mathrm{TV}\left(T, \text{Poisson}\left(\binom{n}{3} p^3\right)\right) = O\left(\min\left(n^4 p^5, n p^2\right)\right)$$

which converges to 0 when $p \ll 1/\sqrt{n}$.

11 Conditional Expectation

Roadmap:

Signed measure $\to$ Radon-Nikodym derivative $\to$ Conditional expectation.

11.1 Signed Measure

Definition 11.1. Let $(\Omega, \mathcal{F})$ be a measurable space. We say φ is a **signed measure** if $\varphi(\varnothing) = 0$ and $\varphi(\cup_n A_n) = \sum_n \varphi(A_n)$ whenever A_n are disjoint.

Proposition 11.2. *For any signed measure φ, we have either $-\infty < \varphi(A) \leq \infty$ for each $A \in \mathcal{F}$, or $-\infty \leq \varphi(A) < \infty$ for each $A \in \mathcal{F}$.*

Proof. If not, there exists $A, B \in \mathcal{F}$ with $\varphi(A) = \infty$ and $\varphi(B) = -\infty$. We have

$$\varphi(A \cup B) = \varphi(A) + \varphi(B \setminus A) = \varphi(B) + \varphi(A \setminus B).$$

The first decomposition forces $\varphi(A \cup B) = \infty$ while the second decomposition forces $\varphi(A \cup B) = -\infty$, a contradiction. □

Theorem 11.3 (Hahn Decomposition)**.** *For any signed measure φ, there exists a partition $\Omega = \Omega^+ \cup \Omega^-$ such that*

$$\varphi\left(A \cap \Omega^+\right) \geq 0 \geq \varphi\left(A \cap \Omega^-\right), \quad A \in \mathcal{F}.$$

This decomposition is unique in the following sense: if $\Omega = \Omega_1^+ \cup \Omega_1^-$ is another such decomposition, then any subset of $\Omega^+ \triangle \Omega_1^+$ or $\Omega^- \triangle \Omega_1^-$ is a null set of φ.

Now for a function $f : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$, we have

$$\varphi(A) = \int_A f \, \mathrm{d}\mu$$

is a signed measure with Hahn decomposition $\Omega^+ = \{f > 0\}$ and $\Omega^- = \{f < 0\}$. The Hahn-decomposition thus gives measures

$$\varphi^+ = \varphi(A \cap \Omega^+), \quad \varphi^- = -\varphi(A \cap \Omega^-).$$

This is called the **Jordan decomposition** of φ. The Jordan decomposition in unique in the sense that each Hahn decomposition gives the same φ^+ and φ^-. Also Jordan decomposition is the only possible singular decomposition of φ into two nonnegative measures (singular means that they are supported on disjoint sets).

Definition 11.4. Let φ and μ be a signed measure and a measure respectively on $(\Omega, \mathcal{F})$. We say **φ is absolutely continuous with respect to μ** and write $\varphi \ll \mu$ if for each $A \in \mathcal{F}$ such that $\mu(A) = 0$, we have $\varphi(A) = 0$.

11.2 Radon-Nikodym Derivative

Theorem 11.5 (Radon-Nikodym)**.** *Suppose φ and μ are, respectively, a signed measure and a σ-finite measure on $(\Omega, \mathcal{F})$ and $\varphi \ll \mu$. Then there exists a μ-a.e. unique function which we denote $\mathrm{d}\varphi/\mathrm{d}\mu$ such that*

$$\varphi(A) = \int_A \frac{\mathrm{d}\varphi}{\mathrm{d}\mu}\,\mathrm{d}\mu, \quad \forall A \in \mathcal{F}.$$

Example 11.6. Consider $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. Let $\mathbb{P} = \mathcal{N}(0,1)$ and λ the Lebesgue measure. Then $\mathrm{d}\mathbb{P}/\mathrm{d}\lambda$ is the density of the standard normal distribution. ☍

Example 11.7. Consider $(\mathbb{N}, 2^{\mathbb{N}})$. If $\mathbb{P}$ is the Poisson distribution and λ is the counting measure, then $\mathrm{d}\mathbb{P}/\mathrm{d}\lambda$ is the probability mass function of the Poisson distribution. ☍

11.3 Conditional Expectation

Example 11.8. Intuitively, if $X = \mathbb{1}_A$ and $Y = \mathbb{1}_B$,

$$\mathbb{E}[X|Y] = \mathbb{P}(A|\mathbb{1}_B) = \mathbb{P}(A|B)\mathbb{1}_B + \mathbb{P}(A|B^{\mathsf{c}})\mathbb{1}_{B^{\mathsf{c}}}.$$

☍

Example 11.9. Let $Z = a\mathbb{1}_B + b\mathbb{1}_{B^{\mathsf{c}}}$ with $a \neq b$.

$$\mathbb{E}[X|Z] = \mathbb{P}(A|Z)$$

☍

From the examples above, we see that $\mathbb{E}[X|Y]$ depends not on Y but only the σ-algebra generated by Y.

Example 11.10. For each $D \in \mathcal{G} := \{\varnothing, \Omega, B, B^{\mathsf{c}}\}$, we have

$$\begin{aligned}
\mathbb{E}\left[\mathbb{E}[X|Y]\mathbb{1}_D\right] &= \mathbb{P}(A|B)\mathbb{P}(B \cap D) + \mathbb{P}(A|B^{\mathsf{c}})\mathbb{P}(B^{\mathsf{c}} \cap D) \\
&= \begin{cases} 0, & D = \varnothing \\ \mathbb{P}(A), & D = \Omega \\ \mathbb{P}(A \cap B), & D = B \\ \mathbb{P}(A \cap B^{\mathsf{c}}), & D = B^{\mathsf{c}} \end{cases} \\
&= \mathbb{E}[X\mathbb{1}_D].
\end{aligned}$$

This is the tower property:

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}]\mathbb{1}_D] = \mathbb{E}\left[\mathbb{E}[X\mathbb{1}_D|\mathcal{G}]\right] = \mathbb{E}[X\mathbb{1}_D].$$

☍

Remark 11.11 (A Wishlist for the Conditional Expectation). $\mathbb{E}[X|Y]$ should be:

- $\mathbb{E}[X|\mathcal{G}]$, where $\mathcal{G} := \sigma(Y)$.

- $\mathbb{E}[X|Y]$ is measurable with respect to $\mathcal{G}$.
- $\mathbb{E}[X|Y]$ satisfies the **tower property**: $\mathbb{E}[\mathbb{E}[X|Y]\mathbb{1}_D] = \mathbb{E}[X\mathbb{1}_D]$ for each $D \in \mathcal{G}$.

It turns out that these conditions uniquely characterize $\mathbb{E}[X|Y]$:

Definition 11.12. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\mathcal{G} \subset \mathcal{F}$ a sub-σ-field, and $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ an integrable random variable. Then, $\mathbb{E}[X|\mathcal{G}]$ is defined as the unique $\mathcal{G}$-measurable function such that

$$\int_A \mathbb{E}[X|\mathcal{G}] \, \mathrm{d}\mathbb{P} = \int_A X \, \mathrm{d}\mathbb{P}, \quad \forall A \in \mathcal{G}.$$

Why do we have existence and uniqueness? Define $\varphi(A) := \int_A X \, \mathrm{d}\mathbb{P}$, $A \in \mathcal{G}$. Note that φ and $\mathbb{P}$ are respectively a signed measure and a measure on $(\Omega, \mathcal{G})$ such that $\varphi \ll \mathbb{P}$. Thus by the Radon-Nikodym theorem, there exists a unique $\mathcal{G}$-measurable function $\mathrm{d}\varphi/\mathrm{d}\mathbb{P}$ such that

$$\int_A \frac{\mathrm{d}\varphi}{\mathrm{d}\mathbb{P}} \, \mathrm{d}\mathbb{P} = \int_A X \, \mathrm{d}\mathbb{P}, \quad \forall A \in \mathcal{G}.$$

This function is exactly $\mathbb{E}[X|\mathcal{G}]$. Thus the uniqueness is only with respect to the sub-σ-field $\mathcal{G}$ and the measure $\mathbb{P}$.

Example 11.13. $\mathcal{G} = \{\Omega, \varnothing\}$. Then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$.

Example 11.14. $\mathcal{G} = \{\varnothing, \Omega, B, B^{\mathsf{c}}\}$; $X = \mathbb{1}_A$. Since the conditional expectation is $\mathcal{G}$-measurable, it must be of the form

$$\mathbb{E}[X|\mathcal{G}] = a\mathbb{1}_B + b\mathbb{1}_{B^{\mathsf{c}}}.$$

It must also satisfy the tower property:

$$\begin{aligned} a\mathbb{P}(B) &= \mathbb{P}(A \cap B), & D &= B \\ b\mathbb{P}(B^{\mathsf{c}}) &= \mathbb{P}(A \cap B^{\mathsf{c}}), & D &= B^{\mathsf{c}} \\ a\mathbb{P}(B) + b\mathbb{P}(B^{\mathsf{c}}) &= \mathbb{P}(A), & D &= \Omega. \end{aligned}$$

Thus,

$$a = \begin{cases} \frac{\mathbb{P}(A\cap B)}{\mathbb{P}(B)} = \mathbb{P}(A|B), & \mathbb{P}(B) > 0 \\ \text{arbitrary}, & \mathbb{P}(B) = 0 \end{cases}, \quad b = \begin{cases} \frac{\mathbb{P}(A\cap B^{\mathsf{c}})}{\mathbb{P}(B^{\mathsf{c}})} = \mathbb{P}(A|B^{\mathsf{c}}), & \mathbb{P}(B^{\mathsf{c}}) > 0 \\ \text{arbitrary}, & \mathbb{P}(B^{\mathsf{c}}) = 0 \end{cases}.$$

This shows also the non-uniqueness of the conditional expectation on a $\mathcal{G}$-null set.

Example 11.15. $\mathcal{G} = \mathcal{F}$. $\mathbb{E}[X|\mathcal{G}] = X$.

Lemma 11.16 (Doob-Dynkin)**.** *Consider $Y : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ and $Z : (\Omega, \sigma(Y)) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ (so $\sigma(Z) \subset \sigma(Y)$). Then there exists $f : (\mathbb{R}, \mathcal{B}_{\mathbb{R}}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ such that $Z = f(Y)$.*

Proof. We crank the standard machine. Let's consider the case where $Z = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ is simple. Since $\sigma(Z) \subset \sigma(Y)$, we must have $A_i \in \sigma(Y) = \{Y^{-1}B : B \in \mathcal{B}_{\mathbb{R}}\}$. There then exists $B_i \in \mathcal{B}_{\mathbb{R}}$ such that $A_i = Y^{-1}B_i$. Then,

$$Z(\omega) = \sum_{i=1}^n a_i \mathbb{1}_{A_i}(\omega) = \sum_{i=1}^n a_i \mathbb{1}_{B_i}(Y(\omega)) = f(Y(\omega)),$$

with $f = \sum_{i=1}^n a_i \mathbb{1}_{B_i}$. □

Corollary 11.17. *The conditional expectation $\mathbb{E}[X|Y]$ is a function of Y.*

From the defining equation of the conditional expectation, we have

$$\mathbb{E}\left((X - \mathbb{E}[X|\mathcal{G}])\mathbb{1}_A\right) = 0, \quad \forall A \in \mathcal{G}.$$

By the standard machine, we have then that

$$\mathbb{E}\left((X - \mathbb{E}[X|\mathcal{G}])Y\right) = 0, \quad \forall Y \text{ s.t. } \sigma(Y) \subset \mathcal{G}.$$

This is an orthogonality result. The conditional expectation $\mathbb{E}[X|\mathcal{G}]$ is thus a projection of X onto the space of $\mathcal{G}$-measurable random variables.

Note also that $\mathbb{E}[X|\mathcal{G}]$ solves

$$\min_{Y:\sigma(Y)\subset\mathcal{G}} \mathbb{E}[(X - Y)^2].$$

Given Y, any random variable Z that is $\sigma(Y)$-measurable is a function of Y by the Doob-Dynkin lemma. Thus $\mathbb{E}[X|Y]$ satisfies

$$\mathbb{E}\left[(X - \mathbb{E}[X|Y])g(Y)\right] = 0, \quad g.$$

In this case, $\mathbb{E}[X|Y]$ solves

$$\min_g \mathbb{E}\left[(X - g(Y))^2\right].$$

Our orthogonality definition, however, is more general than defining conditional expectation as a projection, since the latter requires the existence of a second moment, while the former only requires integrability.

Proposition 11.18 (Properties of Conditional Expectation)**.**

(i) If $\sigma(X) \subset \mathcal{G}$, the $\mathbb{E}[X|\mathcal{G}] = X$.

(ii) If $\sigma(X) \perp\!\!\!\perp \mathcal{G}$, then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}X$.

(iii) If $\mathcal{G}_0 \subset \mathcal{G}$, then

$$\mathbb{E}\left[\mathbb{E}(X|\mathcal{G})|\mathcal{G}_0\right] = \mathbb{E}\left[\mathbb{E}(X|\mathcal{G}_0)|\mathcal{G}\right] = \mathbb{E}[X|\mathcal{G}_0].$$

(iv) $X \leq Y$ implies $\mathbb{E}[X|\mathcal{G}] \leq \mathbb{E}[Y|\mathcal{G}]$.

(v) $\mathbb{E}[aX + bY|\mathcal{G}] = a\mathbb{E}[X|\mathcal{G}] + b\mathbb{E}[Y|\mathcal{G}]$.

(vi) *If* $0 \leq X_n \uparrow X$, *then* $\mathbb{E}[X_n|\mathcal{G}] \uparrow \mathbb{E}[X|\mathcal{G}]$.

(vii) *If* $X_n \geq 0$, *then*

$$\mathbb{E}[\liminf_n X_n|\mathcal{G}] \leq \liminf_n \mathbb{E}[X_n|\mathcal{G}].$$

(viii) *If* $X_n \to X$, $|X_n| \leq Y$, $\mathbb{E}Y < \infty$, *then*

$$\mathbb{E}[X_n|\mathcal{G}] \to \mathbb{E}[X|\mathcal{G}].$$

(ix) *If* $\sigma(Y) \subset \mathcal{G}$, *then*

$$\mathbb{E}[XY|\mathcal{G}] = Y\mathbb{E}[X|\mathcal{G}].$$

Of course, all of these statements are up to $\mathcal{G}$*-null sets.*

Proof. (i) is easy. For (ii), crank the standard machine. For (iii), use the projection interpretation. The rest are left as exercises. □

12 Appendix A: A List of Theorems

Lemma 12.1. *The intersection of any collection of σ-fields is a σ-field.*

Proposition 12.2. *$x \in \liminf A_n$ if and only if there exists N such that for all $n \geq N$, $x \in A_n$. That is, x belongs to all but finitely many of the sets A_n.*

Similarly, $x \in \limsup A_n$ if and only if for each k, $x \in A_n$ for some $n \geq k$. That is, s belongs to infinitely many A_n's.

Corollary 12.3. $\liminf A_n \subset \limsup A_n$.

Theorem 12.4 (Monotone Class Theorem)**.** *If $\mathcal{A}$ is a field, then $m(\mathcal{A}) = \sigma(\mathcal{A})$.*

Theorem 12.5 (Dynkin's π-λ Theorem)**.** *If $\mathcal{P}$ is a π-system, then $\lambda(\mathcal{P}) = \sigma(\mathcal{P})$.*

Proposition 12.6. *Let μ be a measure on a field $\mathcal{F}$. We have the following:*

(i) (Monotonicity): If $A \subset B$, then $\mu(A) \leq \mu(B)$.

(ii) If $A_n \uparrow$, then $\mu(\lim A_n) = \lim \mu(A_n)$.

(iii) If $A_n \downarrow$ and $\mu(A_1) < \infty$, then $\mu(\lim A_n) = \lim \mu(A_n)$.

Theorem 12.7 (Uniqueness of Measure Extension)**.** *If two finite measures μ and ν agree on a field $\mathcal{A}$, then they agree on $\sigma(\mathcal{A})$.*

Theorem 12.8 (Construction of Outer Measures)**.** *Fix $\varnothing \in \mathcal{E} \subset 2^{\Omega}$ and a function $\mu : \mathcal{E} \to [0, \infty]$ such that $\mu(\varnothing) = 0$. For each $A \subset \Omega$, define*

$$\tau(A) := \inf \left\{ \sum \mu(A_n) : A_n \in \mathcal{E}, A \subset \bigcup_n A_n \right\},$$

with the convention that the infimum over an empty set is ∞. Then τ is an outer measure on Ω.

Theorem 12.9 (Carathéodory Extension)**.** *Let τ be an outer measure on Ω. Define*

$$\mathcal{F}_\tau := \left\{ A \subset \Omega : \text{for each } B \subset \Omega, \tau(B) = \tau(B \cap A) + \tau(B \cap A^c) \right\}.$$

Then $\mathcal{F}_\tau$ is a σ-field and $(\Omega, \mathcal{F}_\tau, \tau)$ is a measure space.

Theorem 12.10. *Suppose μ is a finite measure on a field $\mathcal{A}$. Then, there exists a unique extension of μ from $\mathcal{A}$ to $\sigma(\mathcal{A})$.*

Theorem 12.11. *For an arbitrary outer measure τ, the measure space $(\Omega, \mathcal{F}_\tau, \tau)$ constructed from Carathéodory's extension theorem is complete.*

Theorem 12.12 (Completion)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Define $\tilde{\mathcal{F}} := \{A \cup N : A \in \mathcal{F}, \exists B \in \mathcal{F} \text{ st } \mu(B) = 0, N \subset B\}$. For every $(A \cup N) \in \tilde{\mathcal{F}}$, define $\tilde{\mu}(A \cup N) := \mu(A)$. Then $(\Omega, \tilde{\mathcal{F}}, \tilde{\mu})$ is a well-defined complete measure space. Moreover, it extends $(\Omega, \mathcal{F}, \mu)$ in the sense that $\tilde{\mu}$ agrees with μ on $\mathcal{F}$.*

Lemma 12.13. *Suppose μ is a finite measure on a field $\mathcal{A}$ and $(\Omega, \mathcal{F}_\tau, \tau)$ is its Carathéodory extension. Then, for each $A \in \mathcal{F}_\tau$, there exists $B \in \sigma(A)$ such that $A \subset B$ and $\tau(A) = \tau(B)$.*

Theorem 12.14. *Suppose μ is a finite measure on a field $\mathcal{A}$ and $(\Omega, \mathcal{F}_\tau, \tau)$ is its Carathéodory extension. Then, $(\Omega, \mathcal{F}_\tau, \tau)$ is the completion of $(\Omega, \sigma(A), \tau)$.*

Proposition 12.15 (Rules of Preimage)**.**

(i) $f^{-1}\varnothing = \varnothing$.

(ii) $f^{-1}(B^c) = (f^{-1}B)^c$

(iii) $A \subset B$ implies $f^{-1}A \subset f^{-1}B$.

(iv) $f^{-1}(\cup_\lambda A_\lambda) = \cup_\lambda f^{-1}A_\lambda$.

(v) $f^{-1}(\cap_\lambda A_\lambda) = \cap_\lambda f^{-1} A_\lambda$.

Lemma 12.16. *$f : (X, \mathcal{F}) \to (Y, \sigma(\mathcal{E}))$ is measurable if $f^{-1}\mathcal{E} \subset \mathcal{F}$.*

Proposition 12.17.

(i) *If f, g are measurable, so is $f + g$.*

(ii) *If f_k are measurable, then so is $\sup_k f_k$.*

(iii) *Suppose $f : (X, \mathcal{F}) \to (Y, \mathcal{G})$ and $g : (Y, \mathcal{G}) \to (Z, \mathcal{H})$ are measurable. Then, $g \circ f : (X, \mathcal{F}) \to (Z, \mathcal{H})$ is measurable.*

Lemma 12.18. *For each nonnegative measurable function $f \geq 0$, there exists nonnegative simple functions $f_n \geq 0$ such that $f_n \uparrow f$ pointwise everywhere. If additionally, f is bounded, then the convergence is uniform.*

Proposition 12.19. *For a nonnegative simple function f, the value of $\int f \, \mathrm{d}\mu$ does not depend on the representation of f as a linear combination of indicator functions.*

Lemma 12.20. *Let f_n, f, g be nonnegative simple functions. Then*

(i) $\int f \, \mathrm{d}\mu \geq 0$.

(ii) $\int (af) \, \mathrm{d}\mu = a \int f \, \mathrm{d}\mu$, $a \geq 0$.

(iii) $\int (f + g) \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu$.

(iv) *$f \geq g$ implies $\int f \, \mathrm{d}\mu \geq \int g \, \mathrm{d}\mu$.*

(v) *If $f_n \uparrow$ and $\lim f_n \geq g$, then $\lim \int f_n \, \mathrm{d}\mu \geq \int g \, \mathrm{d}\mu$.*

Lemma 12.21. *Assume $f, g \geq 0$ are nonnegative functions.*

(i) *Suppose a sequence of nonnegative simple functions $f_n \uparrow f$. Then, $\int f \, \mathrm{d}\mu = \lim_n \int f_n \, \mathrm{d}\mu$. Note that from this property, the linearity of integral follows.*

(ii)

$$\int f \, \mathrm{d}\mu = \lim_n \sum_{k=0}^{n2^n - 1} \frac{k}{2^n} \mu\left(\frac{k}{2^n} \leq f < \frac{k+1}{2^n}\right) + n\mu\left(f \geq n\right).$$

(iii) $\int (af) \, \mathrm{d}\mu = a \int f \, \mathrm{d}\mu$, $a \geq 0$.

(iv) $\int (f + g) \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu$.

(v) *$f \leq g$ implies $\int f \, \mathrm{d}\mu \leq \int g \, \mathrm{d}\mu$.*

Lemma 12.22. *Let f, g be integrable.*

(i) *If $A \in \mathcal{F}$, $\mu(A) = 0$, then $\int_A f \, \mathrm{d}\mu = 0$.*

(ii) *If $f \leq g$ a.e. ($\mu(f > g) = 0$), then $\int f \, \mathrm{d}\mu = \int g \, \mathrm{d}\mu$.*

(iii) *If $f = g$ a.e., then $\int f \, \mathrm{d}\mu = \int g \, \mathrm{d}\mu$.*

Lemma 12.23. *Let f, g be integrable.*

(i) *$\int af \, \mathrm{d}\mu = a \int f \, \mathrm{d}\mu$ for each $a \in \mathbb{R}$.*

(ii) $\int f + g \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu + \int g \, \mathrm{d}\mu$.

Theorem 12.24 (Levi, Monotone Convergence Theorem)**.** *Let f_n, f be nonnegative measurable functions such that $f_n \uparrow f$ a.e. Then,*

$$\lim_n \int f_n \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu.$$

Theorem 12.25 (Fatou's Lemma)**.** *Suppose $f_n \geq 0$ ae. Then,*

$$\int \liminf_n f_n \, \mathrm{d}\mu \leq \liminf_n \int f_n \, \mathrm{d}\mu.$$

Theorem 12.26 (Lebesgue, Dominated Convergence Theorem)**.** *Suppose $f_n \to f$ and $|f_n| \le g$ ae for some g with $\int |g| \, d\mu < \infty$. Then,*

$$\lim_n \int f_n \, d\mu = \int f \, d\mu.$$

Corollary 12.27 (Bounded Convergence)**.** *Suppose $f_n \to f$ with $|f_n| \le M < \infty$ ae for some constant M and $\mu(\Omega) < \infty$. Then,*

$$\lim_n \int f_n \, d\mu = \int f \, d\mu.$$

Proposition 12.28. *Let f, g be measurable functions. If $X \perp\!\!\!\perp Y$, then $f(X) \perp\!\!\!\perp g(Y)$.*

Proposition 12.29. *If $X \perp\!\!\!\perp Y$ with $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$, then*

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

Lemma 12.30 (Borel-Cantelli)**.** *If $\sum_n \mathbb{P}(A_n) < \infty$, then*

$$\mathbb{P}(A_n \text{ i.o.}) = \mathbb{P}\left(\limsup_n A_n\right) = 0.$$

Here, i.o. means infinitely often. Thus, with probability 1, only finitely many of the events A_n occur.

Lemma 12.31. *If $\mathbb{E}|X|^r < \infty$ then $\mathbb{E}|X|^s < \infty$ for all $0 < s < r$.*

Theorem 12.32 (Strong Law of Large Numbers)**.** *Let $X_1, X_2, \ldots$ be iid.*[6] *If $\mathbb{E}|X_1| < \infty$, then*

$$\frac{X_1 + \cdots + X_n}{n} \longrightarrow \mu := \mathbb{E}[X_1] \quad \text{a.s.}$$

Lemma 12.33 (Maxmimal Inequality)**.** *Consider independent random variables $Z_1, \ldots, Z_N$. Then,*

$$\mathbb{P}\left(\max_{i \le N} |Z_1 + \cdots + Z_i| > \varepsilon_1 + \varepsilon_2\right) \le \beta \mathbb{P}\left(|Z_1 + \cdots + Z_N| > \varepsilon_1\right),$$

where

$$\beta = \frac{1}{\min_{i \le N} \mathbb{P}(|Z_{i+1} + \cdots + Z_N| \le \varepsilon_2)}.$$

Theorem 12.34 (Kolmogorov SLLN)**.** *Let X_i be independent with $\mathbb{E}[X_i] = 0$ and $\sum_i \mathbb{E}X_i^2/i^2 < \infty$. Then $(X_1 + \cdots + X_n)/n \to 0$ a.s.*

Theorem 12.35. *Let X_i be iid (note that we need identical distribution here) with $\mathbb{E}X_i = 0$. Then,*

$$\frac{X_1 + \cdots + X_n}{n} \longrightarrow 0 \quad \text{a.s.}$$

Lemma 12.36 (Approximation of LSC in BL)**.** *Let (S, d) be a metric space. Suppose $g : S \to \mathbb{R}$ is LSC and $g \ge -M$. Then there exists $\ell_n \in \mathrm{BL}(S)$ such that $\ell_n \nearrow g$.*

Lemma 12.37. *Let (S, d) be a metric space. Suppose $P_n \xrightarrow{\mathrm{d}} P$. Then we have the following:*

(i) For any LSC and bounded from below $g : S \to \mathbb{R}$, $\liminf_n P_n g \ge Pg$.

(ii) For any USC and bounded from above $g : S \to \mathbb{R}$, $\limsup_n P_n g \le Pg$.

(iii) For each open B, $\liminf_n P_n(B) \ge P(B)$.

(iv) For each closed C, $\limsup_n P_n(C) \le P(C)$.

(v) For each B with $P(\partial B) = 0$, $\lim_n P_n(B) = P(B)$.

Proposition 12.38. *Let B be a measurable subset of the metric space S. We have $\mathring{\mathbb{1}}_B = \mathbb{1}_{\mathring{B}}$ and $\overline{\mathbb{1}}_B = \mathbb{1}_{\overline{B}}$.*

Theorem 12.39. *Suppose $|f| \le M$. We have $\mathring{f}(x) = \overline{f}(x)$ if and only if f is continuous at x.*

[6] We say X, Y are identically distributed if $\mathbb{P}(X \in A) = \mathbb{P}(Y \in A)$ for each measurable A.

Theorem 12.40. *Suppose $P_n \xrightarrow{\mathrm{d}} P$. Then*

(i) $P_n f \to P f$ *for all* f *that is bounded and continuous a.s.*

(ii) $P_n B \to P B$ *for all* $B \in \mathcal{B}(S)$ *such that* $P(\partial B) = 0$.

Theorem 12.41 (Portmanteau Theorem)**.** *Fix a metric space* (S, d). *The following are all equivalent to* $P_n \xrightarrow{\mathrm{d}} P$*:*

(i) $P_n \ell \to P \ell$ *for all* $\ell \in \mathrm{BL}(S)$.

(ii) $\liminf P_n g \geq P g$ *for all bounded below LSC* g.

(iii) $\limsup P_n g \leq P g$ *for all bounded above USC* g.

(iv) $\liminf P_n(B) \geq P(B)$ *for all open* B.

(v) $\limsup P_n(C) \leq P(C)$ *for all closed* C.

(vi) $P_n f \to P f$ *for all bounded* f *that is continuous a.s.*

(vii) $P_n B \to P B$ *for all* $B \in \mathcal{B}(S)$ *with* $P(\partial B) = 0$.

Theorem 12.42 (Fubini)**.** *Suppose* μ_1, μ_2 *are finite measures in the setting above. Then,*

(i) There exists a unique μ *on* $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$ *such that*

$$\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2), \quad A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2.$$

(ii) Suppose $f : (\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ *is measurable and* $\int |f| \, \mathrm{d}\mu < \infty$. *Then,*

$$\begin{aligned} \int f \, \mathrm{d}\mu &= \int \left(\int f(\omega_1, \omega_2) \, \mathrm{d}\mu_1(\omega_1) \right) \mathrm{d}\mu_2(\omega_2) \\ &= \int \left(\int f(\omega_1, \omega_2) \, \mathrm{d}\mu_2(\omega_2) \right) \mathrm{d}\mu_1(\omega_1). \end{aligned}$$

Theorem 12.43. *Let* P_n, P *be probability measures on* $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. *The following are equivalent:*

(i) $P_n \ell \xrightarrow{\mathrm{d}} P \ell$ *for all* $\ell \in \mathrm{BL}(\mathbb{R})$.

(ii) $P_n(-\infty, t] \to P(-\infty, t]$ *for all* t *such that* $P(\{t\}) = 0$.

(iii) $P_n f \to P f$ *for all* $f \in C^\infty(\mathbb{R})$, *where*

$$C^\infty(\mathbb{R}) := \left\{ f : \mathbb{R} \to \mathbb{R}, \|f\|_\infty < \infty, \|f^{(k)}\|_\infty, \forall k \right\}.$$

Proposition 12.44. *Let* X_i *be iid with mean* 0 *and variance* 1. *Then,*

$$\frac{X_1 + \cdots + X_n}{\sqrt{n}} \xrightarrow{\mathrm{d}} \mathcal{N}(0, 1).$$

Proposition 12.45. *Let* ξ_i, η_i *be independent random variables with* $\mathbb{E}\xi_i = \mathbb{E}\eta_i$, $\mathbb{E}\xi_i^2 = \mathbb{E}\eta_i^2$, *and* $\mathbb{E}|\xi_i|^3 < \infty$, *and such that* η_i*'s are Gaussian. Then, for each* $f \in C^\infty$,

$$\left| \mathbb{E} f\left(\sum_{i=1}^n \xi_i \right) - \mathbb{E} f\left(\sum_{i=1}^n \eta_i \right) \right| \leq C \sum_{i=1}^n \mathbb{E}|\xi_i|^3.$$

Theorem 12.46 (Triangular Array CLT)**.** *Consider a triangular array such that for each row* n,

(i) $\sum_{i=1}^{k(n)} \mathbb{E}\xi_{n,i} \to \mu$,

(ii) $\sum_{i=1}^{k(n)} \mathbb{V}(\xi_{n,i}) \to \sigma^2$,

(iii) $\sum_{i=1}^{k(n)} \mathbb{E}|\xi_{n,i}|^3 \to 0$.

Then as $n \to \infty$,

$$\sum_{i=1}^{k(n)} \xi_{n,i} \xrightarrow{\mathrm{d}} \mathcal{N}(\mu, \sigma^2).$$

Corollary 12.47 (Classical CLT)**.** *Let $X_i \overset{\text{iid}}{\sim} P$ with $\mathbb{E}[X] = 0$ and $\mathbb{V}(X) = 1$. Then,*

$$\frac{X_1 + \cdots + X_n}{\sqrt{n}} \xrightarrow{\text{d}} \mathcal{N}(0, 1).$$

Theorem 12.48 (Lindeberg CLT)**.** *Consider a triangular array $\{X_{n,i}\}$ such that*

(i) $\mathbb{E}X_{n,i} = 0$.

(ii) $\sum \mathbb{V}(X_{n,i}) = 1$.

(iii) ***(Lindeberg condition)*** *For each $\varepsilon > 0$,*

$$L_n(\varepsilon) \coloneqq \sum_i \mathbb{E}\left[X_{n,i}^2 \mathbb{1}\{|X_{n,i}| > \varepsilon\}\right] \longrightarrow 0.$$

Then,

$$\sum_i X_{n,i} \xrightarrow{\text{d}} \mathcal{N}(0, 1).$$

Lemma 12.49. *Suppose H_n is such that for any $\varepsilon > 0$, we have $H_n(\varepsilon) \to 0$. Then, there exists a sequence $\varepsilon_n \to 0$ such that $H_n(\varepsilon_n) \to 0$.*

Theorem 12.50. *Consider independent $X_{ni}, \ldots, X_{nn}$ with $\mathbb{E}X_{ni} = 0$. Write*

$$S_n \coloneqq \sqrt{\sum_i \mathbb{E}X_{ni}^2}.$$

If the Lindeberg condition

$$\frac{1}{S_n^2} \sum_i \mathbb{E}\left[X_{n,i}^2 \mathbb{1}\{|X_{n,i}| > \varepsilon S_n\}\right] \longrightarrow 0, \quad \forall \varepsilon > 0$$

holds, then

$$\frac{X_{n1} + \cdots + X_{nn}}{S_n} \xrightarrow{\text{d}} \mathcal{N}(0, 1).$$

Theorem 12.51 (Lyapunov CLT)**.** *Consider independent $X_{ni}, \ldots, X_{nn}$ with $\mathbb{E}X_{ni} = 0$. Write*

$$S_n \coloneqq \sqrt{\sum_i \mathbb{E}X_{ni}^2}.$$

If for some $\delta > 0$,

$$\frac{1}{S_n^{2+\delta}} \sum_i \mathbb{E}|X_{n,i}|^{2+\delta} \longrightarrow 0,$$

then

$$\frac{X_{n1} + \cdots + X_{nn}}{S_n} \xrightarrow{\text{d}} \mathcal{N}(0, 1).$$

Lemma 12.52 (Stein's Identity)**.** *Let $Z \sim \mathcal{N}(0, 1)$. Let f be absolutely continuous and such that $\mathbb{E}|f'(Z)| < \infty$, $\mathbb{E}|Zf(Z)| < \infty$. Then, $\mathbb{E}f'(Z) = \mathbb{E}Zf(Z)$.*

Proposition 12.53. *If W is such that $\mathbb{E}f'(W) = \mathbb{E}Wf(W)$ for all f such that $\mathbb{E}|f'(Z)| < \infty$ and $\mathbb{E}|Zf(Z)| < \infty$, then $W \sim \mathcal{N}(0, 1)$.*

Lemma 12.54. *Stein's equation $f_h'(w) - wf_h(w) = h(w) - Nh$, where $Nh \coloneqq \mathbb{E}h(\mathcal{N}(0, 1))$ has solution*

$$f_h(w) = e^{w^2/2} \int_{-\infty}^{w} e^{-x^2/2} \left(h(x) - Nh\right) \, \mathrm{d}x.$$

Lemma 12.55. $\|f_h\|_\infty, \|f_h'\|_\infty, \|f_h''\|_\infty \leq 2\|h'\|_\infty$.

Proposition 12.56. *Let $Z \sim \mathcal{N}(0, 1)$. Then, for each absolutely continuous f, $\mathbb{E}f'(Z) = \mathbb{E}Zf(Z)$ provided that both sides exist. Conversely, if W is such that $\mathbb{E}|W| < \infty$ and $\mathbb{E}f'(W) = \mathbb{E}Wf(W)$ for all bounded absolutely continuous f with bounded derivative, then $W \sim \mathcal{N}(0, 1)$.*

Theorem 12.57. *Let $X_1, \dots, X_n$ be independent, each with mean 0 and variance 1. Then,*

$$\mathcal{W}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, \mathcal{N}(0,1)\right) \le \frac{3}{\sqrt{n}} \frac{1}{n} \mathbb{E}|X_1|^3.$$

Thus if X_i has uniformly bounded third moment, then we get CLT.

Proposition 12.58.

$$\mathrm{KS}(W, Z) \le 2\sqrt{\mathcal{W}(W, Z)}.$$

Corollary 12.59. *Let $X_1, \dots, X_n$ be independent, each with mean 0 and variance 1. Then,*

$$\mathrm{KS}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, \mathcal{N}(0,1)\right) \le 2\sqrt{\frac{3}{\sqrt{n}} \frac{1}{n} \sum_i \mathbb{E}|X_i|^3}.$$

Proposition 12.60 (Berry-Esseen)**.** *Let $X_1, \dots, X_n$ be independent, each with mean 0 and variance 1. Then,*

$$\mathrm{KS}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, \mathcal{N}(0,1)\right) \le \frac{C}{\sqrt{n}} \frac{1}{n} \sum_i \mathbb{E}|X_i|^3,$$

for some universal constant C.

Theorem 12.61 (Maximal coupling, discrete case)**.** *Let X, Y be discrete, and wlog, take values in $\mathbb{N}$. Write $p_i \coloneqq \mathbb{P}(X = i)$ and $q_i \coloneqq \mathbb{P}(Y = i)$. The maximal coupling of (X, Y) satisfies*

$$\mathbb{P}(\hat{X} \ne \hat{Y}) = \sum_i \min\{p_i, q_i\} = \sum_i p_i \wedge q_i.$$

Theorem 12.62 (Maximal coupling, continuous case)**.** *Let P and Q be probability measures on $\mathbb{R}$ with densities p and q with respect to Lebesgue measure. Then the maximal coupling $(\hat{X}, \hat{Y})$ satisfies $\mathbb{P}(\hat{X} = \hat{Y}) = \int p(x) \wedge q(x)\, \mathrm{d}x$.*

Proposition 12.63. *Let P, Q be measures and let $(\hat{X}, \hat{Y})$ be a maximal coupling of (P, Q). Then, $\mathrm{TV}(P, Q) = \mathbb{P}(\hat{X} \ne \hat{Y})$.*

Theorem 12.64 (Le Cam)**.** *Let $X_i \sim \mathrm{Bernoulli}(p_i)$ be independent for $i = 1, \dots, n$. Define $\lambda \coloneqq \sum_{i=1}^n p_i$. Then,*

$$\mathrm{TV}\left(\sum_{i=1}^n X_i, \mathrm{Poisson}(\lambda)\right) \le \sum_{i=1}^n p_i^2.$$

Lemma 12.65 (Stein-Chen)**.** *$Z \sim \mathrm{Poisson}(\lambda)$ if and only if $\mathbb{E}Zf(Z) = \lambda \mathbb{E}f(Z+1)$ for all f such that both expectations are finite.*

Lemma 12.66. *For each A, the solution $f_A(k+1) = \frac{k!}{\lambda^{k+1}} \sum_{j=0}^k \frac{\lambda^j}{j!} (\mathbb{1}\{j \in A\} - \mathbb{P}(Z \in A))$ to the Stein-Chen equation is Lipschitz:*

$$|f_A(k) - f_A(\ell)| \le \min\left\{1, \frac{1}{\lambda}\right\} |k - \ell|, \quad k, \ell = 0, 1, \dots.$$

Proposition 12.67. *If $Z \sim \mathrm{Poisson}(\lambda)$, then*

$$\begin{aligned}
\mathrm{TV}(W, Z) &= \sup_A |\mathbb{P}(W \in A) - \mathbb{P}(Z \in A)| \\
&= \sup_A |\mathbb{E}\left(Wf_A(W) - \lambda f_A(W+1)\right)| \\
&\le \sup_{\mathrm{Lip}(f) \le \min\{1, 1/\lambda\}} |\mathbb{E}\left(Wf(W) - \lambda f(W+1)\right)|.
\end{aligned}$$

Theorem 12.68 (Stein-Chen)**.** *Let $X_i \sim \mathrm{Bernoulli}(p_i)$ be independent for $i = 1, \dots, n$ and $\lambda \coloneqq \sum_{i=1}^n p_i$. Then,*

$$\mathrm{TV}\left(\sum_{i=1}^n X_i, \mathrm{Poisson}(\lambda)\right) \le \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^n p_i^2,$$

Theorem 12.69. *Let $X_i \sim \text{Bernoulli}(p_i)$ be random variables with dependency neighborhoods N_i for $i = 1, \ldots, n$. Let $\lambda := \sum_{i=1}^n p_i$. Then,*

$$\text{TV}\left(\sum_{i=1}^n X_i, \text{Poisson}(\lambda)\right) \leq \min\left\{1, \frac{1}{\lambda}\right\}\left(\sum_{i=1}^n \sum_{j \in N_i \setminus \{i\}} \mathbb{E}X_i X_j + \sum_{i=1}^n \sum_{j \in N_i} p_i p_j\right).$$

Proposition 12.70. *For any signed measure φ, we have either $-\infty < \varphi(A) \leq \infty$ for each $A \in \mathcal{F}$, or $-\infty \leq \varphi(A) < \infty$ for each $A \in \mathcal{F}$.*

Theorem 12.71 (Hahn Decomposition)**.** *For any signed measure φ, there exists a partition $\Omega = \Omega^+ \cup \Omega^-$ such that*

$$\varphi\left(A \cap \Omega^+\right) \geq 0 \geq \varphi\left(A \cap \Omega^-\right), \quad A \in \mathcal{F}.$$

This decomposition is unique in the following sense: if $\Omega = \Omega_1^+ \cup \Omega_1^-$ is another such decomposition, then any subset of $\Omega^+ \triangle \Omega_1^+$ or $\Omega^- \triangle \Omega_1^-$ is a null set of φ.

Theorem 12.72 (Radon-Nikodym)**.** *Suppose φ and μ are, respectively, a signed measure and a σ-finite measure on $(\Omega, \mathcal{F})$ and $\varphi \ll \mu$. Then there exists a μ-a.e. unique function which we denote $\mathrm{d}\varphi/\mathrm{d}\mu$ such that*

$$\varphi(A) = \int_A \frac{\mathrm{d}\varphi}{\mathrm{d}\mu}\, \mathrm{d}\mu, \quad \forall A \in \mathcal{F}.$$

Lemma 12.73 (Doob-Dynkin)**.** *Consider $Y : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ and $Z : (\Omega, \sigma(Y)) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ (so $\sigma(Z) \subset \sigma(Y)$). Then there exists $f : (\mathbb{R}, \mathcal{B}_{\mathbb{R}}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ such that $Z = f(Y)$.*

Corollary 12.74. *The conditional expectation $\mathbb{E}[X|Y]$ is a function of Y.*

Proposition 12.75 (Properties of Conditional Expectation)**.**

(i) *If $\sigma(X) \subset \mathcal{G}$, the $\mathbb{E}[X|\mathcal{G}] = X$.*

(ii) *If $\sigma(X) \perp\!\!\!\perp \mathcal{G}$, then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}X$.*

(iii) *If $\mathcal{G}_0 \subset \mathcal{G}$, then*
$$\mathbb{E}\left[\mathbb{E}(X|\mathcal{G})|\mathcal{G}_0\right] = \mathbb{E}\left[\mathbb{E}(X|\mathcal{G}_0)|\mathcal{G}\right] = \mathbb{E}[X|\mathcal{G}_0].$$

(iv) *$X \leq Y$ implies $\mathbb{E}[X|\mathcal{G}] \leq \mathbb{E}[Y|\mathcal{G}]$.*

(v) *$\mathbb{E}[aX + bY|\mathcal{G}] = a\mathbb{E}[X|\mathcal{G}] + b\mathbb{E}[Y|\mathcal{G}]$.*

(vi) *If $0 \leq X_n \uparrow X$, then $\mathbb{E}[X_n|\mathcal{G}] \uparrow \mathbb{E}[X|\mathcal{G}]$.*

(vii) *If $X_n \geq 0$, then*
$$\mathbb{E}[\liminf_n X_n|\mathcal{G}] \leq \liminf_n \mathbb{E}[X_n|\mathcal{G}].$$

(viii) *If $X_n \to X$, $|X_n| \leq Y$, $\mathbb{E}Y < \infty$, then*
$$\mathbb{E}[X_n|\mathcal{G}] \to \mathbb{E}[X|\mathcal{G}].$$

(ix) *If $\sigma(Y) \subset \mathcal{G}$, then*
$$\mathbb{E}[XY|\mathcal{G}] = Y\mathbb{E}[X|\mathcal{G}].$$

Of course, all of these statements are up to $\mathcal{G}$-null sets.

13 Appendix B: Common Distributions

$X \sim$	$\operatorname{supp} \mathbb{P}(X = \cdot)$	$\mathbb{P}(X = x)$	$\mathbb{E}$	$\mathbb{V}$	Notes
Binomial(n, p)	$\{0, 1, \ldots, n\}$	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$	$\mathbb{E}[X^2] = np(1-p+np)$ $\mathbb{E}[X(X-1)] = n(n-1)p^2$
Geometric(p)	$\mathbb{N}$	$(1-p)^{x-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\mathbb{E}[X^2] = \frac{2-p}{p^2}$
Poisson(λ)	$\mathbb{N} \cup \{0\}$	$\frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ	$\mathbb{E}[X^2] = \lambda(\lambda+1)$ $\mathbb{E}[X^{\underline{k}}] = \lambda^k$

$X^{\underline{k}} := X(X-1) \cdots (X-k+1)$ denotes the falling factorial (factorial moment).

$X \sim$	$\operatorname{supp} f$	f	$\mathbb{E}$	$\mathbb{V}$	Notes
Uniform(a, b)	$[a, b]$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\mathbb{E}[X^2] = \frac{a^2+ab+b^2}{3}$
$\mathcal{N}(\mu, \sigma^2)$	$(-\infty, \infty)$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$\mathbb{E}[(X-\mu)^{2k}] = (2k-1)!!\,\sigma^{2k}$ odd central moments = 0 $\mathbb{E}[\lvert X-\mu \rvert] = \sigma\sqrt{2/\pi}$
Exponential(λ) = Gamma$(1, \lambda)$	$[0, \infty)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\mathbb{E}[X^k] = \frac{k!}{\lambda^k}$
Gamma(α, β)	$(0, \infty)$	$\frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)}$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\mathbb{E}[X^k] = \frac{\alpha^{(k)}}{\beta^k}$
Beta(α, β)	$(0, 1)$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$\mathbb{E}[X^k] = \frac{\alpha^{(k)}}{(\alpha+\beta)^{(k)}}$

$\alpha^{(k)} := \alpha(\alpha+1) \cdots (\alpha+k-1)$ denotes the rising factorial.

Definition 13.1. Let $Z_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$.

- Then
$$\sum Z_i^2 \sim \chi_k^2 = \text{Gamma}\left(\frac{k}{2}, \frac{1}{2}\right).$$
- If $W \sim \chi_n^2$ and $Z \perp\!\!\!\perp W$, then
$$\frac{Z}{\sqrt{W/k}} \sim t_k.$$
- If $W_1 \sim \chi_{k_1}^2$, $W_2 \sim \chi_{k_2}^2$, and $W_1 \perp\!\!\!\perp W_2$, then
$$\frac{W_1/k_1}{W_2/k_2} \sim F_{k_1,k_2}.$$

Note that $\operatorname{supp} F = \mathbb{R}_+$, and $\mathbb{E}[F] = k_2/(k_1 + k_2) \approx 1$.

Proposition 13.2 (Approximations of Binomial)**.**

- *If* $np_n \to \lambda$*, then* Binomial$(n, p) \to$ Poisson(λ).
- *Using CLT,* Binomial$(n, p) \approx \mathcal{N}(np, np(1-p))$.

Proposition 13.3 ($\mathcal{N}$)**.** *Let* $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ *and put* $S^2 := \frac{1}{n-1}\sum(X_i - \overline{X})^2$*. Then*

(i) $\mathbb{E}[S^2] = \sigma^2$. $\overline{X} \perp\!\!\!\perp S^2$.

(ii) $\sqrt{n}\frac{\overline{X}_n - \mu}{\sigma} \sim \mathcal{N}(0,1)$.

(iii) $(n-1)\frac{S^2}{\sigma^2} = \frac{1}{\sigma^2}\sum(X_i - \overline{X})^2 \sim \chi^2_{n-1}$.

(iv) $\sqrt{n}\frac{\overline{X} - \mu}{S} \sim t_{n-1}$.

Proposition 13.4 (Multivariate Normal)**.** *Let* $(X, Y) \sim \mathcal{N}_2$.

- $\mathbb{C}(X, Y) = 0$ *if and only if* $X \perp\!\!\!\perp Y$.
- *The MLEs are given by*

$$\hat{\mu}_X = \overline{X}, \quad \hat{\sigma}^2_Y =^{-1} \sum(X_i - \overline{X})^2, \quad R := \hat{\rho} = \frac{\sum(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum(X_i - \overline{X})^2 \sum(Y_i - \overline{Y})^2}}.$$

  R is known as the Pearson correlation coefficient.

- $(X, Y) \sim \mathcal{N}_2$ *if and only if* $aX + bY \sim \mathcal{N}$ *for any* $a, b \in \mathbb{R}$.
- *The distribution of* X *conditional on* $Y = y$ *is given by*

$$X|Y = y \sim \mathcal{N}\left(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y - \mu_Y), \sigma^2_X(1 - \rho^2)\right).$$

Let $\mathbf{X} \sim \mathcal{N}_k(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. *If* $\boldsymbol{\Sigma}$ *is positive definite, then* $\mathbf{X}$ *has density*

$$f(X) = \frac{1}{(2\pi)^{k/2}\det(\boldsymbol{\Sigma})}\exp\left(-\frac{1}{2}(\mathbf{X} - \boldsymbol{\mu})^\intercal \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})\right).$$

Proposition 13.5 (Exponential Distribution, "Memoryless")**.**

$$\mathbb{P}(T \leq x + y | T > x) = \mathbb{P}(T \leq y).$$

Proposition 13.6 (Gamma Distribution)**.** *Let* $X \sim \text{Gamma}(\alpha, \beta)$.

(i) *If* $X_i \overset{\text{iid}}{\sim} \text{Gamma}(\alpha_i, \beta)$, *then* $\sum X_i \sim \text{Gamma}\left(\sum \alpha_i, \beta\right)$.

(ii) *If* $\alpha > 1$, *then* $\mathbb{E}\left[1/X\right] = \frac{\beta}{\alpha - 1}$.

(iii) $\beta X \sim \text{Gamma}\left(\alpha, 1\right)$.

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